arXiv:submit/7437113 [math.AG] 2 Apr 2026

A MOTIVIC POISSON FORMULA FOR SPLIT ALGEBRAIC TORI WITH AN APPLICATION TO MOTIVIC HEIGHT ZETA FUNCTIONS

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Abstract. We prove a motivic version of the Poisson formula on the adelic points of a split algebraic torus and apply it to the study of the motivic height zeta function of split projective toric varieties, in the context of the motivic Manin-Peyre principle.

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Introduction

Harmonic analysis stands as one of the foundational tools in the number theorist's arsenal for addressing counting problems. Within the framework of the Batyrev-Manin-Peyre conjectures—which predict the distribution of rational points on Fano varieties over global fields—this method has been instrumental since the field's inception.

Classically, the counting problem is encoded into a height zeta function on which harmonic analysis is performed. In that context, an important role is played by the Poisson formula.

The goal of the present work is to state and prove a geometric-motivic analogue of the Poisson formula for split algebraic tori and apply it to the study of the motivic height zeta function of toric varieties over function fields of curves. In particular, we recover a motivic stabilisation result for the class of the moduli space of high degree morphisms from a smooth and geometrically irreducible projective curve to a smooth split projective toric variety that was previously obtained by the second author using a different approach [Fai25b, Fai25a]. This work also extends residue-type results of Bourqui [Bou09] and Bilu–Das–Howe [BDH22].

The classical setting. Let us recall the classical Poisson formula in its form used by Batyrev and Tschinkel in their proof of Manin's conjecture for toric varieties over number fields [BT95, BT98] and function fields of curves over finite fields [Bou11a]. Let $\mathscr{G}$ be a locally compact topological commutative group, endowed with a certain Haar measure $\mathrm{d}x$, and let $\mathscr{H} \subset \mathscr{G}$ be a cocompact discrete subgroup of $\mathscr{G}$. We give ourselves a function $f : \mathscr{H} \to \mathbf{R}$ and assume that f can be extended to an L^1-function on the whole of $\mathscr{G}$. Finally, up to normalising $\mathrm{d}x$, one can assume that $\mathscr{H}$ has covolume 1 in $\mathscr{G}$.

Assume that the Fourier transform $\hat{f}$ of f with respect to $\mathrm{d}x$ is an L^1-function on the group $\mathscr{H}^\perp$ of characters of $\mathscr{G}$ which are trivial on $\mathscr{H}$. Then

$$\sum_{x\in\mathscr{H}} f(x) = \sum_{\chi\in\mathscr{H}^\perp} \hat{f}(\chi). \tag{1}$$

Given a global field K, Batyrev–Tschinkel and Bourqui apply this formula to the adelic points

$$\mathscr{G} = U(\mathbb{A}_K)$$

of an algebraic torus U defined over K, with $\mathscr{H} = U(K)$ being its set of K-rational points diagonally embedded in $U(\mathbb{A}_K)$.

Moduli spaces of curves and motivic Batyrev-Manin-Peyre principle. Assume now that K is the function field of a geometrically irreducible smooth projective curve $\mathscr{C}$ defined over an absolute base field k of characteristic zero and let X be a projective variety defined over k. By the valuative criterion of properness, the set of K-rational points of X can be identified with the set of k-morphisms $\mathscr{C} \to X$. Such morphisms are parametrised by a k-scheme $\mathrm{Hom}_k(\mathscr{C}, X)$. Moreover, any morphism $f : \mathscr{C} \to X$ induces a linear form $\deg(f) \in \mathrm{Pic}(X)^\vee$, called the *multidegree*, which sends the class of a line bundle to the degree of its pull-back to the curve $\mathscr{C}$. For any $\delta \in \mathrm{Pic}(X)^\vee$, the condition $\deg(f) = \alpha$ defines a subscheme $\mathrm{Hom}_k^\delta(\mathscr{C}, X)$ of finite type. Finally, if U is a non-empty open subset of X, we denote by $\mathrm{Hom}_k^\delta(\mathscr{C}, X)_U$ the submoduli space whose points correspond to morphisms sending the generic point of $\mathscr{C}$ into U.

Assume that X is a Fano-like variety in the sense of [Fai25b, Definition 1] (for example, a Fano variety or a split smooth projective toric variety). The motivic Batyrev-Manin-Peyre principle is a set of predictions concerning the behaviour of the class of $\mathrm{Hom}_k^\delta(\mathscr{C}, X)_U$ in a suitable localisation of the Grothendieck ring of k-varieties $\mathrm{K}_0\mathbf{Var}_k$. As a group, $\mathrm{K}_0\mathbf{Var}_k$ is defined by generators and relations: generators are isomorphism classes $[Y]$ of k-varieties and relations are of the form

$$[Y] - [Z] - [Y - Z]$$

whenever Z is a closed subscheme of a variety Y. The ring structure of $\mathrm{K}_0\mathbf{Var}_k$ is given by taking Cartesian products of varieties:

$$[Y_1][Y_2] = [Y_1 \times_k Y_2].$$

The class of the affine line plays a special role and is denoted by $\mathbf{L}_k$. It is convenient to invert it and work in the localisation

$$\mathscr{M}_k = \mathrm{K}_0\mathbf{Var}_k[\mathbf{L}_k^{-1}].$$

The ring $\mathscr{M}_k$ admits a filtration by the virtual dimension and a corresponding completion $\widehat{\mathscr{M}_k}^{\dim}$ which allows one to define sums of absolutely convergent series.

The most basic version of the motivic Batyrev-Manin-Peyre principle asks whether the normalised class

$$\left[\mathrm{Hom}_k^\delta(\mathscr{C}, X)_U\right] \mathbf{L}_k^{-\delta\cdot\omega_X^{-1}} \tag{2}$$

stabilises (if necessary, in a well-chosen completion of $\mathscr{M}_k$) as $\delta \in \mathrm{Pic}(X)^\vee$ lies inside the movable cone of X and goes arbitrarily far away from its boundaries. This question is closely related to the study of the following series.

Definition. Let $\mathbf{T}$ be a family of formal variables indexed by a basis of $\mathrm{Pic}(X)^\vee$. The multivariate motivic height zeta function is the formal series

$$\zeta_H^{\mathrm{mot}}(\mathbf{T}) = \sum_{\delta\in\mathrm{Pic}(X)^\vee} \left[\mathrm{Hom}_k^\delta(\mathscr{C}, X)_U\right] \mathbf{T}^\delta.$$

Split toric varieties and main results. In this article we focus on the case of smooth, projective and split toric varieties.

Let $U \simeq \mathbf{G}_m^n$ be a split torus and $X = X_\Sigma$ a split smooth projective variety over a field k compactifying U, defined by a certain fan Σ of the lattice of cocharacters $\mathcal{X}_*(U) = \mathrm{Hom}_{\mathrm{gp}}(\mathbf{G}_m, U)$ of U. Let r be the Picard rank of X_Σ and ω_{X_Σ} its canonical line bundle.

Theorem 1 (Meromorphic continuation)**.** *There exists an $\eta > 0$ and an integer a_{X_Σ} such that the formal series*

$$(1 - (\mathbf{L}T)^{a_{X_\Sigma}})^{\mathrm{rk}(\mathrm{Pic}(X_\Sigma))} \zeta_H^{\mathrm{mot}}\left(T^{\omega_{X_\Sigma}^{-1}}\right)$$

converges for $|T| < \mathbf{L}_k^{-1+\eta}$, with its value at $\mathbf{L}^{-1}$ being a non-zero element of $\widehat{\mathscr{M}_k}^{\dim}$.

We refer to Theorem 6.14 for a more detailed version. The following second statement is compatible with the predictions from [Fai25b].

Theorem 2 (Multi-height motivic stabilisation)**.** *As the class*

$$\delta \in \mathrm{Pic}(X_\Sigma)^\vee \cap \mathrm{Eff}(X_\Sigma)^\vee$$

goes arbitrarily far away from the boundary of $\mathrm{Eff}(X_\Sigma)^\vee$, the normalised class

$$\left[\mathrm{Hom}_k^\delta(\mathscr{C}, X_\Sigma)_U\right] \mathbf{L}_k^{-\delta \cdot \omega_{X_\Sigma}^{-1}}$$

tends to the non-zero motivic Euler product

$$\mathbf{L}_k^{(1-g(\mathscr{C}))\dim(X_\Sigma)} \left(\frac{\left[\mathrm{Pic}^0(\mathscr{C})\right] \mathbf{L}_k^{1-g(\mathscr{C})}}{\mathbf{L}_k - 1}\right)^{\mathrm{rk}(\mathrm{Pic}(X_\Sigma))} \prod_{p \in \mathscr{C}} \left(1 - \mathbf{L}_{\kappa(p)}^{-1}\right)^{\mathrm{rk}(\mathrm{Pic}(X_\Sigma))} \frac{[X_{\kappa(p)}]}{\mathbf{L}_{\kappa(p)}^{\dim(X_\Sigma)}}.$$

Relation with earlier works.

The motivic Poisson formula for additive groups. The first occurrence of harmonic analysis in the motivic setting was the motivic Poisson formula of Hrushovski and Kazhdan [HK09]. While originally formulated in the language of model theory of valued fields, it was translated in [CLL16] into a more geometric context, where it corresponds to a motivic analogue of the formula (1) in the specific case when $\mathscr{G} = \mathbf{G}_a^n(\mathbf{A}_F)$ are the adelic points of the additive group scheme $\mathbf{G}_a^n$ for $F = k(\mathscr{C})$ the function field of a curve, and $\mathscr{H} = \mathbf{G}_a^n(F)$, diagonally embedded into $\mathscr{G}$. In the motivic setting there is no local compactness, and the classical theory of integration is replaced by a form of motivic integration, with the motivic Poisson formula being an equality between two classes in the so-called *Grothendieck ring of varieties with exponentials*. The first ingredient of the construction is the definition of *motivic Schwartz-Bruhat functions*, which are motivic analogues of locally constant functions with compact support, and which in this setting are elements of appropriate relative Grothendieck rings with exponentials. Integration and Fourier transformation operators are defined as pushforwards between such Grothendieck rings. Thanks to the self-duality properties of the additive group, the objects appearing on both sides of the Poisson formula are of the same nature, given as the image of a summation operator applied to a motivic Schwartz-Bruhat function and its Fourier transform, respectively. The Poisson formula itself is proved by first reducing to (analogues of) characteristic functions of balls, where it follows from the Riemann-Roch theorem and Serre duality of the curve $\mathscr{C}$. This Poisson formula was used successively in [CLL16], [Bil23] and [Fai23] to study moduli spaces of curves on equivariant compactifications of vector groups.

Previous work for toric varieties in the motivic setting. The first one to study Manin's problem for split toric varieties in the motivic setting was Bourqui [Bou09], obtaining a convergence result for the motivic height zeta function in the Grothendieck ring of Chow motives. Combining Bourqui's work with the notion of motivic Euler product from [Bil23] led to a lift of Bourqui's result to the Grothendieck ring of varieties (and even its generalisation to so-called Hadamard convergence) in [BDH22], and to the full motivic Batyrev-Manin-Peyre principle in [Fai25b] (for rational curves) and [Fai25a] (in arbitrary genus). All of these results were based on the universal torsor method, and the present paper is the first one to produce a functioning Fourier theory for algebraic tori in the motivic setting.

Main ideas. As for the motivic Poisson formula in the additive setting, the first step is an appropriate variant of the Grothendieck ring of varieties. To explain our choice, let us first describe the objects that we will need to work with.

Let $F = k(\mathscr{C})$ be the function field of a smooth projective curve $\mathscr{C}/k$, and for every closed point $v \in \mathscr{C}$, denote by F_v the corresponding completion. For the sake of simplicity, we specialize in this paragraph to the case where our torus U is one-dimensional: $U = \mathbf{G}_m$, so that the local height functions which we want to build motivic analogues of are functions $f : F_v^\times \to \mathbf{C}$ which are invariant modulo the group of units $\mathcal{O}_v^\times$, and thus may be thought of as functions on $F_v^\times/\mathcal{O}_v^\times \simeq \mathbf{Z}$ (where the isomorphism is given by the valuation), which also turn out to be finitely supported. Thus, our families of local functions will naturally arise as elements of a relative Grothendieck ring over the constant group scheme $\underline{\mathbf{Z}}_{\mathscr{C}}$. For a more general torus U, $\mathbf{Z}$ will be replaced by the lattice of cocharacters $\mathcal{X}_*(U)$.

As opposed to the additive setting described above, there is no self-duality in the multiplicative setting, and Fourier transforms are *a priori* objects of a different nature: they are motivic functions over the Cartier dual $D(\underline{\mathbf{Z}}_{\mathscr{C}})$, which is isomorphic to $\mathbf{G}_{m,\mathscr{C}}$. We introduce in a *Grothendieck ring of varieties with characters* where characters are encoded into a separate component. More precisely, given an abstract commutative group M that is finitely generated, a typical generator of the ring $\mathrm{K}_0\mathbf{Char}_M\mathbf{Var}_{\mathscr{C}}$ is of the form

$$[X \times_{\mathscr{C}} D(\underline{L}_{\mathscr{C}}), \chi \mapsto \chi(g(m))]$$

where X is a variety over $\mathscr{C}$, $g : M \to L$ is a morphism of groups, $m \in M$ and the quotient by cut-and-paste relations is only on the first factor. This ring comes with a corresponding "integration over characters" $\int_{D(M)}$ operation which kills the second component when $g(m) \neq 0$, compatibly with the usual orthogonality relations. It is interesting to note that this setup bears some similarities with the one of [CLNV24], though a precise comparison is yet to be made.

We start by building a local Fourier theory using this framework, before achieving the passage from local to global via the notion of motivic Euler product as in [Bil23]. The role of the idèles in this setting is played by the group of divisors $\mathrm{Div}(\mathscr{C})$ on the curve $\mathscr{C}$, with the multiplicative group of the field being represented by the subgroup of principal divisors $\mathrm{PDiv}(\mathscr{C})$. Both of these groups are obtained as limits, as S varies, of the corresponding subgroups $\mathrm{Div}_{\mathsf{S}}(\mathscr{C})$ (resp. $\mathrm{PDiv}_{\mathsf{S}}(\mathscr{C})$) with support restricted to a finite set of places S. In this context, our local Fourier theory applies, giving local integration operators $\int_{D(\mathrm{Div}_{\mathsf{S}}(\mathscr{C})/\,\mathrm{PDiv}_{\mathsf{S}}(\mathscr{C}))}$ satisfying a Poisson formula. From these local integration operators, we then build a global integration operator $\int_{\mathrm{PDiv}^\perp}$ which corresponds to integrating over characters trivial on $\mathrm{PDiv}(\mathscr{C})$ and prove a global Poisson formula. The latter may be applied to the motivic height zeta function, rewriting it analogously to the proofs of Batyrev-Tschinkel and Bourqui.

Organisation of the paper. In Section 1, we start with recalling the classical definitions of rings of varieties before introducing our new Grothendieck ring of varieties with characters.

We proceed the same way in Section 2 where we first recall the definition of motivic Euler products from [Bil23] before extending the notion to motivic functions on constant group schemes and on their Cartier duals.

We define Fourier transforms of motivic functions on constant group schemes in Section 3, where we obtain Fourier inversion and Poisson formulas. This framework is adapted to the toric setting in Section 4, resulting in a global Poisson formula for split tori.

We eventually apply our theory in the two final sections: in Section 5 we apply the motivic Poisson formula to the motivic height zeta function and the final analysis of the contributions of the Fourier transforms is performed in Section 6.

Notations and conventions. To avoid confusion, in this article U will always be a split n-dimensional torus over the absolute base field k, while we will use a capital T for indeterminates and its bold version $\mathbf{T}$ for sets of indeterminates. On the other hand, lowercase t will usually denote a local parameter.

The k-curve $\mathscr{C}$ is assumed to be smooth, projective and geometrically irreducible. Moreover, we assume that it admits a k-divisor of degree one (this additional assumption is automatically satisfied, for example, if $k = \mathbf{F}_q$ or if k is algebraically closed). Its function field is denoted by F and its set $|\mathscr{C}|$ of closed points is identified with the set of discrete valuations of F. For each $v \in |\mathscr{C}|$, we denote by F_v the corresponding completion of F, with ring of integers $\mathcal{O}_v$. The adèle ring $\mathbf{A}_F$ is the restricted product of the F_v with respect to the $\mathcal{O}_v$, and the idèle ring $\mathbf{A}_F^\times$ is the restricted product of the multiplicative groups $F_v^\times$ with respect to their subgroups $\mathcal{O}_v^\times$.

We use bold letters to differentiate schemes and group schemes from respectively sets and abstract groups, with the following exception: if M is an abstract group and S a base scheme, the constant group scheme above S will be denoted by $\underline{M}_S$.

As an abstract group, the group of divisors $\mathrm{Div}(\mathscr{C})$ on the curve $\mathscr{C}$ is filtered by the subgroups $\mathrm{Div}_{\mathsf{S}}(\mathscr{C})$ of divisors with support in finite subsets S of the set $|\mathscr{C}|$ of closed points of $\mathscr{C}$, giving

$$\mathrm{Div}(\mathscr{C}) = \varinjlim_{\mathsf{S}} \mathrm{Div}_{\mathsf{S}}(\mathscr{C}) = \varinjlim_{\mathsf{S}} \bigoplus_{x \in \mathsf{S}} \mathbf{Z}_{\kappa(x)} = \bigoplus_{x \in |\mathscr{C}|} \mathbf{Z}_{\kappa(x)}.$$

This filtration restricts to degree-zero divisors

$$\mathrm{Div}^0(\mathscr{C}) = \varinjlim_{\mathsf{S}} \mathrm{Div}^0_{\mathsf{S}}(\mathscr{C})$$

and to principal divisors

$$\mathrm{PDiv}(\mathscr{C}) = \varinjlim_{\mathsf{S}} \mathrm{PDiv}_{\mathsf{S}}(\mathscr{C}).$$

Let I be a set. Elements of the free abelian monoid

$$\mathbf{N}^{(I)} = \{(n_i)_{i \in I} \in \mathbf{N},\ n_i = 0 \text{ for almost all } i\}$$

are called *generalised partitions*. If I is of the form $I_0 \setminus \{0\}$ where I_0 is a commutative monoid, and $\pi = (n_i)_{i \in I} \in \mathbf{N}^{(I)}$, we may define

$$\sum \pi = \sum_{i \in I} i n_i \in I_0.$$

If $i \in I_0$ is such that $\sum \pi = i$, we say that π is a partition of i and write $\pi \vdash i$.

Acknowledgments. MB was partially supported by the ANR projects Cyclades (ANR-23-CE40-0011) and AdAnAr (ANR-24-CE40-6184). LF acknowledges partial funding from the European Union's Horizon 2020 research and innovation programme under the Marie Skłodowska-Curie grant agreement No 101034413, as well as partial support by the KU Leuven IF grant C16/23/010.

The authors thank A. Chambert-Loir who first proposed this project to MB, as well as D. Bourqui, R. Cluckers, M. Raibaut and E. Peyre for useful conversations.

1. Grothendieck rings

This section aims to define a new variant of the ring of varieties that will allow us to handle certain families of multiplicative characters. We first start with recalling the classical definition of ring of varieties over an arbitrary base scheme.

1.1. **Rings of varieties.** Let S be a scheme. An S-variety is a finitely presented scheme morphism $X \to S$.

Definition 1.1. The *Grothendieck group of S-varieties*

$$\mathrm{K}_0\mathbf{Var}_S$$

is defined as the abelian group generated by the isomorphism classes of *S-varieties*, with relations given by the *cut-and-paste relations*

$$X - Y - U$$

whenever X is an S-variety, Y is a closed subscheme of X and U is its open complement in X. The class of an S-variety X in this ring is denoted by $[X \to S]$, $[X]_S$ or simply $[X]$ if the structural morphism is clear from the context. Let $\mathbf{L}_S$ be the class of the affine line.

A ring structure on $\mathrm{K}_0\mathbf{Var}_S$ is defined by

$$[X][Y] = [X \times_S Y]$$

for which $[\mathrm{id} : S \to S]$ is the unitary element.

A constructible subset of an S-variety is a finite union of locally closed subsets. By [CLNS18, p. 59], any constructible subset X of an S-variety admits a class $[X]$ in $\mathrm{K}_0\mathbf{Var}_S$.

Lemma 1.2 ([CLL16, Lemma 1.1.8] or [CH22, Theorem 1])**.** *Let φ be a motivic function on S, that is to say an element of $\mathrm{K}_0\mathbf{Var}_S$. If $s^*\varphi = 0$ in $\mathrm{K}_0\mathbf{Var}_{\kappa(s)}$ for all point $s \in S$ then $\varphi = 0$ in $\mathrm{K}_0\mathbf{Var}_S$.*

Definition 1.3. The localised Grothendieck ring of varieties $\mathscr{M}_S$, sometimes called *ring of motivic integration* is the localisation of $\mathrm{K}_0\mathbf{Var}_S$ at $\mathbf{L}_S$.

Definition 1.4. The ring $\mathscr{M}_S$ admits a decreasing filtration by the virtual dimension: for $m \in \mathbf{Z}$, let $\mathcal{F}^m\mathscr{M}_S$ be the subgroup of $\mathscr{M}_S$ generated by elements of the form

$$[X]\mathbf{L}_S^{-i}$$

where X is an S-variety and i an integer such that $\dim_S(X) - i \leqslant -m$. The completion of $\mathscr{M}_S$ with respect to this decreasing dimensional filtration is the projective limit

$$\widehat{\mathscr{M}_S}^{\dim} = \varprojlim \mathscr{M}_S/\mathcal{F}^m\mathscr{M}_S.$$

It comes with a morphism $\mathscr{M}_S \to \widehat{\mathscr{M}_S}^{\dim}$.

1.2. **Biduality for constant commutative group schemes.** A reference for this subsection is [DGA+62, Exp. VIII]. Let S be a scheme. Recall that the Cartier dual of a group scheme G above S is the S-group scheme

$$D(G) = \mathrm{Hom}_{S-\mathrm{gp}}(G, \mathbf{G}_{m,S})$$

of multiplicative characters of G. It is a closed subscheme of $\mathrm{Hom}_{S-\mathrm{sch}}(G, \mathbf{G}_{m,S})$ and comes with a canonical pairing of group schemes

$$\mathrm{ev} : \begin{cases} G \times_S D(G) & \longrightarrow & \mathbf{G}_m \\ (g, \chi) & \longmapsto & \chi(g). \end{cases}$$

A constant commutative group scheme Γ above S is an S-group scheme of the form

$$\Gamma = \underline{M}_S = \coprod_M S$$

where M is an abstract commutative group. A group scheme $G \to S$ is said to be *diagonalisable* if it is of the form $G = D(\underline{M}_S)$ where M is an abstract commutative group, and *locally diagonalisable* if every point of S admits an open neighborhood U such that $G_{|U}$ is diagonalisable.

Let Γ be a constant commutative group scheme above S. Then the canonical morphism

$$\Gamma \longrightarrow D(D(\Gamma))$$

is an isomorphism [DGA+62, Exp. VIII, Théorème 1.2].

Proposition 1.5 ([DGA+62, Exp. VIII, Prop. 2.1])**.** *Let $G = D(\underline{M}_S)$ be a diagonalisable group scheme over S. Then G is faithfully flat and affine over S. Moreover,*

- *M is of finite type as an abstract group if and only if $G \to S$ is of finite type ;*
- *M is finite if and only if $G \to S$ is finite, and M is torsion if and only if $G \to S$ is integral ;*
- *M is of finite type, with torsion of order coprime to the residual characteristics of S, if and only if $G \to S$ is smooth.*

Proposition 1.6. *Let M, L, L' be abstract commutative groups such that there exist group morphisms $M \to L$ and $M \to L'$. We denote by $L \times_M L'$ the corresponding pushout in the category of abelian groups, i.e. the universal object fitting into the diagram*

$$\begin{array}{ccc} M & \longrightarrow & L \\ \downarrow & & \downarrow \\ L' & \longrightarrow & L \times_M L'. \end{array}$$

Given a scheme S, the canonical morphisms $D(\underline{L \times_M L'}_S) \to D(\underline{L}_S)$ and $D(\underline{L \times_M L'}_S) \to D(\underline{L'}_S)$ induce an isomorphism

$$D(\underline{L \times_M L'}_S) \to D(\underline{L}_S) \times_{D(\underline{M}_S)} D(\underline{L'}_S)$$

of $D(\underline{M}_S)$-schemes, explicitly given by $\chi \mapsto (\chi_{|L}, \chi_{|L'})$. Moreover, for all $l \in L, l' \in L'$, this isomorphism makes the diagram

$$\begin{array}{ccc} D(\underline{L \times_M L'}_S) & \longrightarrow & D(\underline{L}_S) \times_{D(\underline{M}_S)} D(\underline{L'}_S) \\ \Big\downarrow{\scriptstyle \mathrm{ev}((l,l'),\cdot)} & & \Big\downarrow{\scriptstyle (\mathrm{ev}(l,\cdot),\mathrm{ev}(l',\cdot))} \\ \mathbf{G}_m & \xleftarrow[\text{group law}]{} & \mathbf{G}_m^2 \end{array}$$

commute.

1.3. **Ring of varieties with multiplicative characters.** Let S be a scheme.

Remark 1.7. By [DGA+62, Exp. VIII, Corollaire 1.3], any S-group morphism

$$D(\underline{L}_S) \to \mathbf{G}_{m,S}$$

is defined by a unique section of $\underline{L}_S$, that is to say a locally constant map $S \to L$. In particular, if S is connected, $D(\underline{L}_S) \to \mathbf{G}_{m,S}$ is uniquely determined by an element of L.

More generally, if M and N are abstract commutative groups, the natural morphism

$$\mathrm{Hom}_{\mathrm{gp}}(M,N) \longrightarrow \mathrm{Hom}_{S-\mathrm{gp}}(D(\underline{N}_S), D(\underline{M}_S))$$

is an isomorphism. If moreover M is of finite type, the morphism of S-functors

$$\underline{\mathrm{Hom}_{\mathrm{gp}}(M,N)}_S \longrightarrow \mathbf{Hom}_{S-\mathrm{gp}}(\underline{M}_S, \underline{N}_S)$$

is an isomorphism, from which one deduces that $\underline{\mathrm{Hom}_{\mathrm{gp}}(M,N)}_S$, represents the functor

$$\mathbf{Hom}_{S-\mathrm{gp}}(D(\underline{N}_S), D(\underline{M}_S)).$$

Again, if moreover S is connected, $\mathrm{Hom}_{S-\mathrm{gp}}(D(\underline{N}_S), D(\underline{M}_S))$ is isomorphic to

$$\mathrm{Hom}_{\mathrm{gp}}(M,N).$$

Definition 1.8. Let M be an abstract commutative group. The Grothendieck group of S-varieties with characters on M is the $\mathbf{Z}$-module

$$\mathrm{K}_0\mathbf{Char}_M\mathbf{Var}_S$$

generated by isomorphism classes

$$[X \times_S D(\underline{L}_S), h]_{S \times D(\underline{M}_S)}$$

of diagrams

$$(X \times_S D(\underline{L}_S), h : D(\underline{L}_X) \to \mathbf{G}_m) = \begin{array}{ccccc} X \times_S D(\underline{L}_S) & = & D(\underline{L}_X) & \xrightarrow{h} & \mathbf{G}_m \\ \downarrow{\scriptstyle (f,g)} & & & & \\ S \times_S D(\underline{M}_S) & & & & \end{array} \tag{1.1}$$

where

- $f : X \to S$ is an S-variety,
- g corresponds by Cartier duality to a morphism $\underline{M}_S \to \underline{L}_S$ of constant S-groups,
- $h \in \mathrm{Hom}_{\mathrm{gp}}(D(\underline{L}_X), \mathbf{G}_{m,X})$ is given by the evaluation at an element belonging to the image of $\underline{M}_S \to \underline{L}_S$.

$$\left[\begin{array}{ccccc} X \times_S D(\underline{L}_S) & = & D(\underline{L}_X) & \xrightarrow{h} & \mathbf{G}_m \\ \downarrow{\scriptstyle (f,g)} & & & & \\ S \times_S D(\underline{M}_S) & & & & \end{array}\right]$$

$$= \left[\begin{array}{ccccc} U \times_S D(\underline{L}_S) & = & D(\underline{L}_U) & \xrightarrow{h_{|U}} & \mathbf{G}_m \\ \downarrow{\scriptstyle (f_{|U},g)} & & & & \\ S \times_S D(\underline{M}_S) & & & & \end{array}\right] + \left[\begin{array}{ccccc} Z \times_S D(\underline{L}_S) & = & D(\underline{L}_Z) & \xrightarrow{h_{|Z}} & \mathbf{G}_m \\ \downarrow{\scriptstyle (f_{|Z},g)} & & & & \\ S \times_S D(\underline{M}_S) & & & & \end{array}\right]$$

for every Zariski-closed subscheme Z of X with open complement U. A product law is defined relatively to $S \times_S D(\underline{M}_S)$:

$$\left[\begin{array}{ccc} X_1 \times_S D(\underline{L_1}_S) & \xrightarrow{h_1} & \mathbf{G}_m \\ \downarrow & & \\ S \times_S D(\underline{M}_S) & & \end{array}\right] \times \left[\begin{array}{ccc} X_2 \times_S D(\underline{L_2}_S) & \xrightarrow{h_2} & \mathbf{G}_m \\ \downarrow & & \\ S \times_S D(\underline{M}_S) & & \end{array}\right]$$

$$= \left[\begin{array}{ccccc} (X_1 \times_S X_2) \times_S (D(\underline{L_1}_S) \times_{D(\underline{M}_S)} D(\underline{L_2}_S)) & \xrightarrow{(h_1,h_2)} & \mathbf{G}_m^2 & \xrightarrow{\text{group law}} & \mathbf{G}_m \\ \downarrow & & & & \\ S \times_S D(\underline{M}_S) & & & & \end{array}\right]$$

where the last diagram is a well-defined element of $\mathrm{K}_0\mathbf{Char}_M\mathbf{Var}_S$ by Proposition 1.6. In particular, this endows $\mathrm{K}_0\mathbf{Char}_M\mathbf{Var}_S$ with a structure of $\mathrm{K}_0\mathbf{Var}_S$-module.

Remark 1.9. When M is the trivial group, then all of the diagrams of the form (1.1) for fixed X over S become isomorphic, so that $\mathrm{K}_0\mathbf{Char}_M\mathbf{Var}_S = \mathrm{K}_0\mathbf{Var}_S$.

Notation 1.10 (Pullback and pushforward on the first factor)**.** Let $p : S \to S'$ be a morphism of schemes. Then, as usual with Grothendieck rings, p induces a ring morphism:

$$p^* : \mathrm{K}_0\mathbf{Char}_M\mathbf{Var}_{S'} \to \mathrm{K}_0\mathbf{Char}_M\mathbf{Var}_S$$

given by

$$\left[\begin{array}{ccccc} X \times_{S'} D(\underline{L}_{S'}) & = & D(\underline{L}_X) & \xrightarrow{h} & \mathbf{G}_m \\ \downarrow{\scriptstyle (f,g)} & & & & \\ S' \times_{S'} D(\underline{M}_{S'}) & & & & \end{array}\right] \mapsto$$

$$\left[\begin{array}{ccccc} (X \times_{S'} S) \times_S D(\underline{L}_S) & = & D(\underline{L}_{X \times_{S'} S}) & \xrightarrow{p^*h} & \mathbf{G}_m \\ \downarrow{\scriptstyle (p^*f,p^*g)} & & & & \\ S \times_S D(\underline{M}_S) & & & & \end{array}\right]$$

and, if p is of finite presentation, a group morphism $p_!$:

$$p_! : \mathrm{K}_0\mathbf{Char}_M\mathbf{Var}_S \to \mathrm{K}_0\mathbf{Char}_M\mathbf{Var}_{S'}$$

given by

$$\left[\begin{array}{ccccc} X \times_S D(\underline{L}_S) = X \times_S (S \times_{S'} D(\underline{L}_{S'})) & = & D(\underline{L}_X) & \xrightarrow{h} & \mathbf{G}_m \\ \downarrow{\scriptstyle (f,g)} & & & & \\ S \times_S D(\underline{M}_S) & & & & \end{array}\right] \mapsto$$

$$\left[\begin{array}{ccccc} X \times_{S'} D(\underline{L}_{S'}) & = & D(\underline{L}_{X_{S'}}) & \xrightarrow{h_{S'}} & \mathbf{G}_m \\ \downarrow{\scriptstyle (p \circ f, g_{S'})} & & & & \\ S' \times_{S'} D(\underline{M}_{S'}) & & & & \end{array}\right]$$

Remark 1.11. More generally, if $p : S \to S'$ is a morphism of schemes and $\mathfrak{a} \in \mathrm{K}_0\mathbf{Char}_M\mathbf{Var}_S$ which is supported over a subscheme of S of finite presentation over S', we may make sense of $p_!\mathfrak{a} \in \mathrm{K}_0\mathbf{Char}_M\mathbf{Var}_{S'}$. We will use this repeatedly in the special case where p is the projection $\underline{M}_S \to S$ and $\mathfrak{a}$ has finite support.

Notation 1.12 (Evaluation map)**.** For any $m \in M$, we write

$$\mathrm{ev}(m, \cdot) := [S \times_S D(\underline{M}_S), \chi \mapsto \chi(m)]_{S \times_S D(\underline{M}_S)} \in \mathrm{K}_0\mathbf{Char}_M\mathbf{Var}_S.$$

Note that by definition of the product in $\mathrm{K}_0\mathbf{Char}_M\mathbf{Var}_S$, we have $\mathrm{ev}(m, \cdot)\,\mathrm{ev}(n, \cdot) = \mathrm{ev}(m + n, \cdot)$ for all $m, n \in M$.

We may also define the evaluation pairing

$$\mathrm{ev} := [\underline{M}_S \times_S D(\underline{M}_S), (m, \chi) \mapsto \chi(m)] \in \mathrm{K}_0\mathbf{Char}_M\mathbf{Var}_{\underline{M}_S}.$$

Orthogonality relations are encoded by the following morphism of integration with respect to the character variables.

Definition 1.13 (Integration with respect to the second factor)**.** There exists a unique $\mathrm{K}_0\mathbf{Var}_S$-linear group morphism

$$\int_{D(\underline{M}_S)} : \mathrm{K}_0\mathbf{Char}_M\mathbf{Var}_S \to \mathrm{K}_0\mathbf{Var}_S$$

sending a class $[X \times_S D(\underline{L}_S), \chi \mapsto \chi(l)]$ to $[X]_S$ if $l = 0$ and zero otherwise.

For $\psi \in \mathrm{K}_0\mathbf{Char}_M\mathbf{Var}_S$ we write the image via this morphism either as $\int_{D(\underline{M}_S)} \psi$ or

$$\int_{D(\underline{M}_S)} \psi(\chi)\mathrm{d}\chi$$

to highlight the interpretation as an integral over characters.

Remark 1.14. In particular, we get the formal identity

$$\int_{D(\underline{M}_S)} \mathrm{ev}(m, \chi)\mathrm{d}\chi = \begin{cases} 1 & \text{if } m = 0 \\ 0 & \text{otherwise} \end{cases} \tag{1.2}$$

for all $m \in \underline{M}_S$.

Let us now consider the following situation: let $N \subset M$ be commutative abstract groups. We identify $D(M/N)$ with the kernel of the restriction morphism $\mathrm{res}_{|N} : D(M) \to D(N)$ so that we have a short exact sequence

$$0 \to D(M/N) \xrightarrow{i} D(M) \xrightarrow{j} D(N) \to 0.$$

and a pushforward morphism

$$i_! : \mathrm{K}_0\mathbf{Char}_{M/N}\mathbf{Var}_S \to \mathrm{K}_0\mathbf{Char}_M\mathbf{Var}_S.$$

Definition 1.15. We define an integration morphism $\int_{D(\underline{M/N}_S)} : \mathrm{K}_0\mathbf{Char}_M\mathbf{Var}_S \to \mathrm{K}_0\mathbf{Var}_S$ by

$$\int_{D(\underline{M/N}_S)} = \int_{D(\underline{M}_S)} \mathbf{1}_{D(\underline{M/N}_S)}$$

where

$$\mathbf{1}_{D(\underline{M/N}_S)} = [D(\underline{M/N}_S) \hookrightarrow D(\underline{M}_S), 1]_{D(\underline{M}_S)}.$$

In other words, given a generator $[X \times_S D(\underline{L}_S), h]$

$$\int_{D(\underline{M/N}_S)} [X \times_S D(\underline{L}_S), h] = \int_{D(\underline{M}_S)} [X \times_S (D(\underline{L}_S) \times_{D(\underline{M}_S)} D(\underline{M/N}_S)), h].$$

Remark 1.16. It is straightforward to check that the operator $\int_{D(\underline{M/N}_S)}$ is compatible with the operator

$$\int_{D(\underline{M/N}_S)} : \mathrm{K}_0\mathbf{Char}_{M/N}\mathbf{Var}_S \to \mathrm{K}_0\mathbf{Var}_S$$

defined in Definition 1.13, in the sense that one recovers the latter by composing the former with the pushforward map $i_!$ on the second factor.

Remark 1.17. In particular, we can check that we get the formal identity

$$\int_{D(\underline{M/N}_S)} \mathrm{ev}(m, \chi)\mathrm{d}\chi = \begin{cases} 1 & \text{if } m \in N \\ 0 & \text{otherwise} \end{cases} \tag{1.3}$$

for all $m \in \underline{M}_S$.

Definition 1.18. Assume that M splits as $M = M' \oplus M''$, so that $D(M) = D(M') \times D(M'')$. We define a *partial integration operator*

$$\int_{D(M')} : \mathrm{K}_0\mathbf{Char}_M\mathbf{Var}_S \to \mathrm{K}_0\mathbf{Char}_{M''}\mathbf{Var}_S$$

in the following way. Given a generator

$$[X \times_S D(\underline{L}_S), \chi \mapsto \chi(l)]_{S \times_S D(\underline{M' \oplus M''}_S)},$$

with $g : M' \oplus M'' \to L$ and $l = g(m', m'')$ in the image of g, we define

$$\int_{D(M')} [X \times_S D(\underline{L}_S), \chi \mapsto \chi(l)] = \begin{cases} 0 & \text{if } m' \neq 0 \\ [X \times_S D(\underline{L}_S), \chi \mapsto \chi(l)]_{S \times_S D(\underline{M''}_S)} & \text{otherwise,} \end{cases}$$

where in the latter case the morphism $D(\underline{L}_S) \to D(\underline{M''}_S)$ comes from $g_{|M''} : M'' \to L$. If moreover we are given an subgroup $N \subset M'$, we may more generally define

$$\int_{D(M'/N)} : \mathrm{K}_0\mathbf{Char}_M\mathbf{Var}_S \to \mathrm{K}_0\mathbf{Char}_{M''}\mathbf{Var}_S$$

given by

$$\int_{D(M'/N)} [X \times_S D(\underline{L}_S), \chi \mapsto \chi(l)] = \begin{cases} 0 & \text{if } m' \notin N \\ [X \times_S D(\underline{L}_S), \chi \mapsto \chi(l)]_{S \times_S D(\underline{M''}_S)} & \text{otherwise.} \end{cases}$$

Remark 1.19. Let $M = M' \oplus M''$ with N a subgroup of M' as in Definition 1.18. Then it is straightforward to check that the operator

$$\int_{D(M/N)} : \mathrm{K}_0\mathbf{Char}_M\mathbf{Var}_S \to \mathrm{K}_0\mathbf{Var}_S$$

is equal to the composition $\int_{D(M'')} \circ \int_{D(M'/N)}$.

Definition 1.20. We define the localisation

$$\mathcal{C}har_M\mathscr{M}_S = \mathrm{K}_0\mathbf{Char}_M\mathbf{Var}_S\left[\mathbf{L}_S^{-1}\right].$$

1.4. **A local integration operator.** Let $(M_{\mathsf{S}})_{\mathsf{S}}$ be a directed system of finite type abstract commutative groups, i.e., for every $\mathsf{S} \subset \mathsf{S}'$ there is a group morphism

$$i_{\mathsf{S},\mathsf{S}'} : M_{\mathsf{S}} \to M_{\mathsf{S}'},$$

which we additionally assume to be injective. We also fix directed systems $(M'_{\mathsf{S}})_{\mathsf{S}}$ and $(M''_{\mathsf{S}})_{\mathsf{S}}$, assuming that for every S

$$M_{\mathsf{S}} = M'_{\mathsf{S}} \oplus M''_{\mathsf{S}}$$

and that $M''_{\mathsf{S}} = M''$ is constant in S. Finally, we fix a family $(N_{\mathsf{S}})_{\mathsf{S}}$ of groups such that for every S, N_{S} is a subgroup of M'_{S}, and such that for every $\mathsf{S} \subset \mathsf{S}'$, $i_{\mathsf{S},\mathsf{S}'}(N_{\mathsf{S}}) \subset N_{\mathsf{S}'}$, so that the restrictions of the $i_{\mathsf{S},\mathsf{S}'}$ make the family $(N_{\mathsf{S}})_{\mathsf{S}}$ into a directed system. We define $M = \varinjlim_{\mathsf{S}} M_{\mathsf{S}}$ and $M' = \varinjlim_{\mathsf{S}} M'_{\mathsf{S}}$.

We wish to define an operator

$$\int_{D(M'/N)} : \mathrm{K}_0\mathbf{Char}_M\mathbf{Var}_X \to \mathrm{K}_0\mathbf{Char}_{M''}\mathbf{Var}_X.$$

For this, let $[Y \times_X D(\underline{L}_X), \chi \mapsto \chi(l)]$ be a generator, with a group morphism $g : M \to L$ and $l \in L$ in the image of g, i.e. $l = g(m)$ where $m = m' + m'' \in M = M' \oplus M''$. We pick S such that $m \in M_{\mathsf{S}}$ and define

$$\int_{D(M'/N)} [Y \times_X D(\underline{L}_X), \chi \mapsto \chi(l)] := \int_{D(M'_{\mathsf{S}}/N_{\mathsf{S}})} [Y \times_X D(\underline{L}_X), \chi \mapsto \chi(l)] \in \mathrm{K}_0\mathbf{Char}_{M''}\mathbf{Var}_X$$

in the sense of Definition 1.18. We claim that this definition does not depend on the choice of such an S. Take $\mathsf{S} \subset \mathsf{S}'$. We have

$$\int_{D(M'_{\mathsf{S}}/N_{\mathsf{S}})} [Y \times_X D(\underline{L}_X), \chi \mapsto \chi(l)] = \begin{cases} 0 & \text{if } m' \notin N_{\mathsf{S}} \\ [Y \times_X D(\underline{L}_X), \chi \mapsto \chi(l)]_{D(M'')} & \text{otherwise.} \end{cases}$$

in $\mathrm{K}_0\mathbf{Char}_{M''}\mathbf{Var}_X$. Now, viewing m' as an element of $M'_{\mathsf{S}'}$, we have $m' \in N_{\mathsf{S}'}$ if and only if $m' \in N_{\mathsf{S}'} \cap M'_{\mathsf{S}} = N_{\mathsf{S}}$. This shows that

$$\int_{D(M'_{\mathsf{S}}/N_{\mathsf{S}})} [Y \times_X D(\underline{L}_X), \chi \mapsto \chi(l)] = \int_{D(M'_{\mathsf{S}'}/N_{\mathsf{S}'})} [Y \times_X D(\underline{L}_X), \chi \mapsto \chi(l)],$$

proving the claim.

1.5. **Dimensional filtration and convergence of series.**

Definition 1.21. For any integer i, we define $\mathrm{Fil}^i \mathscr{M}_S$ as the subgroup of $\mathscr{M}_S$ generated by classes of the form $[X]_S \mathbf{L}_S^{-m}$ where X is a variety over S and m is an integer such that $\dim_S X - m \leq -i$. This defines a decreasing filtration $(\mathrm{Fil}^i \mathscr{M}_S)_{i \in \mathbf{Z}}$ on $\mathscr{M}_S$, endowing $\mathscr{M}_S$ with the structure of a filtered ring.

For any class $\mathfrak{a} \in \mathscr{M}_S$, we define

$$\dim_S(\mathfrak{a}) = \inf\{i \in \mathbf{Z}, \mathfrak{a} \in \mathrm{Fil}^i \mathscr{M}_S\}$$

We define the completed Grothendieck ring

$$\widehat{\mathscr{M}_S} = \varprojlim \mathscr{M}_S / \mathrm{Fil}^i \mathscr{M}_S.$$

The $\dim_S$ function extends naturally to $\widehat{\mathscr{M}_S}$.

Definition 1.22. Let $F(T) = \sum_{i\geq 0} A_i T^i \in \widehat{\mathscr{M}_S}[[T]]$ be a power series with coefficients in $\widehat{\mathscr{M}_S}$. The radius of convergence of F is defined to be

$$\sigma_F = \limsup_{i\geq 1} \frac{\dim_S(A_i)}{i}.$$

We say that F converges for $|T| < \mathbf{L}^{-r}$ if $r \geq \sigma_F$.

If F converges for $|T| < \mathbf{L}^{-r}$, then for every element $\mathfrak{a} \in \widehat{\mathscr{M}_S}$ such that $\dim_S(\mathfrak{a}) < -r$, the value $F(\mathfrak{a})$ exists as an element of $\widehat{\mathscr{M}_S}$.

We now turn to convergence of multivariate series. The place ourselves in the following general setting: let N be a finitely generated free $\mathbf{Z}$-module, and consider

$$\langle \cdot, \cdot \rangle : N^\vee \times N \to \mathbf{Z}$$

the natural pairing with the dual $N^\vee = \mathrm{Hom}(N, \mathbf{Z})$. For a rational polyhedral cone $\Lambda \subset N_{\mathbf{R}}$, the intersection $\Lambda \cap N$ is a finitely generated monoid to which we may associate the corresponding completed monoid algebra $\widehat{\mathscr{M}_S}[[\Lambda \cap N]]$ with coefficients in $\widehat{\mathscr{M}_S}$. The elements of this algebra are viewed as power series with coefficients in $\widehat{\mathscr{M}_S}$ in a set of indeterminates $\mathbf{T} = (T_i)_{i\in I}$ indexed by a basis of $\Lambda \cap N$. The relative interior of a cone Λ is denoted by $\overset{\circ}{\Lambda}$.

Definition 1.23. Let $\Lambda \subset N_{\mathbf{R}}$ be a cone. Let $F(\mathbf{T}) = \sum_{\mathbf{m}\in\Lambda\cap N} A_{\mathbf{m}} \mathbf{T}^{\mathbf{m}} \in \widehat{\mathscr{M}_S}[[\Lambda \cap N]]$ be a power series with coefficients in $\widehat{\mathscr{M}_S}$. Let λ be an element of $N^\vee \cap \overset{\circ}{\Lambda}{}^\vee$. The radius of convergence of F in the direction λ is defined to be

$$\sigma_{F,\lambda} = \limsup_{\mathbf{m}\in\Lambda} \frac{\dim_S(A_{\mathbf{m}})}{\langle \lambda, \mathbf{m} \rangle}.$$

We say that F converges for $|\mathbf{T}| < \mathbf{L}^{-s\lambda}$ (or, equivalently, for $|T_i| < \mathbf{L}^{-s\lambda_i}$ for all $i \in I$) if $s \geq \sigma_{F,\lambda}$.

If F converges for $|\mathbf{T}| < \mathbf{L}^{-s\lambda}$, then for every tuple of elements $\mathfrak{a} = (\mathfrak{a}_i)_{i\in I} \in \widehat{\mathscr{M}_S}^I$ such that $\dim_S(\mathfrak{a}) < -s\lambda_i$, the value $F(\mathfrak{a})$ exists as an element of $\widehat{\mathscr{M}_S}$.

1.6. **Motivic stabilisation.** The following is a motivic variant of [Bou11b, Lemme 12].

Lemma 1.24. *Let $(\mathfrak{a}_{\boldsymbol{d}})_{\boldsymbol{d}\in\mathbf{N}^r}$ be a family of classes in $\mathscr{M}_S$ and*

$$F(\mathbf{T}) = \sum_{\boldsymbol{d}\in\mathbf{N}^r} \mathfrak{a}_{\boldsymbol{d}} \mathbf{T}^{\boldsymbol{d}}.$$

Assume that $F(\mathbf{T})$ converges absolutely for $|T_i| < \mathbf{L}^{-\rho_i+\varepsilon}$ and consider the power series

$$\frac{F(\mathbf{T})}{\prod_{i=1}^r (1 - \mathbf{L}^{\rho_i} T_i)} = \sum_{\boldsymbol{d}\in\mathbf{N}^r} \mathfrak{b}_d \mathbf{T}^{\boldsymbol{d}}.$$

Then for all $0 < \eta < \varepsilon$

$$\mathfrak{b}_{\boldsymbol{d}} \mathbf{L}^{-\langle \boldsymbol{\rho}, \boldsymbol{d} \rangle} - F(\mathbf{L}^{-\boldsymbol{\rho}})$$

has virtual dimension bounded above by

$$-\eta \min(\rho_i d_i).$$

Proof. It goes as for [Bou11b, Lemme 12]. The difference $\mathfrak{b}_{\boldsymbol{d}} - F(\mathbf{L}^{-\boldsymbol{\rho}})\mathbf{L}^{\langle \boldsymbol{\rho}, \boldsymbol{d} \rangle}$ is the coefficient of degree $\boldsymbol{d} \in \mathbf{N}^r$ of $\frac{F(\mathbf{T}) - F(\mathbf{L}^{-\boldsymbol{\rho}})}{(1-\mathbf{L}^{\rho_1} T_1)\cdots(1-\mathbf{L}^{\rho_r} T_r)}$. We can write

$$\frac{F(\mathbf{T}) - F(\mathbf{L}^{-\boldsymbol{\rho}})}{\prod_{i=1}^r (1 - \mathbf{L}^{\rho_i} T_i)} = \sum_{j=1}^r \underbrace{\frac{F(T_1, ..., T_j, \mathbf{L}^{-\rho_{j+1}}, ..., \mathbf{L}^{-\rho_r}) - F(T_1, ..., T_{j-1}, \mathbf{L}^{-\rho_j}, ..., \mathbf{L}^{-\rho_r})}{\prod_{i=1}^r (1 - \mathbf{L}^{\rho_i} T_i)}}_{=: G_j(\mathbf{T})}$$

where the degree $\boldsymbol{d} \in \mathbf{N}^r$ coefficient of $G_j(\mathbf{T})$ is

$$\mathfrak{b}_{j,\boldsymbol{d}} = - \sum_{\substack{0 \leqslant \delta_1 \leqslant d_1 \\ \cdots \\ 0 \leqslant \delta_{j-1} \leqslant d_{j-1}}} \sum_{\delta \in \mathbf{N}} \sum_{\substack{\delta_{j+1} \in \mathbf{N} \\ \cdots \\ \delta_r \in \mathbf{N}}} \mathfrak{a}_{\delta_1,\ldots,\delta_{j-1},d_j+\delta+1,\delta_{j+1},\ldots\delta_r} \mathbf{L}^{-(\delta+1)\rho_j + \sum_{\ell \neq j}(d_\ell - \delta_\ell)\rho_\ell}.$$

By assumption, we have

$$\dim(\mathfrak{a}_{\delta_1,\ldots,\delta_{j-1},d_j+\delta+1,\delta_{j+1},\ldots\delta_r}) < \langle (\delta_1, \ldots, \delta_{j-1}, d_j + \delta + 1, \delta_{j+1}, \ldots \delta_r), (1-\eta)\boldsymbol{\rho} \rangle$$

for $0 < \eta < \varepsilon$ and $\boldsymbol{d}$ large enough, hence the claim. □

2. Symmetric products and motivic Euler products

The aim of this section is to recall the definitions of symmetric products and motivic Euler products from [Bil23], and explain how one can interpret them as functions on the scheme of effective divisors of the curve $\mathscr{C}$.

2.1. **Symmetric products.** Following for example [BH21, §6.1], we start with a few basic definitions concerning symmetric products of varieties. These products are the first building blocks leading to the notion of motivic Euler product.

Symmetric powers. Let S be a scheme, and let X be a quasi-projective variety over S. If n is a non-negative integer, the n-th symmetric power of the variety $X \to S$ is the quotient

$$\mathbf{Sym}^n_{/S}(X) = \underbrace{X \times_S \ldots \times_S X}_{n \text{ times}} / \mathfrak{S}_n.$$

If $Y \to X$ is a variety over X, quasi-projective over S, the symmetric power $\mathbf{Sym}^n_{/S}(Y)$ naturally comes with a morphism

$$\mathbf{Sym}^n_{/S}(Y) \longrightarrow \mathbf{Sym}^n_{/S}(X).$$

The incidence algebra. We now explain how to define symmetric products of classes in $\mathrm{K}_0\mathbf{Var}_X$. The product

$$\mathrm{K}_0\mathbf{Var}_{\mathbf{Sym}^\bullet_{/S}(X)} := \prod_{n \geq 0} \mathrm{K}_0\mathbf{Var}_{\mathbf{Sym}^n_{/S}(X)}$$

has the structure of a $\mathrm{K}_0\mathbf{Var}_S$-algebra, where addition is termwise, and the product is given by the Cauchy product

$$((ab)_n)_{n \in \mathbf{N}} = \left(\sum_{i=0}^{n} a_i b_{n-i} \right)_{n \in \mathbf{N}}$$

where for every $n \in \mathbf{N}$ and $i \in \{0, \ldots, n\}$, the product comes from the exterior product

$$a_i \boxtimes b_{n-i} \in \mathrm{K}_0\mathbf{Var}_{\mathbf{Sym}^i_{/S}(X) \times_k \mathbf{Sym}^{n-i}_{/S}(X)}$$

composed with the pushforward via the natural arrows

$$\mathbf{Sym}^i_{/S}(X) \times_k \mathbf{Sym}^{n-i}_{/S}(X) \longrightarrow \mathbf{Sym}^n_{/S}(X).$$

We denote by $\mathrm{K}_0\mathbf{Var}^1_{\mathbf{Sym}^\bullet_{/S}(X)} \subset \mathrm{K}_0\mathbf{Var}_{\mathbf{Sym}^\bullet_{/S}(X)}$ the subset of families $(a_n)_{n \geq 0}$ with "constant term" a_0 equal to 1. It is a subgroup of the multiplicative units in $\mathrm{K}_0\mathbf{Var}_{\mathbf{Sym}^\bullet_{/S}(X)}$.

Then there is a unique group morphism (see [Bil23, Lemma 3.5.1.2])

$$\mathrm{K}_0\mathbf{Var}_X \to \mathrm{K}_0\mathbf{Var}^1_{\mathbf{Sym}^\bullet_{/S}(X)}$$

which to the class $[Y \to X]$ of an X-variety Y associates the family

$$([\mathbf{Sym}^i_{/S}(Y) \to \mathbf{Sym}^i_{/S}(X)])_{i\geq 1}$$

of classes of its symmetric powers; for every $a \in \mathrm{K}_0\mathbf{Var}_X$ we then denote by $\mathbf{Sym}^i_{X/S}(a)$ the i-th coordinate of its image via this morphism.

Generalized symmetric products. Let I be a set and let $\pi \in \mathbf{N}^{(I)}$ be a generalized partition. Let X be a quasi-projective variety over S. As before, all products are taken relatively to S. We denote

The product of permutation groups $\prod_{i\in I} \mathfrak{S}_{n_i}$ acts on the product $\prod_{i\in I} X^{n_i}$, inducing a quotient morphism

$$\prod_{i\in I} X^{n_i} \to \prod_{i\in I} \mathbf{Sym}^{n_i}_{/S} X \tag{2.1}$$

We denote by

$$\left(\prod_{i\in I} X^{n_i}\right)_{*,X/S}$$

the complement of the big diagonal in $\prod_{i\in I} X^{n_i}$, that is, the open subset consisting of points with coordinates all distinct.

Definition 2.1. The symmetric product $\mathbf{Sym}^\pi_{/S}(X)_*$ of X relatively to S, corresponding to the partition π, is defined as

$$\left(\prod_{i\in I} \mathbf{Sym}^{n_i} X\right)_{*,X/S},$$

that is, the image of the open subset $(\prod_{i\in I} X^{n_i})_{*,X/S}$ via the morphism (2.1).

Definition 2.2. Let now $\mathscr{X} = (X_i)_{i\in I}$ be a family of X-varieties, quasi-projective over S. We denote by $\mathbf{Sym}^\pi_{X/S}(\mathscr{X})_*$ the variety fitting into the top left corner of the following diagram:

$$\begin{array}{ccc} \mathbf{Sym}^\pi_{X/S}(\mathscr{X})_* & \longrightarrow & \prod_{i\in I} \mathbf{Sym}^{n_i}(X_i) \\ \downarrow & & \downarrow \\ \mathbf{Sym}^\pi_{/S}(X)_* & \longrightarrow & \prod_{i\in I} \mathbf{Sym}^{n_i}_{/S} X \end{array}$$

where the bottom horizontal arrow is the natural inclusion, and the rightmost vertical arrow is induced by the given morphisms $X_i \to X$.

Generalized symmetric product classes. We now extend the above construction to general classes in $\mathrm{K}_0\mathbf{Var}_X$. Let $\mathscr{A} = (a_i)_{i\in I}$ be a family of classes in $\mathrm{K}_0\mathbf{Var}_X$ and $\pi = (n_i)_{i\in I} \in \mathbf{N}^{(I)}$. We set

$$\mathbf{Sym}^\pi_{X/S}(\mathscr{A})_* = \left(\boxtimes_{i\in I} \mathbf{Sym}^{n_i}_{X/S}(a_i)\right)_{*,X/S}$$

in $\mathrm{K}_0\mathbf{Var}_{\mathbf{Sym}^\pi_{/S}(X)_*}$, where $\boxtimes_{i\in I}$ is the exterior product and the subscript "$*, X/S$" indicates application of the restriction morphism

$$\mathrm{K}_0\mathbf{Var}_{\prod_{i\in I} \mathbf{Sym}^{n_i}_{/S}(X)} \to \mathrm{K}_0\mathbf{Var}_{\mathbf{Sym}^\pi_{/S}(X)_*}$$

induced by the inclusion $\mathbf{Sym}^\pi_{/S}(X)_* \to \prod_{i\in I} \mathbf{Sym}^{n_i}_{/S}(X)$.

Remark 2.3 (Points of symmetric products). We will repeatedly use the following description of points of symmetric products, explained in detail in [Bil23, Section 3.2.2.].

Let I be a set, and $\pi = (n_i)_{i\in I}$ a partition. Following in [Bil23, Section 3.2.2.], a schematic point $E \in \mathbf{Sym}^{\pi}_{/S}(X)_*$ can be written as an effective zero-cycle

$$\sum_{i\in I} i(v_{i,1} + \ldots + v_{i,n_i})$$

where the $v_{i,j}$ are distinct geometric points of X (lying in the same fibre above S). More generally, given a family $\mathscr{A} = (A_i)_{i\in I}$ of varieties over X, a point of $\mathbf{Sym}^{\pi}(\mathscr{A})$ lying above E is written in the form

$$\sum_{i\in I} i(a_{i,1} + \ldots + a_{i,n_i})$$

where for every i, the $a_{i,j}$ are geometric points of A_i such that $a_{i,j}$ maps to $v_{i,j}$.

2.2. **Mixed symmetric products.** We also need the more general notion of mixed symmetric products, introduced in [Bil23, Section 3.3.2]. As before, let X be a variety over S. All products of varieties in this sections are taken relatively to S.

Let $p \geq 1$ be an integer, and let I be a set. For every $j \in \{1, \ldots, p\}$, we fix a generalised partition $\pi_j = (n_{i,j})_{i\in I}$. We may define the variety $\mathbf{Sym}^{\pi_1,\ldots,\pi_p}_{/S}(X)_*$ to be the image of the complement of the big diagonal in

$$\prod_{i\in I} \mathbf{Sym}^{n_{i,1}}_{/S}(X) \times \ldots \times \mathbf{Sym}^{n_{i,p}}_{/S}(X).$$

More generally, let $\mathscr{Y}_1 = (Y_{i,1})_{i\in I}, ..., \mathscr{Y}_p = (Y_{i,p})_{i\in I}$ be families of varieties over X, all quasi-projective over S.

For every p-tuple $(\pi_1, ..., \pi_p) = ((n_{i,1})_{i\in I}, ..., (n_{i,p})_{i\in I})$ of generalised partitions of I, we consider the product

$$\prod_{i\in I} Y_{i,1}^{n_{i,1}} \times \ldots \times Y_{i,p}^{n_{i,p}} \to \prod_{i\in I} \mathbf{Sym}^{n_{i,1}}_{/S}(X) \times \ldots \times \mathbf{Sym}^{n_{i,p}}_{/S}(X).$$

and restrict to $\mathbf{Sym}^{\pi_1,\ldots,\pi_p}_{/S}(X)_*$, to get the *mixed symmetric product*

$$\mathbf{Sym}^{\pi_1,\ldots,\pi_p}_{X/S}(\mathscr{Y}_1, ..., \mathscr{Y}_p)_* \to \mathbf{Sym}^{\pi_1,\ldots,\pi_p}_{/S}(X)_*$$

of the families $\mathscr{Y}_1, \ldots \mathscr{Y}_p$.

Now let, for every $j \in \{1, \ldots, p\}$, $\mathscr{A}_j = (a_{i,j})_{i\in I}$ be a family of classes in $\mathrm{K}_0\mathbf{Var}_X$ indexed by the set I_j. The class

$$\mathbf{Sym}^{\pi_1,\ldots,\pi_p}_{X/S}(\mathscr{A}_1, \ldots, \mathscr{A}_p)_* \in \mathrm{K}_0\mathbf{Var}_{\mathbf{Sym}^{\pi_1,\ldots,\pi_p}_{/S}(X)_*}$$

is defined as the image of the class

$$\prod_{i\in I} \mathbf{Sym}^{n_{i,1}}_{X/S}(a_{i,1}) \times \ldots \times \mathbf{Sym}^{n_{i,p}}_{X/S}(a_{i,p})$$

via the restriction morphism

$$\mathrm{K}_0\mathbf{Var}_{\prod_{i\in I} \mathbf{Sym}^{n_{i,1}}_{/S}(X)\times\ldots\times\mathbf{Sym}^{n_{i,p}}_{/S}(X)} \to \mathrm{K}_0\mathbf{Var}_{\mathbf{Sym}^{\pi_1,\ldots,\pi_p}_{/S}(X)_*}.$$

Remark 2.4. We will be in a situation where there is the following extra structure: the set I is partitioned into subsets:

$$I = \bigsqcup_{j=1}^{p} I_j$$

and in fact every family $\mathscr{A}_j$ is supported above I_j. Then each partition π_j may in fact be viewed as an element of $\mathbf{N}^{(I_j)}$.

2.3. **Motivic Euler products.** Let us now recall the definition of motivic Euler products introduced in [Bil23, Chap. 3]. We will consider the following set-up, which will fit our purposes.

Let I be a set which is of the form $I_0 \setminus \{0\}$ where I_0 is a commutative monoid. Given $\iota_1, \iota_2 \in I$, the natural morphism

$$\mathbf{Sym}^{\iota_1}_{/S}(X) \times_S \mathbf{Sym}^{\iota_2}_{/S}(X) \to \mathbf{Sym}^{\iota_1+\iota_2}_{/S}(X)$$

induces, by composition with the exterior product, a morphism

$$\mathrm{K}_0\mathbf{Var}_{\mathbf{Sym}^{\iota_1}_{/S}(X)} \times \mathrm{K}_0\mathbf{Var}_{\mathbf{Sym}^{\iota_2}_{/S}(X)} \to \mathrm{K}_0\mathbf{Var}_{\mathbf{Sym}^{\iota_1+\iota_2}_{/S}(X)}. \tag{2.2}$$

We define

$$\mathrm{K}_0\mathbf{Var}_{\mathbf{Sym}^{\bullet}_{/S}X} := \prod_{\iota \in I_0} \mathrm{K}_0\mathbf{Var}_{\mathbf{Sym}^{\iota}_{/S}(X)}$$

the $\mathrm{K}_0\mathbf{Var}_S$-algebra with addition given by coordinatewise addition, and multiplication defined using the morphisms (2.2). We also define $\mathrm{K}_0\mathbf{Var}^1_{\mathbf{Sym}^{\bullet}_{/S}X} \subset \mathrm{K}_0\mathbf{Var}_{\mathbf{Sym}^{\bullet}_{/S}X}$ the subset of families $(a_\iota)_{\iota \in I_0}$ with $a_0 = 1$. It is a subgroup of the group of multiplicative units of $\mathrm{K}_0\mathbf{Var}_{\mathbf{Sym}^{\bullet}_{/S}X}$. An element of $\mathrm{K}_0\mathbf{Var}^1_{\mathbf{Sym}^{\bullet}_{/S}X}$ may be viewed as a power series

$$\sum_{\iota \in I} A_\iota T_\iota$$

with $A_0 = 1$, where each A_ι is viewed as an element of $\mathrm{K}_0\mathbf{Var}_{\mathbf{Sym}^{\iota}_{/S}(X)}$ and where the indeterminates T_i satisfy relations $T_{\iota_1}T_{\iota_2} = T_{\iota_1+\iota_2}$

Our motivic Euler products will be defined as such power series.

Definition 2.5. Let $\mathscr{A} = (a_i)_{i \in I}$ be a family of classes above X. The formal motivic Euler product

$$\prod_{x \in X/S} \left(1 + \sum_{i \in I} a_{i,x} T_i\right)$$

associated to $\mathscr{A}$ is by definition the formal series

$$\sum_{i \in I} \left(\sum_{\pi \vdash i} \mathbf{Sym}^{\pi}_{X/S}(\mathscr{A})_*\right) T_i \in \mathrm{K}_0\mathbf{Var}_{\mathbf{Sym}^{\bullet}_{/S}(X)}.$$

We now define a refined setting. Assume additionally that $I = \bigsqcup_{j=1}^{p} I_j$ for some commutative semigroups I_j. We pick sets of indeterminates $(T_{i,j})_{i \in I_j, 1 \le j \le p}$ such that $T_{i_1,j}T_{i_2,j} = T_{i_1+i_2,j}$ for all j and all $i_1, i_2 \in I_j$.

We denote for every j, $\bar{I}_j = I_j \cup \{0\}$ and define

$$\mathrm{K}_0\mathbf{Var}_{\mathbf{Sym}^{\bullet}_{/S}(X)} = \prod_{(i_1,\dots,i_p) \in \bar{I}_1 \times \dots \times \bar{I}_j} \mathrm{K}_0\mathbf{Var}_{\mathbf{Sym}^{i_1,\dots,i_p}_{/S}(X)}$$

again with coordinatewise addition, and appropriate multiplication operations.

Definition 2.6. For every $j \in \{1, \dots, p\}$ let $\mathscr{A}_j = (a_{i,j})_{i \in I}$ be a family of classes in $\mathrm{K}_0\mathbf{Var}_X$ The formal motivic Euler product

$$\prod_{x \in X/S} \left(1 + \sum_{j=1}^{p} \sum_{i \in I_j} a_{i,j,x} T_{i,j}\right)$$

is by definition the formal series

$$\sum_{(i_1,\dots,i_p)\in \bar{I}_1\times\dots\times\bar{I}_j}\left(\sum_{\pi_1\vdash i_1,\dots,\pi_p\vdash i_j}\mathbf{Sym}^{\pi_1,\dots,\pi_p}_{X/S}(\mathscr{A}_1,\dots,\mathscr{A}_p)_*\right)T_{i_1,1}\dots T_{i_p,p}.$$

Remark 2.7. All of these definitions work analogously for localised Grothendieck rings.

2.4. **Symmetric products of varieties with characters.** In this section we extend the notion of motivic Euler product from [Bil23] to the Grothendieck ring of varieties with multiplicative characters. In particular, this will allow us to make sense of products of the specific form

$$\prod_{v\in\mathscr{C}}\left(1+\sum_{m\in\mathbf{Z}^n_{\neq 0}}X_{m,v}\chi_v(m)\mathbf{T}^m\right)$$

appearing later as certain Fourier transforms. The approach is completely analogous to how one defines motivic Euler products for varieties with exponentials in [Bil23, Section 3.6.1].

Let I be a set. Let S be a scheme, X a variety over S, $(M_i)_{i\in I}$ a family of commutative groups, and

$$(\mathscr{Y},h)=(Y_i\times_X D(\underline{L_i}_X),h_i:D(\underline{L_i}_{Y_i})\to\mathbf{G}_m)$$

a family of quasiprojective varieties with characters over X. Let $\pi=(n_i)_{i\in I}\in\mathbf{N}^{(I)}$ be a generalised partition.

We consider the product

$$\prod_{i\in I}(Y_i\times_X D(\underline{L_i}_X))^{n_i}$$

(taken relatively to the base scheme S), endowed with the map

$$h^\pi:\prod_{i\in I}D(\underline{L_i}_{Y_i})^{n_i}\to\mathbf{G}_m$$

above $\prod_{i\in I}Y_i^{n_i}$ given by

$$(\chi_{i,1},\dots,\chi_{i,n_i})\mapsto\prod_{i\in I}\prod_{j=1}^{n_i}h_i(\chi_{i,j}).$$

The aforementioned product comes with a morphism

$$\prod_{i\in I}Y_i^{n_i}\times_X D(\underline{L_i}_X)^{n_i}\to\prod_{i\in I}X^{n_i}$$

(all products are taken relatively to S) and we denote by

$$\left(\prod_{i\in I}Y_i^{n_i}\times_X D(\underline{L_i}_X)^{n_i}\right)_{*,X/S}$$

the inverse image of the complement of the big diagonal in $\prod_{i\in I}X^{n_i}$. The map h^π restricts to $\left(\prod_{i\in I}Y_i^{n_i}\times_X D(\underline{L_i}_X)^{n_i}\right)_{*,X/S}$, and is invariant modulo the permutation action of $\prod_{i\in I}\mathfrak{S}_{n_i}$, giving us a diagram

$$\begin{array}{ccc}\mathbf{Sym}^\pi(Y_i\times_X D(\underline{L_i}_X))_{i\in I} & \xrightarrow{\mathbf{Sym}^\pi(h_i)_{i\in I}} & \mathbf{G}_m\\ \downarrow & & \\ \mathbf{Sym}^\pi_{X/S}(Y_i)_{i\in I} & & \end{array}$$

denoted $\mathbf{Sym}^\pi(\mathscr{Y},h)$.

2.5. **Higher order Grothendieck rings.**

Definition 2.8 (Grothendieck ring of varieties with characters of level π)**.** We define the Grothendieck group $\mathrm{K}_0\mathbf{Char}_{\pi,(M_i)_{i\in I}}\mathbf{Var}_{\mathbf{Sym}^\pi_{/S}(X)_*}$ as the free abelian group generated by isomorphism classes

$$[Y \times_{\mathbf{Sym}^\pi_{/S}(X)_*} \mathbf{Sym}^\pi_{X/S}((D(L_i))_{i\in I})_*, h]_{\mathbf{Sym}^\pi_{/S}(X)_*\times_{\mathbf{Sym}^\pi_{/S}(X)_*}\mathbf{Sym}^\pi_{X/S}(D(\underline{M_i}_X))_{i\in I}}$$

of diagrams

$$\begin{array}{c} Y \times_{\mathbf{Sym}^\pi_{/S}(X)_*} \mathbf{Sym}^\pi_{X/S}((D(L_i))_{i\in I})_* \xrightarrow{\quad h \quad} \mathbf{G}_m \\ \downarrow \\ \mathbf{Sym}^\pi_{/S}(X)_* \times_{\mathbf{Sym}^\pi_{/S}(X)_*} \mathbf{Sym}^\pi_{X/S}\, D(\underline{M}_X) \end{array}$$

such that h is of the form $\mathbf{Sym}^\pi(h_i)_{i\in I}$ as in Section 2.4, modulo cut-and-paste relations on the first factor. When all the M_i are equal to the same M, we simply write $\mathrm{K}_0\mathbf{Char}_{\pi,M}\mathbf{Var}_{\mathbf{Sym}^\pi_{/S}(X)_*}$.

As for classical Grothendieck rings, it will be convenient to combine Grothendieck rings with characters into incidence algebras, which will be natural receptacles for the notion of motivic Euler product with characters.

Let I be a set which is of the form $I_0 \setminus \{0\}$ where I_0 is a commutative monoid.

Definition 2.9. We denote by $\mathrm{K}_0\mathbf{Char}_{\iota,M}\mathbf{Var}_{\mathbf{Sym}^\iota_{/S}(X)}$ the free abelian group generated by isomorphism classes

$$[Y \times_{\mathbf{Sym}^\iota_{/S}(X)} \mathbf{Sym}^\iota_{X/S}(D(L_i))_{i\in I}, h]_{\mathbf{Sym}^\iota_{/S}(X)\times_{\mathbf{Sym}^\iota_{/S}(X)}\mathbf{Sym}^\iota_{X/S}\, D(\underline{M}_X)}$$

of diagrams

$$\begin{array}{c} Y \times_{\mathbf{Sym}^\iota_{/S}(X)} \mathbf{Sym}^\iota_{X/S}(D(L_i))_{i\in I} \xrightarrow{\quad h \quad} \mathbf{G}_m \\ \downarrow \\ \mathbf{Sym}^\iota_{/S}(X) \times_{\mathbf{Sym}^\iota_{/S}(X)} \mathbf{Sym}^\iota_{X/S}\, D(\underline{M}_X) \end{array}$$

such that the restriction of h to $\mathbf{Sym}^\pi_{X/S}((D(L_i))_{i\in I})_*$ for every $\pi \vdash \iota$ is as in Definition 2.8, modulo cut-and-paste relations on the first factor.

For every partition π of ι, the inclusion $i_{\pi,\iota} : \mathbf{Sym}^\pi_{X/S}(\underline{M}_X) \to \mathbf{Sym}^\iota_{X/S}(\underline{M}_X)$ induces a restriction morphism

$$\mathrm{K}_0\mathbf{Char}_{\iota,M}\mathbf{Var}_{\mathbf{Sym}^\iota_{/S}(X)} \to \mathrm{K}_0\mathbf{Char}_{\pi,M}\mathbf{Var}_{\mathbf{Sym}^\pi_{/S}(X)_*}.$$

Lemma 2.10. *These restriction morphisms induce an isomorphism*

$$\mathrm{K}_0\mathbf{Char}_{\iota,M}\mathbf{Var}_{\mathbf{Sym}^\iota_{/S}(X)} \simeq \prod_{\pi\vdash\iota} \mathrm{K}_0\mathbf{Char}_{\pi,M}\mathbf{Var}_{\mathbf{Sym}^\pi_{/S}(X)_*}.$$

Proof. Let $([Y_\pi \times_{\mathbf{Sym}^\pi_{X/S}(\underline{M}_X)} \mathbf{Sym}^\pi_{X/S}(D(L_{\pi,i})_{i\in I})_*, h^\pi])_{\pi\vdash\iota}$ be a generator on the right-hand side (beware that different partitions π here a priori correspond to different families $(L_{\pi,i})_{i\in I}$ of groups with morphisms $M \to L_{\pi,i}$). We define $Y_{\pi,\iota}$ to be the variety Y_π viewed above $\mathbf{Sym}^\iota_{/S}(X)$ via composition with the inclusion $\mathbf{Sym}^\pi_{/S}(X)_* \to \mathbf{Sym}^\iota_{/S}(X)$. Then we send our tuple to

$$\sum_{\pi\vdash\iota}([Y_{\pi,\iota} \times_{\mathbf{Sym}^\iota_{X/S}(\underline{M}_X)} \mathbf{Sym}^\iota_{X/S}(D(L_{\pi,i})_{i\in I}), h^{\pi,\iota}])_{\pi\vdash\iota}$$

where $h^{\pi,\iota}$ is defined to be h^π extended by 1 outside of $\mathbf{Sym}^\pi_{X/S}(D(L_{\pi,i})_{i\in I})$. $\square$

Given $\iota_1, \iota_2 \in I$, the natural morphisms

$$\mathbf{Sym}^{\iota_1}_{/S}(X) \times_S \mathbf{Sym}^{\iota_2}_{/S}(X) \to \mathbf{Sym}^{\iota_1+\iota_2}_{/S}(X)$$

and

$$\mathbf{Sym}^{\iota_1}_{/S}\, D(\underline{M}_X) \times_S \mathbf{Sym}^{\iota_2}_{/S}\, D(\underline{M}_X) \to \mathbf{Sym}^{\iota_1+\iota_2}_{/S}\, D(\underline{M}_X)$$

induce maps

$$\mathrm{K}_0\mathbf{Char}_{\iota_1,M}\mathbf{Var}_{\mathbf{Sym}^{\iota_1}_{/S}(X)} \times \mathrm{K}_0\mathbf{Char}_{\iota_2,M}\mathbf{Var}_{\mathbf{Sym}^{\iota_2}_{/S}(X)} \to \mathrm{K}_0\mathbf{Char}_{\iota_1+\iota_2,M}\mathbf{Var}_{\mathbf{Sym}^{\iota_1+\iota_2}_{/S}(X)}.$$

These in turn allow to define a multiplication map on

$$\prod_{\iota} \mathrm{K}_0\mathbf{Char}_{\iota,M}\mathbf{Var}_{\mathbf{Sym}^{\iota}_{/S}(X)}.$$

Definition 2.11. We denote by

$$\mathrm{K}_0\mathbf{Char}_{\mathbf{Sym}^\bullet(M)}\mathbf{Var}_{\mathbf{Sym}^\bullet_{/S}(X)} = \prod_{\iota} \mathrm{K}_0\mathbf{Char}_{\iota,M}\mathbf{Var}_{\mathbf{Sym}^{\iota}_{/S}(X)}.$$

the $\mathrm{K}_0\mathbf{Var}_S$-algebra with addition given by coefficientwise addition, and with multiplication given by the above multiplication map.

We can also do the same construction with localised Grothendieck rings, giving us the following definition.

Definition 2.12. We denote by

$$\mathcal{C}har_{\mathbf{Sym}^\bullet(M)}\mathscr{M}_{\mathbf{Sym}^\bullet_{/S}(X)} = \prod_{\iota} \mathcal{C}har_{\iota,M}\mathscr{M}_{\mathbf{Sym}^{\iota}_{/S}(X)}.$$

the $\mathscr{M}_S$-algebra with addition given by coefficientwise addition, and with multiplication given by the analogue for localised Grothendieck rings of the above multiplication map.

Now that we have access to incidence algebras also in the setting of Grothendieck rings of varieties with characters, we may define, in the same way as for classical Grothendieck rings, (mixed) symmetric products of classes in the Grothendieck ring of varieties with characters.

2.6. **Motivic Euler products and characters.** The definition of motivic Euler products in the setting of the Grothendieck rings of varieties with characters then works exactly as in 2.3. Again, we give ourselves a set $I = I_0 \setminus \{0\}$ where I_0 is a commutative monoid and indeterminates $(T_i)_{i\in I}$ satisfying $T_{i_1}T_{i_2} = T_{i_1+i_2}$.

Definition 2.13. Let S be a scheme, X a quasi-projective variety over S, and M an abstract commutative group. Let $\mathscr{A} = (a_i)_{i\in I}$ be a family of classes in $\mathrm{K}_0\mathbf{Char}_M\mathbf{Var}_X$. The motivic Euler product

$$\prod_{x\in X/S}\left(1 + \sum_{i\in I} a_{i,x}T_i\right)$$

is a notation for the series

$$\sum_{i\in I}\left(\sum_{\pi\vdash i}\mathbf{Sym}^{\pi}_{X/S}(\mathscr{A})_*\right)T_i \in \mathrm{K}_0\mathbf{Char}_{\mathbf{Sym}^\bullet(M)}\mathbf{Var}_{\mathbf{Sym}^\bullet_{/S}(X)}.$$

This notion of motivic Euler product with characters in particular allows us to define a notion of *motivic L-function*.

Definition 2.14 (Motivic L-function)**.** The (universal) motivic L-function is the formal series

$$L(\cdot, T) = \prod_{v \in \mathscr{C}} \left(\sum_{n \geqslant 0} \mathrm{ev}_n T^n \right) \in \mathrm{K}_0 \mathbf{Char}_{\mathbf{Sym}^\bullet \mathbf{Z}} \mathbf{Var}_{\mathbf{Sym}^\bullet \mathscr{C}}$$

which should be understood as the motivic function sending χ to the formal series

$$\prod_{v \in \mathscr{C}} (1 - \chi_v T)^{-1}.$$

2.7. **Symmetric products and Hilbert 90.** We generalise [Bil23, Proposition 3.4.0.1] to algebraic groups satisfying Hilbert 90. We say that an algebraic group G satisfies Hilbert 90 if a G-torsor which is locally trivial for the étale topology is also Zariski-locally trivial. By [Mil13, Theorem 11.4], the multiplicative group $\mathbf{G}_m$ (and, thus, more generally, split tori) satisfies Hilbert 90.

Proposition 2.15. *Let X be a variety over S, $\mathscr{Y} = (Y_i)_{i \in I}$ a family of varieties above X and $\mathscr{G} = (G_i)_{i \in I}$ a family of algebraic groups over S satisfying Hilbert 90. Let $\mathscr{Y} \times \mathscr{G}$ be the family $(Y_i \times G_i)_{i \in I}$, each element of this family being viewed above X via the first projection.*

Then for any generalised partition $\pi = (n_i)_{i \in I}$, the symmetric product

$$\mathbf{Sym}^\pi_{X/S}(\mathscr{Y} \times \mathscr{G})_*$$

is endowed with the structure of a $\prod_i G_i^{n_i}$-torsor (for the Zariski topology) above $\mathbf{Sym}^\pi_{X/S}(\mathscr{Y})_*$.

Proof. The argument used to prove [Bil23, Proposition 3.4.0.1] works here almost verbatim. Since

$$\left(\prod_{i \in I} Y_i^{n_i} \times \prod_{i \in I} G_i^{n_i} \right)_* \simeq \left(\prod_{i \in I} Y_i^{n_i} \times \prod_{i \in I} G_i^{n_i} \right) \times_{\prod_{i \in I} X^{n_i}} \left(\prod_{i \in I} X_i^{n_i} \right)_* \simeq \left(\prod_{i \in I} Y_i^{n_i} \right)_* \times \prod_{i \in I} G_i^{n_i}$$

we have the Cartesian square:

$$\begin{array}{ccc} (\prod_{i \in I} Y_i^{n_i})_* \times \prod_{i \in I} G_i^{n_i} & \longrightarrow & (\prod_{i \in I} Y_i^{n_i})_* \\ \downarrow{\scriptstyle q'} & & \downarrow{\scriptstyle q} \\ \mathbf{Sym}^\pi_{X/S}(\mathscr{Y} \times \mathscr{G})_* & \xrightarrow{p} & \mathbf{Sym}^\pi_{X/S}(\mathscr{Y})_* \end{array}$$

where q' is an étale-local trivialisation of p and thus the claim follows by Hilbert 90 for G_i, $i \in I$. □

Example 2.16. Let S be the spectrum of a field k. Take $\mathscr{Y}$ to be the constant family $\mathscr{C} \to \mathscr{C}$ and $(G_i)_{i \in I}$ to be the constant family $\mathbf{G}_{m,k}$. Then the previous proposition tells us that

$$\mathbf{Sym}^\pi_{\mathscr{C}/k} (\mathscr{C} \times_k \mathbf{G}_m)_* = \mathbf{Sym}^\pi_{\mathscr{C}/k} (\mathbf{G}_{m,\mathscr{C}})_*$$

is a $\mathbf{G}_m^{\sum_{i \in I} n_i}$-torsor above $\mathbf{Sym}^\pi_{/k} (\mathscr{C})_*$, a Zariski trivialisation being canonically given by the Cartesian square

$$\begin{array}{ccc} (\prod_i \mathscr{C}^{n_i})_* \times \mathbf{G}_m^{\sum_{i \in I} n_i} & \xrightarrow{\mathrm{pr}_1} & (\prod_i \mathscr{C}^{n_i})_* \\ \downarrow{\scriptstyle q'} & & \downarrow{\scriptstyle q} \\ \mathbf{Sym}^\pi_{\mathscr{C}/k} (\mathbf{G}_{m,\mathscr{C}})_* & \xrightarrow{p} & \mathbf{Sym}^\pi_{/k} (\mathscr{C})_* . \end{array}$$

2.8. **Symmetric products of functions on constant group schemes.** In what follows, we will need to work with symmetric products in the following setting: let X be a quasi-projective variety over S, let $(M_i)_{i\in I}$ be sets indexed by a set I and consider, for every $i \in I$, $Y_i = \underline{M_i}_X$ the corresponding constant X-scheme (which, by definition, is the coproduct $\bigsqcup_{m\in M_i} X$, and thus is locally algebraic). In this section, products are taken relatively to S.

2.8.1. *Domain of definition.* Let $\pi \in \mathbf{N}^{(I)}$ be a partition. The product of symmetric groups $\prod_{i\in I} \mathfrak{S}_{n_i}$ acts on the product

$$\prod_{i\in I} (\underline{M_i}_X)^{n_i},$$

and we want to check that the corresponding quotient exists as a scheme. For this, by [Gro71, Exposé V, Proposition 1.8] it suffices to check that the orbit of every point is contained in an affine open. Let $x \in \prod_{i\in I}(\underline{M_i}_X)^{n_i}$. Then the orbit of x is contained in a finite union of products $X \times_S \ldots \times_S X$ of copies of X. Thus, for the sake of proving our statement we may assume that the M_i are finite, and then it follows from quasi-projectivity of X over S.

Thus, the quotient $\prod_{i\in I} \mathbf{Sym}^{n_i}_{/S}(\underline{M_i}_X)$ exists as a (locally algebraic) scheme, and it is by construction endowed with a morphism to $\prod_{i\in I} \mathbf{Sym}^{n_i}_{/S}(X)$. By pulling back via the inclusion

$$\mathbf{Sym}^{\pi}_{X/S}(X)_* \to \prod_{i\in I} \mathbf{Sym}^{n_i}_{/S}(X),$$

we also get a scheme $\mathbf{Sym}^{\pi}_{X/S}((\underline{M_i}_X)_{i\in I})_*$.

Assume that each M_i is endowed with a commutative group structure. Then the group scheme structures $\underline{M_i}_X$ over X canonically induce a group scheme structure on

$$\prod_{i\in I} (\underline{M_i}_X)^{n_i},$$

which passes to the quotient and gives $\prod_{i\in I} \mathbf{Sym}^{n_i}_{X/S}(\underline{M_i}_X)$, respectively $\mathbf{Sym}^{\pi}_{X/S}((\underline{M}_X)_{i\in I})_*$, a group scheme structure over $\prod_{i\in I} \mathbf{Sym}^{\pi}_{/S}(X)$, resp. $\mathbf{Sym}^{\pi}_{/S}(X)_*$. The construction also extends to mixed symmetric products in the obvious way.

2.8.2. *Symmetric products of functions.* Let again $X \to S$ be a quasi-projective variety above a base scheme S.

Let I and $(M_i)_{i\in I}$ be sets and $(X_{m_i})_{(m_i)\in\prod_{i\in I} M_i}$ a family of varieties above X, which we see as a motivic function on the constant group $\prod_{i\in I} \underline{M_i}_X$ above X. We consider the family

$$(\varphi_i)_{i\in I} = \left(\coprod_{m_i\in M_i} (X_{m_i} \to X) \right)_{i\in I}.$$

For any generalised partion $\pi = (n_i)_{i\in I}$, we have canonical isomorphisms

$$\begin{aligned}\prod_{i\in I} \left(\coprod_{m_i\in M_i} X_{m_i} \right)^{n_i} &\simeq \prod_{i\in I} \left(\coprod_{m_{i,1}\in M_i} X_{m_{i,1}} \right) \times_S \cdots \times_S \left(\coprod_{m_{i,n_i}\in M_i} X_{m_{i,n_i}} \right) \\ &\simeq \coprod_{(m_{i,j})\in\prod_{i\in I} M_i^{n_i}} \prod_{i\in I} \left(X_{m_{i,1}} \times_S \ldots \times_S X_{m_{i,n_i}} \right).\end{aligned}$$

For every $i \in I$, the action of $\sigma_i \in \mathfrak{S}_{n_i}$ on X^{n_i} permuting the factors induces an action on $\left(\coprod_{m_{i,1}\in M_i} X_{m_{i,1}}\right) \times_S \cdots \times_S \left(\coprod_{m_{i,n_i}\in M_i} X_{m_{i,n_i}}\right)$ which does the following: for $(m_{i,j})_{j=1}^{n_i} \in M_i^{n_i}$ identifies the component $X_{m_{i,1}} \times_S \ldots \times_S X_{m_{i,n_i}}$ with the component $X_{m_{i,\sigma(1)}} \times_S \ldots \times_S X_{m_{i,\sigma_i(n_i)}}$

given by $(m_{i,\sigma(j)})_{j=1}^{n_i} \in M_i^{n_i}$ (the factors are the same, they are simply reordered) relatively to X^{n_i}. Taking quotients with respect to this action leads to the commutative diagram

$$\begin{tikzcd}
\coprod_{(m_{i,j})\in\prod_{i\in I} M_i^{n_i}} \prod_{i\in I} X_{m_{i,1}} \times_X \ldots \times_X X_{m_{i,n_i}} \arrow[r] \arrow[d, "/\prod_{i\in I}\mathfrak{S}_{n_i}"] & \prod_{i\in I} X^{n_i} \arrow[d, "/\prod_{i\in I}\mathfrak{S}_{n_i}"] \\
\mathbf{Sym}^{\pi}_{X/S}((\varphi_i)_{i\in I}) \arrow[r] & \mathbf{Sym}^{\pi}_{/S}(X)
\end{tikzcd}$$

and one restricts to $\mathbf{Sym}^{\pi}_{/S}(X)_*$ to get $\mathbf{Sym}^{\pi}_{X/S}((\varphi_i)_{i\in I})_*$.

Example 2.17. Take $M_i = \mathbf{Z}$ and $X_i = X$ for every $i \in I$. Then

$$\left(\coprod_{m_i\in M_i} X_{m_i}\right)_{i\in I} = \left(\coprod_{\mathbf{Z}} X\right)_{i\in I} = (\underline{\mathbf{Z}}_X)_{i\in I}$$

and the previous diagram becomes

$$\begin{tikzcd}
\prod_{i\in I} \underbrace{\underline{\mathbf{Z}}_X \times_S \ldots \times_S \underline{\mathbf{Z}}_X}_{n_i \text{ times}} \arrow[r, "\simeq"] & \coprod_{\prod_{i\in I}\mathbf{Z}^{n_i}} \prod_{i\in I} X^{n_i} \arrow[r] \arrow[d, "/\prod_{i\in I}\mathfrak{S}_{n_i}"] & \prod_{i\in I} X^{n_i} \arrow[d, "/\prod_{i\in I}\mathfrak{S}_{n_i}"] \\
& \mathbf{Sym}^{\pi}_{X/S}((\underline{\mathbf{Z}}_X)_{i\in I}) \arrow[r] & \mathbf{Sym}^{\pi}_{/S}(X).
\end{tikzcd}$$

The fibre above a point $D = \sum_{i\in I} D_i$ where $D_i = [p_{i,1}] + \ldots + [p_{i,n_i}]$ is an unordered product

$$\oplus_{i\in I} \oplus_{i=1}^{n_i} \underline{\mathbf{Z}}_{\kappa(p_i)}.$$

3. Multiplicative motivic harmonic analysis

In this section we develop an elementary version of motivic harmonic analysis which in practice works well with abstract commutative groups and their Cartier duals, classically called *diagonalisable algebraic groups*. The setting we adopt is sometimes more general than needed and we included alternative simpler proofs that work in the specific situation we consider in our application to the height zeta function.

3.1. Fourier transforms and inversion formulas.

Let S be a scheme and M be an abstract commutative group. In this paragraph, by *motivic function* we will simply mean an element of the relative Grothendieck ring $\mathcal{C}har_M\mathscr{M}_{\underline{M}_S}$, which will be interpreted as a function on $\underline{M}_S$.

Recall from Notation 1.12 that we have the class

$$\mathrm{ev} = [\underline{M}_S \times_S D(\underline{M}_S), (m,\chi) \mapsto \chi(m)] \in \mathcal{C}har_M\mathscr{M}_{\underline{M}_S}.$$

Definition 3.1. Let M be an abstract commutative group and let $\psi \in \mathscr{M}_{\underline{M}_S}$ be a motivic function supported over finitely many points of $\underline{M}_S$. The motivic Fourier transform of ψ is defined as the class

$$\mathscr{F}\,\psi = \psi \cdot \mathrm{ev} \in \mathcal{C}har_M\mathscr{M}_S$$

where the product is taken in $\mathcal{C}har_M\mathscr{M}_{\underline{M}_S}$ and then viewed in $\mathcal{C}har_M\mathscr{M}_S$. We can see this as the motivic function associating

$$(\mathscr{F}\,\psi)(\chi) = \sum_{m\in M} \psi(m)\chi(m)$$

to any $\chi \in D(\underline{M}_S)$.

Assume ψ is of the form $[X \to \underline{M}_S, 1]$. Then, explicitly,

$$\mathscr{F}\,\psi = [X \times_{\underline{M}_S} \underline{M}_S \times_S D(\underline{M}_S), (x, m, \chi) \mapsto \chi(m)]_{S \times_S D(\underline{M}_S)}$$

$$\begin{tikzcd}
X \arrow[d] & \underline{M}_S \times_S D(\underline{M}_S) \arrow[r, "\mathrm{ev}"] \arrow[dl, "\mathrm{pr}_1"'] \arrow[dr, "\mathrm{pr}_2"] & \mathbf{G}_{m,S} \\
\underline{M}_S \arrow[dr] & & D(\underline{M}_S) \arrow[dl] \\
& S &
\end{tikzcd} \tag{3.1}$$

Example 3.2. Taking $\psi = [\{m\} \times S \subset \underline{M}_S, 1]$ for some $m \in M$ (interpreted as the characteristic function $\mathbf{1}_{\{m\}}$ of m), we obtain

$$\mathscr{F}\,\psi = [S \times_S D(\underline{M}_S), \chi \mapsto \chi(m)] = \mathrm{ev}(m, \cdot).$$

Proposition 3.3 (Motivic Fourier inversion formula)**.** *Let $\psi \in \mathscr{M}_{\underline{M}_S}$ be a motivic function supported over finitely many points of M. Then the motivic Fourier transform of ψ is well-defined and*

$$\psi(x) = \int_{D(\underline{M}_s)} (\mathscr{F}\,\psi)(\chi) \cdot \mathrm{ev}(-x, \chi)$$

for any $x \in \underline{M}_S$.

Proof. By Lemma 1.2 we can assume $S = \mathrm{Spec}(k)$. The function ψ can be written as a finite linear combination of indicator functions of points of $M = \underline{M}_{\mathrm{Spec}(k)}$, so it is enough to prove the statement for an elementary function of the form

$$\mathbf{1}_{\{x_0\}} = [\{x_0\} \hookrightarrow M]_M$$

for some point x_0. Then by Example 3.2,

$$\mathscr{F}\,\mathbf{1}_{\{x_0\}} = \mathrm{ev}(x_0, \cdot)\mathrm{d}\chi$$

and

$$\int_{D(\underline{M})} \mathscr{F}\,\mathbf{1}_{\{x_0\}}(\chi) \cdot \mathrm{ev}(-x, \chi) = \int_{D(\underline{M})} \mathrm{ev}(x_0 - x, \chi)\mathrm{d}\chi = \begin{cases} 1 & \text{if } x = x_0 \\ 0 & \text{otherwise} \end{cases}$$

by (1.2), hence the identity. □

3.2. **Local motivic Poisson formula.** Our Proposition 3.3 is actually a special case of the following.

Theorem 3.4 (Local motivic Poisson formula)**.** *Let $H \subset G$ be commutative groups. We identify $D(G/H)$ with the kernel of the restriction morphism $\mathrm{res}_{|H} : D(G) \to D(H)$.*

Then for any motivic function $\psi \in \mathscr{M}_{\underline{G}_S}$ supported over finitely many points of G,

$$\sum_{h \in H} \psi(g + h) = \int_{D(G/H)} \mathscr{F}\,\psi(\chi) e(-g, \chi)\mathrm{d}\chi$$

in $\mathscr{M}_S$. In particular, taking $g = 0_G$, one gets

$$\sum_{h \in H} \psi(h) = \int_{D(G/H)} \mathscr{F}\,\psi(\chi)\mathrm{d}\chi.$$

Here by $\sum_{h\in H}\psi(h)$ we mean the sum of the values of ψ on the finite number of points in the intersection of H with the support of ψ, and $\int_{D(G/H)} : \mathcal{C}har_G\mathscr{M}_S \to \mathscr{M}_S$ is the operator defined in Definition 1.15.

Proof. By Lemma 1.2 again we can assume $S = \mathrm{Spec}(k)$. We can modify the proof of Proposition 3.3 by replacing the functional

$$\psi \mapsto \psi(x) = \sum_{g\in G} \mathbf{1}_{\{0\}}(g)\psi(x+g)$$

by

$$\psi \mapsto \sum_{h\in H} \psi(x+h) = \sum_{g\in G} \mathbf{1}_H(g)\psi(x+g).$$

Up to replacing ψ by a translate, it is sufficient to prove the second identity. Moreover, by linearity we may assume that ψ is supported on on a single point g_0, so that in fact $\psi = \psi(g_0)\mathbf{1}_{\{g_0\}}$. Then the left-hand side is equal to $\psi(g_0)$ if $g_0 \in H$, and to 0 otherwise. Using (1.3), we get

$$\int_{D(G/H)} \mathscr{F}\psi(\chi)\mathrm{d}\chi = \psi(g_0)\int_{D(G/H)} \mathrm{ev}(g_0,\chi)\mathrm{d}\chi = \begin{cases} \psi(g_0) & \text{if } g_0 \in H \\ 0 & \text{otherwise,} \end{cases}$$

hence the result.

□

3.3. **Application to the abstract group of divisors.** The (abstract) group $\mathrm{Div}_{\mathsf{S}}(\mathscr{C})$ of divisors supported on a finite set S of points of $\mathscr{C}$ may be identified with $\mathbf{Z}^{\mathsf{S}}$. If a function ψ is defined on $\mathrm{Div}_{\mathsf{S}}(\mathscr{C})$, it can be seen, via extension by zero, as a function on $\mathrm{Div}_{\mathsf{S}'}(\mathscr{C})$ for every S' containing S. Then for every such S', the associated Fourier transform $\mathscr{F}\psi(\chi)$ will not depend on $\chi_v \in D(\underline{\mathbf{Z}}_{\kappa(v)})$ for $v \notin \mathsf{S}$.

Under the same hypothesis, if ψ is seen as a function on $\mathrm{Div}(\mathscr{C}) = \varinjlim_{\mathsf{S}} \mathrm{Div}_{\mathsf{S}}(\mathscr{C})$, then for any subgroup $H \subset \mathrm{Div}(\mathscr{C})$, ψ is H-equivariant if and only if ψ is $(H \cap \mathrm{Div}_{\mathsf{S}}(\mathscr{C}))$-equivariant for any S such that ψ is supported on $\mathrm{Div}_{\mathsf{S}}(\mathscr{C})$.

Typically, in our application we will take

$$H = \mathrm{PDiv}_{\mathsf{S}}(\mathscr{C})$$

to be the group of principal divisors of $\mathscr{C}$ having support contained in a finite set $\mathsf{S} \subset |\mathscr{C}|$ of closed points of the curve $\mathscr{C}$ (or of $\mathscr{C}\otimes k'$ for some extension k'/k). In that case, H is contained in the subgroup

$$H_0 = \mathrm{Div}^0_{\mathsf{S}}(\mathscr{C}) = (1,...,1)^{\perp} \cap \mathrm{Div}_{\mathsf{S}}(\mathscr{C})$$

of degree-zero divisors having support in S. Then the commutative diagram of exact sequences

$$\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & & \\
 & H \arrow[d] & H \arrow[d] & & \\
0 \arrow[r] & H_0 \arrow[r] \arrow[d] & G \arrow[r] \arrow[d] & G/H_0 \arrow[r] \arrow[d, equal] & 0 \\
0 \arrow[r] & H_0/H \arrow[r] \arrow[d] & G/H \arrow[r] \arrow[d] & (G/H)/(H_0/H) \arrow[r] & 0 \\
 & 0 & 0 & &
\end{tikzcd}$$

induces by duality a diagram of exact sequences of diagonalisable algebraic groups.

By choosing once and for all a k-divisor $\mathfrak{D}_1$ of degree one on the curve and enlarging S if necessary, we fix a section of the quotient morphism

$$\deg : G \to G/H_0$$

and a splitting of the first horizontal sequence of $\mathbf{Z}$-modules by writing any divisor D as

$$(D - \deg(D)\mathfrak{D}_1) + \deg(D)\mathfrak{D}_1.$$

This splitting induces a splitting of the second horizontal sequence and by duality, splittings of $D(G)$ and $D(G/H)$.

4. Toric harmonic analysis

4.1. **Local invariant functions and their Fourier transforms.** We will now specialize a bit more to the concrete setup that we need for our purposes. For this, we place ourselves in the setting described in the Notations section, with a curve $\mathscr{C}$ over a base field k. Let U be a split algebraic torus of dimension n defined over k. We denote by $\mathcal{X}^*(U) = \mathrm{Hom}(U, \mathbf{G}_m)$ its group of characters and $\mathcal{X}_*(U) = \mathrm{Hom}(\mathbf{G}_m, U)$ its group of cocharacters. There is a natural pairing

$$\langle \cdot, \cdot \rangle : \mathcal{X}^*(U) \times \mathcal{X}_*(U) \to \mathbf{Z}.$$

The aim of this section is to define motivic incarnations of families of local functions $(f_v : U(F_v) \to \mathbf{C})_{v \in \mathscr{C}}$ which are $U(\mathcal{O}_v)$-invariant. The starting point here is that the local degree mapping

$$\deg_{F,v} : U(F_v) \to \mathcal{X}_*(U)$$

induces an isomorphism

$$U(F_v)/U(\mathcal{O}_v) \simeq \mathcal{X}_*(U),$$

so that such a local function may be viewed as a function on the free $\mathbf{Z}$-module $\mathcal{X}_*(U)$ of rank n. Moreover, in our setting, we only need to consider such functions with finite support in $\mathcal{X}_*(U)$. This motivates the following definition:

Definition 4.1. A family of local motivic $U(\mathcal{O})$-invariant functions is an element of the ring $\mathscr{M}_{\underline{\mathcal{X}_*(U)}_{\mathscr{C}}}$ supported over finitely many points of $\mathcal{X}_*(U)$. These elements form a subring of $\mathscr{M}_{\underline{\mathcal{X}_*(U)}_{\mathscr{C}}}$ denoted by $\mathrm{Fun}(U(F_v))$.

Notation 4.2. Let $\mathbf{a} \in \mathcal{X}_*(U)$ be a cocharacter of U. We denote by

$$\mathbf{1}_{\mathbf{t}^{\mathbf{a}}U(\mathcal{O})}$$

the family of local motivic $U(\mathcal{O})$-invariant functions given by the class of the subscheme

$$\{\mathbf{a}\} \times \mathscr{C} \hookrightarrow \mathcal{X}_*(U) \times_k \mathscr{C} = \underline{\mathcal{X}_*(U)}_{\mathscr{C}}$$

viewed in $\mathrm{Fun}(U(F_v))$. When picking identifications $U \simeq \mathbf{G}_m^n$ and $\mathcal{X}_*(U) \simeq \mathbf{Z}^n$, so that $\mathbf{a}$ is viewed as a tuple of integers $(a_1, \ldots, a_n)$, it is the motivic incarnation of the family of characteristic functions of the subsets

$$t_1^{a_1}\mathcal{O}_v^\times \times \ldots \times t_n^{a_n}\mathcal{O}_v^\times \subset (F_v^\times)^n.$$

as v runs over points of $\mathscr{C}$.

Remark 4.3. The forgetful morphism

$$\mathrm{Fun}(U(F_v)) \to \mathscr{M}_{\mathscr{C}}$$

may be interpreted as integration over $U(F_v)$ for each v.

By the theory of Section 3.1 applied to the constant group scheme $\underline{\mathcal{X}_*(U)}_{\mathscr{C}}$ over $\mathscr{C}$, we then have access to a notion of Fourier transformation for such a family of functions, which naturally lives in $\mathcal{C}har_{\mathcal{X}_*(U)}\mathscr{M}_{\mathscr{C}}$.

Definition 4.4. A family of local motivic functions with characters on U is an element of $\mathcal{C}har_{\mathcal{X}_*(U)}\mathscr{M}_{\mathscr{C}}$.

Thus, given $f = (f_v)_v \in \mathrm{Fun}(U(F_v))$ a family of local motivic $U(\mathcal{O})$-invariant functions, its Fourier transform $\mathscr{F}f$ in the sense of Definition 3.1 is a family of local motivic functions with characters on U.

Example 4.5. Let $\mathbf{a} \in \mathcal{X}_*(U)$. By example 3.2, we have

$$\mathscr{F}\mathbf{1}_{\mathbf{t}^{\mathbf{a}}U(\mathcal{O})} = \mathrm{ev}(\mathbf{a}, \cdot) \in \mathcal{C}har_{\mathcal{X}_*(U)}\mathscr{M}_{\mathscr{C}}.$$

4.2. **Symmetric powers and divisors on the curve.** Before proceeding to describing the global situation, we insert a quick discussion of the way that symmetric products of the curve $\mathscr{C}$ will allow us to do Fourier analysis on the group of divisors of the curve.

Let I be a set, and $\pi = (n_i)_{i\in I}$ a partition. A schematic point $E \in \mathbf{Sym}^\pi_{/k}(\mathscr{C})_*$ can be written

$$\sum_{i\in I} i(v_{i,1} + \ldots + v_{i,n_i})$$

where the $v_{i,j}$ are geometric points of $\mathscr{C}$. Fixing an algebraic closure K of the residue field $\kappa(E)$, and defining S_i to be the set of $\mathrm{Gal}(K/\kappa(E))$-orbits of elements in $\{v_{i,1}, \ldots, v_{i,n_i}\}$, so that S_i gives a finite set of closed points on $\mathscr{C}_{\kappa(E)}$.

4.3. **Global $\mathbf{K}(U)$-invariant functions and global Fourier transforms.** In this paragraph we define motivic incarnations of global adelic functions. More precisely, denoting by $\mathbf{K}(U)$ the subset $\prod_v U(\mathcal{O}_v) \subset U(\mathbf{A}_F)$, we define motivic incarnations of $\mathbf{K}(U)$-invariant global functions of the form

$$f = \prod_v f_v,$$

where the f_v are local $U(\mathcal{O}_v)$-invariant functions. For this to make sense motivically, we assume that the f_v are chosen among a finite number of different possibilities.

Concretely, let I be a set and let $(f_i)_{i\in I}$ a collection of families of local motivic $U(\mathcal{O})$-invariant functions. Then the number of factors n_i where f_v is given by $f_{i,v}$ is encoded by a partition $\pi = (n_i)_{i\in I}$.

Definition 4.6. Let I be a set and let $\pi = (n_i)_{i\in I}$ be a partition. A family of global motivic $\mathbf{K}(U)$-invariant functions of level π is a finite linear combination, with integral coefficients, of elements of the form

$$\mathbf{Sym}^{\pi}_{\mathscr{C}/k}((f_i)_{i\in I})_* \in \mathscr{M}_{\mathbf{Sym}^{\pi}_{\mathscr{C}/k}(\underline{\mathcal{X}_*(U)}_{\mathscr{C}})_*}$$

(in the sense of Section 2.8.2) where for every $i \in I$, f_i is a family of local motivic $U(\mathcal{O})$-invariant functions (having finite support, by definition).

Remark 4.7. A family of global motivic $\mathbf{K}(U)$-invariant functions of level π naturally lives above

$$\mathbf{Sym}^{\pi}_{\mathscr{C}/k}(\underline{\mathcal{X}_*(U)}_{\mathscr{C}})_* \to \mathbf{Sym}^{\pi}_{/k}(\mathscr{C})_*.$$

The bottom symmetric product $\mathbf{Sym}^{\pi}_{/k}(\mathscr{C})_*$ parametrises supports, and the intermediate symmetric product $\mathbf{Sym}^{\pi}_{\mathscr{C}/k}(\underline{\mathcal{X}_*(U)}_{\mathscr{C}})_*$ parametrises domains of definition.

Concretely, the fibre of $\mathbf{Sym}^{\pi}_{\mathscr{C}/k}((f_i)_{i\in I})$ above $E = \sum i(v_{i,1} + \ldots + v_{i,n_i}) \in \mathbf{Sym}^{\pi}_{/k}(\mathscr{C})_*$ is interpreted as the adelic function given by f_i at the places $v_{i,1}, \ldots, v_{i,n_i}$ for each $i \in I$ (and just equal to 1 at all other places), and thus defined on

$$\prod_{i\in I} U(F_{v_{i,1}}) \times \ldots \times U(F_{v_{i,n_i}})$$

which, since the functions are $U(\mathcal{O})$-invariant, is modeled by the fiber $(\mathbf{Sym}^{\pi}(\underline{\mathcal{X}_*(U)}_{\mathscr{C}}))_E = \prod_{v\in|E|} \mathcal{X}_*(U)_v$.

Example 4.8 (Family of characteristic functions)**.** Assume that there exists $\mathbf{a}_i$ such that

$$f_i = \mathbf{1}_{t^{\mathbf{a}_i}U(\mathcal{O})}$$

for every $i \in I$. Then the fibre of $\mathbf{Sym}^{\pi}_{\mathscr{C}/k}((f_i)_{i\in I})_*$ above $E = \sum i(v_{i,1}+\ldots+v_{i,n_i}) \in \mathbf{Sym}^{\pi}_{/k}(\mathscr{C})_*$ is the global function

$$\prod_{i\in I}\prod_{j=1}^{n_i} \mathbf{1}_{t^{\mathbf{a}_i}U(\mathcal{O}_{v_{i,j}})}.$$

For example, if we are given a map $\gamma^\vee : I \to \mathcal{X}_*(U)$, as it will be the case latter, then we can take $\boldsymbol{a}_i = \gamma^\vee(i)$ for all $i \in I$. Its provides a natural section

$$\gamma^\vee : \mathbf{Sym}^{\pi}_{/k}(\mathscr{C})_* \to \mathbf{Sym}^{\pi}_{\mathscr{C}/k}(\underline{\mathcal{X}_*(U)}_{\mathscr{C}})_*$$

given by

$$\sum_{i\in I} i(v_{i,1} + \ldots + v_{i,n_i}) \mapsto \sum_{i\in I} i((\gamma^\vee(i), v_{i,1}) + \ldots + (\gamma^\vee(i), v_{i,n_i})).$$

4.4. **Summation over rational points.** In this section, we define a motivic analogue of the operation which, to an appropriately integrable function $f : U(\mathbf{A}_F) \to \mathbf{C}$ on the idèles associates

$$\sum_{x\in U(F)} f(x).$$

More specifically, since in the previous paragraph we have defined our functions in families (parametrised by symmetric products), we will explain how to perform this operation simultaneously for all functions in such a family.

Proposition 4.9. *There is a morphism*

$$\mathbf{Sym}^{\pi}_{\mathscr{C}/k}(\underline{\mathcal{X}_*(U)}_{\mathscr{C}})_* \to \mathrm{Hom}_{\mathrm{gp}}(\mathcal{X}^*(U), \mathbf{Pic}(\mathscr{C}))$$

given by

$$\sum_{i\in I} i((\mathbf{b}_{i,1}, v_{i,1}) + \ldots + (\mathbf{b}_{i,n_i}, v_{i,n_i})) \mapsto \left(\mathbf{a} \mapsto \left[\sum_{i,j} \langle \mathbf{a}, \mathbf{b}_{i,j}\rangle v_{i,j}\right]\right).$$

Proof. To simplify notation, we will fix an isomorphism $\mathcal{X}_*(U) \simeq \mathbf{Z}^n$, so that our map becomes

$$\begin{array}{ccc} \mathbf{Sym}^{\pi}_{\mathscr{C}/k}(\underline{\mathbf{Z}^n}_{\mathscr{C}})_* & \to & \mathbf{Pic}(\mathscr{C})^n \\ \sum_{i\in I} i((\mathbf{b}_{i,1}, v_{i,1}) + \ldots + (\mathbf{b}_{i,n_i}, v_{i,n_i})) & \mapsto & \left[\sum_{i,j} \mathbf{b}_{i,j} v_{i,j}\right] \end{array}$$

where the square brackets mean that we take linear equivalence classes in each component. For every $\mathbf{b} = (b_1, \ldots, b_n) \in \mathbf{Z}^n$ we have a morphism

$$\mathscr{C} \to \mathbf{Pic}^{b_1}(\mathscr{C}) \times \ldots \times \mathbf{Pic}^{b_n}(\mathscr{C})$$

sending a point v to the tuple of linear equivalence classes $(b_1[v], \ldots, b_n[v])$. Combining these all together gives a morphism

$$\underline{\mathbf{Z}^n}_{\mathscr{C}} \to \mathbf{Pic}(\mathscr{C})^n.$$

Let now $\pi = (n_i)_{i\in I}$ be a partition. Taking products and restricting to the complement of the diagonal in $\prod_{i\in I} \mathscr{C}^{n_i}$ we obtain a morphism

$$\left(\prod_{i\in I} (\underline{\mathbf{Z}^n}_{\mathscr{C}})^{n_i}\right)_* \to \mathbf{Pic}(\mathscr{C})^n,$$

given by $(\mathbf{b}_{i,j}, v_{i,j})_{\substack{i\in I \\ j\in\{1,\ldots,n_i\}}} \mapsto \sum_{i,j} \mathbf{b}_{i,j}[v_{i,j}]$. This passes to the quotient with respect to the $\prod_{i\in I} \mathfrak{S}_{n_i}$–action, giving the morphism we want. □

Notation 4.10. We denote by $\mathbf{PDiv}^{\pi}(\mathscr{C})$ the fiber above 0 of the morphism of Proposition 4.9.

Recall that we also have a morphism $\mathbf{Sym}^{\pi}_{\mathscr{C}/k}(\underline{\mathcal{X}_*(U)}_{\mathscr{C}})_* \to \mathbf{Sym}^{\pi}(\mathscr{C})$.

Definition 4.11. Given $f \in \mathscr{M}_{\mathbf{Sym}^{\pi}_{\mathscr{C}/k}(\underline{\mathcal{X}_*(U)}_{\mathscr{C}})_*}$ a family of global motivic functions of level π, we denote by

$$\sum_{\mathrm{PDiv}^{\pi}} f$$

the image in $\mathscr{M}_{\mathbf{Sym}^{\pi}_{/k}(\mathscr{C})_*}$ of the pullback of f to $\mathbf{PDiv}^{\pi}(\mathscr{C})$.

Notation 4.12. Using the formalism of incidence algebras from Section 2.3, we may then combine the operators $\sum_{\mathrm{PDiv}^{\pi}}$ into an operator

$$\sum_{\mathrm{PDiv}} : \mathscr{M}_{\mathbf{Sym}^{\bullet}_{\mathscr{C}/k}(\underline{\mathcal{X}_*(U)}_{\mathscr{C}})} \to \mathscr{M}_{\mathbf{Sym}^{\bullet}_{/k}(\mathscr{C})}.$$

4.5. **Global characters and character functions.** Analogously to Definition 4.6, we define:

Definition 4.13. A family of global functions with characters of level π is a finite linear combination, with integral coefficients, of elements of the form

$$\mathbf{Sym}^{\pi}_{\mathscr{C}/k}((g_i)_{i\in I})_* \in \mathcal{C}har_{\pi,\mathcal{X}_*(U)}\mathscr{M}_{\mathbf{Sym}^{\pi}_{/k}(\mathscr{C})_*}$$

where $(g_i)_{i\in I}$ is a family of local motivic character functions on U.

A family of global functions with characters of level π naturally lives above the symmetric product $\mathbf{Sym}^{\pi}_{\mathscr{C}/k}(D(\underline{\mathcal{X}_*(U)}_{\mathscr{C}})) \to \mathbf{Sym}^{\pi}_{/k}(\mathscr{C})_*$, which we may think of as parameterizing families of global characters with support of size $|\pi|$. The fibre of $\mathbf{Sym}^{\pi}_{\mathscr{C}/k}((g_i)_{i\in I})$ above $E = \sum i(v_{i,1} + \ldots + v_{i,n_i}) \in \mathbf{Sym}^{\pi}_{/k}(\mathscr{C})_*$ represents, through the local isomorphism $D(\underline{\mathcal{X}_*(U)}_{\mathscr{C}})_{v_{i,j}} \simeq U_{v_{i,j}}$, functions defined on the character domain

$$\prod_{i\in I} U_{v_{i,1}} \times \ldots \times U_{v_{i,n_i}}.$$

The natural evaluation pairing $\underline{\mathcal{X}_*(U)}_{\mathscr{C}} \times_{\mathscr{C}} D(\underline{\mathcal{X}_*(U)}_{\mathscr{C}}) \to \mathbf{G}_m$ induces a pairing

$$\mathbf{Sym}^{\pi}_{\mathscr{C}/k}(\mathrm{ev}) : \mathbf{Sym}^{\pi}_{\mathscr{C}/k}(\underline{\mathcal{X}_*(U)}_{\mathscr{C}})_* \times_{\mathbf{Sym}^{\pi}_{/k}(\mathscr{C})_*} \mathbf{Sym}^{\pi}_{\mathscr{C}/k}(D(\underline{\mathcal{X}_*(U)}_{\mathscr{C}}))_* \to \mathbf{G}_m,$$

perfect relatively to $\mathbf{Sym}^{\pi}_{/k}(\mathscr{C})_*$, in the sense that for every $E \in \mathbf{Sym}^{\pi}_{/k}(\mathscr{C})_*$, it induces a perfect pairing

$$\mathbf{Sym}^{\pi}_{\mathscr{C}/k}(\underline{\mathcal{X}_*(U)}_{\mathscr{C}})_{*,E} \times_{\kappa(E)} \mathbf{Sym}^{\pi}_{\mathscr{C}/k}(D(\underline{\mathcal{X}_*(U)}_{\mathscr{C}}))_{*,E} \to \mathbf{G}_m,$$

between the corresponding fibres.

Remark 4.14 (Compatibility between Fourier transformation and symmetric products). Let $(f_i)_{i\in I}$ be a collection of families of local motivic $U(\mathcal{O})$-invariant functions, i.e. each $f_i \in \mathscr{M}_{\underline{\mathcal{X}_*(U)}_{\mathscr{C}}}$, supported over a finite number of points of $\mathcal{X}_*(U)$. One may check that

$$\mathbf{Sym}^{\pi}_{\mathscr{C}/k}(\mathscr{F} f_i)_{i\in I} = \mathbf{Sym}^{\pi}_{\mathscr{C}/k}((f_i)_{i\in I})_* \cdot \mathbf{Sym}^{\pi}_{\mathscr{C}/k}(\mathrm{ev})$$

where the product is taken in $\mathcal{C}har_{\pi,\mathcal{X}_*(U)}\mathscr{M}_{\mathbf{Sym}^{\pi}_{\mathscr{C}/k}(\underline{\mathcal{X}_*(U)}_{\mathscr{C}})_*}$ and then viewed in $\mathcal{C}har_{\pi,\mathcal{X}_*(U)}\mathscr{M}_{\mathbf{Sym}^{\pi}_{/k}(\mathscr{C})_*}$.

4.6. **From global to local.** We explain in this section how, looking at the fibre above a point $E \in \mathbf{Sym}^{\pi}_{/k}(\mathscr{C})_*$ we come back to a local situation like the one in Section 3.

Fix an identification $\mathcal{X}_*(U) \simeq \mathbf{Z}^n$. Let $E \in \mathbf{Sym}^{\pi}_{/k}(\mathscr{C})_*$, written $E = \sum_i i(v_{i,1} + \ldots + v_{i,n_i})$ and let, for every $i \in I$, S_i be the set of Galois orbits in $\{v_{i,1}, \ldots, v_{i,n_i}\}$. Then the map

$$\begin{array}{ccc} \mathbf{Sym}^{\pi}(\underline{\mathbf{Z}^n}_{\mathscr{C}}) & \to & \prod_{i\in I} \mathrm{Div}^n(\mathscr{C}) \\ \sum_{i\in I} i((\mathbf{b}_{i,1}, v_{i,1}) + \ldots + (\mathbf{b}_{i,n_i}, v_{i,n_i})) & \mapsto & (\sum_j \mathbf{b}_{i,j} v_{i,j})_{i\in I} \end{array}$$

induces group isomorphisms

$$\mathbf{Sym}^{\pi}(\underline{\mathbf{Z}^n}_{\mathscr{C}})_E \to \prod_{i\in I} \mathrm{Div}^n_{\mathsf{S}_i}(\mathscr{C}_{\kappa(E)}).$$

Through this isomorphism, the subgroup $\mathrm{PDiv}^{\pi}(\mathscr{C})_E \subset \mathbf{Sym}^{\pi}(\underline{\mathbf{Z}^n}_{\mathscr{C}})_E$ corresponds to the subgroup

$$H_{\mathsf{S}} = \left\{ \left(\sum_j \mathbf{b}_{i,j} v_{i,j} \right)_{i\in I}, \ \sum_{i,j} \mathbf{b}_{i,j} v_{i,j} \text{ is principal in each component} \right\}$$

inside

$$G_{\mathsf{S}} = \prod_{i\in I} \mathrm{Div}^n_{\mathsf{S}_i}(\mathscr{C}_{\kappa(E)})$$

As for the dual side, the fibre $\mathbf{Sym}^{\pi}_{\mathscr{C}/k}(D(\underline{\mathcal{X}_*(U)}_{\mathscr{C}}))_E$ is identified with the group of characters $D(G_{\mathsf{S}})$.

4.7. **Integration over invariant characters.** Let $A \in \mathcal{C}har_{\pi,\mathcal{X}_*(U)}\mathscr{M}_{\mathbf{Sym}^\pi_{/k}(\mathscr{C})_*}$. Then via Section 4.6, for every $E \in \mathbf{Sym}^\pi_{/k}(\mathscr{C})_*$, the fibre A_E naturally defines an element of $\mathcal{C}har_{G_\mathsf{S}}\mathscr{M}_{\kappa(E)}$.

We then define

$$\int_E A_E := \int_{D(G_\mathsf{S}/H_\mathsf{S})} A_E.$$

Definition 4.15. We say that $A \in \mathcal{C}har_{\pi,\mathcal{X}_*(U)}\mathscr{M}_{\mathbf{Sym}^\pi_{/k}(\mathscr{C})_*}$ is *integrable* if there exists $B \in \mathscr{M}_{\mathbf{Sym}^\pi_{/k}(\mathscr{C})_*}$ such that for all $E \in \mathbf{Sym}^\pi_{/k}(\mathscr{C})_*$

$$\int_E A_E = B_E.$$

In this case we write

$$\int_{(\mathrm{PDiv}^n)^\perp} A = B.$$

4.8. **The motivic multiplicative Poisson formula.** Let $f = \mathbf{Sym}^\pi((f_i)_{i\in I})_*$ be a family of global motivic $\mathbf{K}(U)$-invariant functions of level π, with corresponding family of Fourier transforms

$$\mathscr{F} f := \mathbf{Sym}^\pi((\mathscr{F} f_i)_{i\in I})_* \in \mathcal{C}har_{\pi,\mathcal{X}_*(U)}\mathscr{M}_{\mathbf{Sym}^\pi_{/k}(\mathscr{C})_*}.$$

Theorem 4.16. *Let f be a family of global motivic $\mathbf{K}(U)$-invariant functions of level π. Then $\mathscr{F} f$ is integrable and we have the equality*

$$\sum_{\mathrm{PDiv}^\pi} f = \int_{(\mathrm{PDiv}^\pi)^\perp} \mathscr{F} f \tag{4.1}$$

in $\mathscr{M}_{\mathbf{Sym}^\pi_{/k}(\mathscr{C})_*}$.

Proof. First of all, we may assume that $f = \mathbf{Sym}^\pi_{\mathscr{C}/k}(f_i)_{i\in I}$ where f_i are families of local motivic $U(\mathscr{O})$-invariant functions. We fix a support divisor $E = \sum_{i\in I} i(v_{i,1} + \ldots + v_{i,n_i}) \in \mathbf{Sym}^\pi_{/k}(\mathscr{C})_*$ and we define S, as well as the groups $H_\mathsf{S} \subset G_\mathsf{S}$ as in Section 4.6. We note that $f_E \in \mathscr{M}_G$ is supported over finitely many points of G, so we may apply Theorem 3.4, getting:

$$\sum_{h\in H_\mathsf{S}} f_E(h) = \int_{D(G_\mathsf{S}/H_\mathsf{S})} \mathscr{F} f_E.$$

On the other hand, we have, by definition,

$$\int_E (\mathscr{F} f)_E = \int_{D(G_\mathsf{S}/H_\mathsf{S})} (\mathscr{F} f)_E$$

Since $\mathscr{F} f_E = (\mathscr{F} f)_E$ by Remark 4.14, this shows integrability and, using Lemma 1.2, also equation (4.1).

□

5. Application to the motivic height zeta function

5.1. **An introductory example: the projective line.** We explain without any proof what are the objects coming into play in the very specific case of $X = \mathbf{P}^1_F$ viewed as a toric variety. Define for every non-zero $m \in \mathbf{Z}$

$$\mathbf{1}_{t^m\mathcal{O}^\times} = \{m\} \times \mathscr{C}$$

as a subscheme of the constant group scheme $\underline{\mathbf{Z}}_\mathscr{C}$. Its fibre above $v \in \mathscr{C}$ is denoted $\mathbf{1}_{t^m\mathcal{O}_v^\times}$ and is the motivic incarnation of the $\mathcal{O}_v^\times$-invariant function $\mathbf{1}_{t^m\mathcal{O}_v^\times}$ on the completion F_v of F at v. Thus we get a family

$$(\mathbf{1}_{t^m\mathcal{O}^\times})_{m\in\mathbf{Z}}$$

of subschemes of $\underline{\mathbf{Z}}_{\mathscr{C}}$ which can be used to define a generalised motivic Euler product like those that are described above.

In this notation, denoting by T_-, T_+ our indeterminates, the local height is the motivic function on $\underline{\mathbf{Z}}_{\mathscr{C}}$

$$H_v(\cdot, \mathbf{T}) = 1 + \sum_{m>0} \mathbf{1}_{t^{-m}\mathcal{O}_v^\times} T_-^m + \sum_{m>0} \mathbf{1}_{t^m\mathcal{O}_v^\times} T_+^m \in \mathrm{K}_0\mathbf{Var}_{\underline{\mathbf{Z}}_{\mathscr{C}}}[\mathbf{T}]$$

fitting into a motivic Euler product

$$H(\mathbf{T}, \cdot) = \prod_{v\in\mathscr{C}} \left(1 + \sum_{m>0} \mathbf{1}_{t^{-m}\mathcal{O}_v^\times} T_-^m + \sum_{m>0} \mathbf{1}_{t^m\mathcal{O}_v^\times} T_+^m\right).$$

The expansion of this motivic Euler product is indexed by pairs of non-negative integers (n_+, n_-), and the coefficient corresponding to (n_+, n_-) lives in the Grothendieck ring above the variety $\mathbf{Sym}^{(n_+,n_-)}_{/k}(\mathscr{C})_*$ parametrising zero-cycles D on $\mathscr{C}$ of the form

$$D_+ - D_-$$

where D_+, D_- are effective with disjoint supports (hence the subscript $*$), of respective degrees n_+ and n_-. Each coefficient may be viewed as a family of characteristic functions of balls inside the idèles $\mathbb{A}_F^\times$, parametrised by $\mathbf{Sym}^{(n_+,n_-)}(\mathscr{C})_*$, the fibre above $D = \sum_v m_v v$ being the (motivic incarnation of the) function $\mathbf{1}_D := \prod_v \mathbf{1}_{t^{m_v}\mathcal{O}_v^*}$.

In Section 4.4 we defined a summation operator

$$\sum_{\mathrm{PDiv}(\mathscr{C})}$$

as a pullback to principal divisors, so that applying this operator to $H(\cdot, \mathbf{T})$ kills the coefficients with $n_+ \neq n_-$, and for those with $n_+ = n_- = n$, only the divisors D which are principal contribute. This way

$$(\mathbf{L}-1) \sum_{\mathrm{PDiv}(\mathscr{C})} H(\cdot, \mathbf{T}) = \sum_{n\geq 0} \left[\mathrm{Hom}^n(\mathscr{C}, \mathbf{P}^1)_{\mathbf{G}_m}\right] \mathbf{T}^{\langle\cdot,n\rangle_\Sigma}.$$

The relative Cartier dual of $\underline{\mathbf{Z}}_{\mathscr{C}}$ is $\mathbf{G}_{m,\mathscr{C}}$ and the local Fourier transforms are

$$\mathscr{F}\, \mathbf{1}_{t^m\mathcal{O}^\times}(\cdot) = \mathrm{ev}_{\mathscr{C}}(m, \cdot),$$

where

$$\mathrm{ev}_{\mathscr{C}} : \mathbf{Z}_{\mathscr{C}} \times_{\mathscr{C}} \mathbf{G}_{m,\mathscr{C}} \to \mathbf{G}_{m,\mathscr{C}}$$

is the evaluation map (relative to $\mathscr{C}$). Explicitly, $\mathscr{F}\, \mathbf{1}_{t^m\mathcal{O}^\times}(\chi_v) = \chi_v(m)$. Thus,

$$\begin{aligned}
\mathscr{F}\, H_v(T, \cdot)(\chi_v) &= 1 + \sum_{m<0} \chi_v(m) T_-^m + \sum_{m>0} \chi_v(m) T_+^m \\
&= \frac{1}{1-\chi_v(1)T_+} + \frac{1}{1-\chi_v^{-1}(1)T_-} - 1 \\
&= \frac{1 - T_-T_+}{(1-\chi_v(1)T_+)(1-\chi_v^{-1}(1)T_-)}.
\end{aligned}$$

Note that this special case has the peculiarity that the numerator of the local Fourier transform does not depend on the character. Taking motivic Euler products with characters in the sense

of Section 2 provides the motivic Euler product with characters

$$\prod_{v\in\mathscr{C}} \mathscr{F}\, H_v(T,\cdot)(\chi_v) = \prod_{v\in\mathscr{C}} \frac{1-T_-T_+}{(1-\chi_v(1)T_+)(1-\chi_v^{-1}(1)T_-)}$$
$$= Z_{\mathscr{C}}^{\mathrm{Kap}}(T_-T_+)^{-1}L(\chi,T_+)L(\chi^{-1},T_-),$$

where (χ_v) is a family of local characters all living in $\mathbf{G}_m$, and L denotes the motivic L-functions from Definition 2.14.

Applying the motivic Poisson formula to each of the coefficients of this series leads to the expression

$$\sum_{d\in\mathbf{N}} \left[\mathrm{Hom}_k(\mathscr{C},\mathbf{P}^1)_U\right] T^d = (\mathbf{L}-1)\int_{\mathrm{PDiv}(\mathscr{C})^\perp} \mathscr{F}\, H(T,\chi)\mathrm{d}\chi.$$

5.2. **General geometric setting.** Let k be a field of characteristic zero and let U be an algebraic torus of dimension n defined over k. We denote by $\mathcal{X}^*(U) = \mathrm{Hom}(U,\mathbf{G}_m)$ its group of characters and $\mathcal{X}_*(U) = \mathrm{Hom}(\mathbf{G}_m,U)$ its group of cocharacters. There is a natural pairing

$$\langle\cdot,\cdot\rangle : \mathcal{X}^*(U)\times\mathcal{X}_*(U)\to\mathbf{Z}. \tag{5.1}$$

Let Σ be a projective and regular fan of the $\mathbf{Z}$-module $\mathcal{X}_*(U)$. We denote by $\Sigma(1)$ the set of its rays (that is, its one-dimensional faces), and for every cone $\sigma\in\Sigma$ we denote by $\sigma(1)$ the set of rays of σ. For every $\alpha\in\Sigma(1)$, we denote by ρ_α a generator of the ray α. We denote by $\mathrm{PL}(\Sigma)$ the group of piecewise linear functions on Σ. The pairing (5.1) naturally extends to a pairing

$$\langle\cdot,\cdot\rangle_\Sigma : \mathrm{PL}(\Sigma)\times\mathcal{X}_*(U)\to\mathbf{Z}. \tag{5.2}$$

given by $\langle\varphi,\boldsymbol{n}\rangle_\Sigma = \varphi(\boldsymbol{n})$, in the sense that the diagram

$$\begin{array}{ccc} \mathcal{X}^*(U)\times\mathcal{X}_*(U) & \xrightarrow{\langle\cdot,\cdot\rangle} & \mathbf{Z} \\ \downarrow & & \downarrow \\ \mathrm{PL}(\Sigma)\times\mathcal{X}_*(U) & \xrightarrow{\langle\cdot,\cdot\rangle_\Sigma} & \mathbf{Z} \end{array}$$

commutes.

We now assume that $U=\mathbf{G}_m^n$ is split. Then the fan Σ defines a smooth projective split toric variety X_Σ with open orbit U. Each ρ_α gives rise to a U-invariant divisor $\mathfrak{D}_\alpha$ on X_Σ, and the natural map

$$\varphi\mapsto\sum_\alpha \varphi(\rho_\alpha)\mathfrak{D}_\alpha$$

defines an isomorphism $\mathrm{PL}(\Sigma)\simeq\bigoplus_\alpha \mathbf{Z}\mathfrak{D}_\alpha$. We will often use this identification in what follows: in particular, $\mathfrak{D}_\alpha$ represents the piecewise linear function which sends ρ_α to 1 and all other $\rho_\beta,\ \beta\neq\alpha$ to 0.

We then have an exact sequence

$$0\longrightarrow\mathcal{X}^*(U)\xrightarrow{\gamma}\mathrm{PL}(\Sigma)\longrightarrow\mathrm{Pic}(X_\Sigma)\longrightarrow 0 \tag{5.3}$$

where γ sends an element $\boldsymbol{m}\in\mathcal{X}^*(U)$ to the unique φ in $\mathrm{PL}(\Sigma)$ such that $\varphi(\rho_\alpha)=\langle\boldsymbol{m},\rho_\alpha\rangle$ for every $\alpha\in\Sigma(1)$, while the second arrow sends each $\mathfrak{D}_\alpha$ to its class in $\mathrm{Pic}(X_\Sigma)$.

In what follows, we will work with a family of variables $(T_\alpha)_{\alpha\in\Sigma(1)}$ indexed by $\Sigma(1)$. We will write

$$\mathbf{T}^{\langle\cdot,\boldsymbol{n}\rangle_\Sigma} := \prod_\alpha T_\alpha^{\langle\mathfrak{D}_\alpha,\boldsymbol{n}\rangle_\Sigma}.$$

Let σ be the cone of Σ containing $\boldsymbol{n}$. Then, by definition, there is a unique representation $\boldsymbol{n} = \sum_{\alpha\in\sigma(1)} n_\alpha \rho_\alpha$ for some positive integers $n_\alpha > 0$, and $\mathbf{T}^{\langle\cdot,\boldsymbol{n}\rangle_\Sigma} = \prod_{\alpha\in\sigma(1)} T_\alpha^{n_\alpha}$.

For any $\sigma \in \Sigma$, let I_σ be the subset of $\mathbf{N}^{\Sigma(1)}$ given by

$$I_\sigma = \{\boldsymbol{n} \in \mathbf{N}^{\Sigma(1)} \mid n_\alpha > 0 \Leftrightarrow \alpha \in \sigma(1)\}.$$

In particular, $I_{\{\mathbf{0}\}} = \{\mathbf{0}\}$ and elements of I_σ have at most n non-zero coordinates and equality holds if and only if σ is maximal. Moreover, there is a bijection

$$\mathcal{X}_*(U) \longrightarrow \sqcup_{\sigma\in\Sigma} I_\sigma \tag{5.4}$$

sending $\mathbf{0} \in \mathcal{X}_*(U)$ to $\mathbf{0} \in I_{\{\mathbf{0}\}}$ and any $\boldsymbol{m}$ to its unique representation $\sum_\alpha n_\alpha \rho_\alpha$ with $(n_\alpha) \in I_\sigma$ for σ containing $\boldsymbol{m}$ in its relative interior.

Notation 5.1. In the rest of the paper, for every $\sigma \in \Sigma$ we identify I_σ with its image in $\mathcal{X}_*(U)$ (as commutative semi-groups) via (5.4). Therefore we are in the situation described in Remark 2.4.

This convention will allows us to make use of the mixed variant of motivic Euler product from Definition 2.6, in particular when defining our motivic height function.

5.3. **Local and global degrees.** Before introducing the motivic height zeta function of the toric variety associated to Σ, we discuss the construction of the height at the level of rational and adelic points. This construction guides us towards the parameterisation we use next.

5.3.1. *Degree on* $\mathbf{G}_m$. Let $\mathscr{C}$ be a smooth projective geometrically irreducible curve over k, and let $F = k(\mathscr{C})$ its function field. For every closed point $v \in \mathscr{C}$, the completion F_v is a valued field, and the corresponding valuation v induces a local degree map:

$$\begin{aligned} \deg_{F,v} : \mathbf{G}_m(F_v) &\to \mathbf{Z} \\ x_v &\mapsto v(x_v) \end{aligned}$$

Combining these for all places v, we get the global degree

$$\mathrm{div}_F : \left\{ \begin{array}{rcl} \mathbf{G}_m(\mathbb{A}_F) & \longrightarrow & \mathrm{Div}(\mathscr{C}) \\ (x_v) & \longmapsto & \sum_{v\in|\mathscr{C}|} \deg_{F,v}(x_v)[v] \end{array} \right.$$

whose fiber above a $D = \sum_v d_v[v]$ is nothing else but $\prod_v t_v^{d_v} \mathcal{O}_v^\times$.

5.3.2. *Degree on an algebraic torus.* Let U be a split algebraic torus of dimension n. One can compose div_F with any $\boldsymbol{m} \in \mathcal{X}^*(U) = \mathrm{Hom}(U, \mathbf{G}_m)$. This provides a new morphism

$$\mathrm{div}_U : \left\{ \begin{array}{rcl} U(\mathbb{A}_F) & \longrightarrow & \mathrm{Hom}_{\mathrm{gp}}(\mathcal{X}^*(U), \mathrm{Div}(\mathscr{C})) = \oplus_{v\in|\mathscr{C}|}\mathcal{X}_*(U) \\ (x_v) & \longmapsto & [\boldsymbol{m} \mapsto \mathrm{div}_F(\boldsymbol{m}((x_v)))]. \end{array} \right.$$

By composing with the Abel-Jacobi morphism $\mathrm{Div}(\mathscr{C}) \to \mathrm{Pic}(\mathscr{C})$, we also get a map

$$\overline{\mathrm{div}_U} : \left\{ \begin{array}{rcl} U(\mathbb{A}_F) & \longrightarrow & \mathrm{Hom}_{\mathrm{gp}}(\mathcal{X}^*(U), \mathrm{Pic}(\mathscr{C})) \\ (x_v) & \longmapsto & [\boldsymbol{m} \mapsto [\mathrm{div}_F(\boldsymbol{m}((x_v)))]]. \end{array} \right.$$

The fiber of $\overline{\mathrm{div}_U}$ above $0 \in \mathrm{Hom}_{\mathrm{gp}}(\mathcal{X}^*(U), \mathrm{Pic}(\mathscr{C}))$ is exactly the set $U(F)$. Composing div_U with the degree morphism

$$\left\{ \begin{array}{rcl} \oplus_{v\in|\mathscr{C}|}\mathcal{X}_*(U) & \longrightarrow & \mathcal{X}_*(U) \\ \sum_{v\in|\mathscr{C}|} \boldsymbol{m}_v[v] & \longmapsto & \sum_{v\in|\mathscr{C}|} \boldsymbol{m}_v \deg(\kappa(v):k), \end{array} \right.$$

we also obtain a degree morphism $\mathbf{deg}_U : U(\mathbb{A}_F) \to \mathcal{X}_*(U)$. The logarithmic multi-height of a point $x \in U(\mathbb{A}_F)$ is defined to be

$$h(x, \mathbf{T}) = \langle \cdot, \mathbf{deg}_U(x) \rangle_\Sigma \in \mathrm{PL}(\Sigma)^\vee.$$

5.3.3. *Parameterisation of* $\oplus_{v \in |\mathscr{C}|} \mathcal{X}_*(U)$. Using the bijection given by (5.4) and Notation 5.1, we can parameterise elements of

$$\mathrm{Hom}(\mathcal{X}^*(U), \mathrm{Div}(\mathscr{C})) \simeq \oplus_{v \in |\mathscr{C}|} \mathcal{X}_*(U)$$

by the family of mixed symmetric products of the form

$$\mathbf{Sym}_{/k}^{(\pi_\sigma)_{\sigma \neq \mathbf{0}}}(\mathscr{C})$$

where for all $\sigma \neq \mathbf{0}$, π_σ is a partition (which can be trivial) indexed by I_σ. Similarly, the algebro-geometric analogue of

$$\mathrm{Hom}(\mathcal{X}^*(U), \mathrm{Pic}(\mathscr{C}))$$

is

$$\mathbf{Hom}_{k-\mathrm{gp}}(\underline{\mathcal{X}^*(U)}_k, \mathbf{Pic}(\mathscr{C}))$$

which is isomorphic to a product of n copies of $\mathbf{Pic}(\mathscr{C})$. Using the splitting

$$\mathbf{Pic}(\mathscr{C}) \simeq \mathbf{Pic}^0(\mathscr{C}) \oplus \underline{\mathbf{Z}}_k$$

given by $\mathscr{O}_{\mathscr{C}}(\mathfrak{D}_1)$, we see that a point of this latter group scheme is determined by an element of $\mathcal{X}_*(U)$ (its tuple of degrees) and an n-tuple of linear classes in $\mathbf{Pic}^0(\mathscr{C})$. Finally, the morphism

$$\mathrm{Hom}(\mathcal{X}^*(U), \mathrm{Div}(\mathscr{C})) \longrightarrow \mathrm{Hom}(\mathcal{X}^*(U), \mathrm{Pic}(\mathscr{C}))$$

also admits an algebro-geometric incarnation:

$$\begin{array}{rcl} \mathbf{Sym}_{/k}^{(\pi_\sigma)_{\sigma \neq \mathbf{0}}}(\mathscr{C})_* & \longrightarrow & \mathbf{Pic}(\mathscr{C})^n \simeq \mathbf{Hom}_{k-\mathrm{gp}}(\underline{\mathcal{X}^*(U)}_k, \mathbf{Pic}(\mathscr{C})) \\ (E_\sigma = \sum_v \boldsymbol{m}_{\sigma,v}[v]) & \longmapsto & [\sum_\sigma E_\sigma] \end{array} \tag{5.5}$$

whose fibre above $\mathbf{0}$ encodes n-tuples of principal divisors.

5.3.4. *Effective degrees.* We also have (abstract) group morphisms

$$\gamma^\vee : \mathrm{Hom}(\mathrm{PL}(\Sigma), \mathrm{Div}(\mathscr{C})) \simeq \oplus_{v \in |\mathscr{C}|} \mathrm{PL}(\Sigma)^\vee \longrightarrow \mathrm{Hom}(\mathcal{X}^*(U), \mathrm{Div}(\mathscr{C})) \simeq \oplus_{v \in |\mathscr{C}|} \mathcal{X}_*(U)$$

and

$$\gamma^\vee : \mathrm{Hom}(\mathrm{PL}(\Sigma), \mathrm{Pic}(\mathscr{C})) \longrightarrow \mathrm{Hom}(\mathcal{X}^*(U), \mathrm{Pic}(\mathscr{C}))$$

induced by the composition by the group morphism $\gamma : \mathcal{X}^*(U) \to \mathrm{PL}(\Sigma)$ from (5.3). In our algebro-geometric setting, elements of $\mathrm{Hom}(\mathrm{PL}(\Sigma), \mathrm{Div}(\mathscr{C}))$ can be parameterised by symmetric products with partitions indexed by $\mathrm{PL}(\Sigma)^\vee - \{\mathbf{0}\}$. In practice these partitions will always be supported on

$$\mathrm{PL}(\Sigma)^\vee_+ = \mathrm{PL}(\Sigma)^\vee \cap \mathbf{N}^{\Sigma(1)}$$

since the local intersection degrees of a morphism $\mathscr{C} \to X$ with the effective divisors $\mathfrak{D}_\alpha$ will always be non-negative. Applying γ to the two different index sets $\mathrm{PL}(\Sigma)^\vee - \{\mathbf{0}\}$ and $\mathcal{X}^*(U) - \{\mathbf{0}\}$ gives us a collection of piecewisely defined morphisms

$$\gamma^\vee : \mathbf{Sym}_{/k}^{\boldsymbol{d}}(\mathscr{C}) := \coprod_{\varpi \vdash \boldsymbol{d}} \mathbf{Sym}_{/k}^{\varpi}(\mathscr{C})_* \longrightarrow \mathbf{Sym}_{/k}^{\gamma^\vee(\boldsymbol{d})}(\mathscr{C}) := \coprod_{\pi \vdash \gamma^\vee(\boldsymbol{d})} \mathbf{Sym}_{/k}^{\pi}(\mathscr{C})_* \tag{5.6}$$

for any $\boldsymbol{d} \in \mathrm{PL}(\Sigma)^\vee$. Again, for our application our partitions for the left hand side will be indexed by $\mathrm{PL}(\Sigma)^\vee_+ - \{\mathbf{0}\}$ and therefore in that case $\boldsymbol{d} \in \mathrm{PL}(\Sigma)^\vee_+$. Eventually (5.6) can be refined

once more by pulling it back to the mixed products of the form $\mathbf{Sym}_{/k}^{(\pi_\sigma)_{\sigma\neq\mathbf{0}}}(\mathscr{C})_*$. Moreover these morphisms are compatible with their counterpart at the level of linear classes

$$\gamma^\vee : \mathbf{Hom}_{k-\mathrm{gp}}(\underline{\mathrm{PL}(\Sigma)}_k, \mathbf{Pic}(\mathscr{C})) \longrightarrow \mathbf{Hom}_{k-\mathrm{gp}}(\underline{\mathcal{X}^*(U)}_k, \mathbf{Pic}(\mathscr{C})). \tag{5.7}$$

The fibre of (5.7) above any point of $\mathbf{Hom}_{k-\mathrm{gp}}(\underline{\mathcal{X}^*(U)}_k, \mathbf{Pic}(\mathscr{C}))$ is isomorphic to a product of n copies of $\mathbf{Pic}^0(\mathscr{C})$. Indeed, it is given by the subgroup of linear classes $(L_\alpha)_{\alpha\in\Sigma(1)}$ satisfying

$$\otimes_\alpha L_\alpha^{n_\alpha} = \mathscr{O}_{\mathscr{C}}$$

for every $(n_\alpha) \in \mathcal{X}^*(U)$.

Remark 5.2. Since the fan Σ is complete, the morphism γ is already surjective when the index set is $\mathrm{PL}(\Sigma)_+^\vee - \{\mathbf{0}\}$.

Remark 5.3. Two $\boldsymbol{d}$ and $\boldsymbol{d}'$ have same image by $\gamma^\vee$ if and only if there exists $\delta \in \mathrm{Pic}(X)^\vee$ such that $\boldsymbol{d}' = \boldsymbol{d} + \delta$.

5.4. **Motivic height zeta function.**

Definition 5.4. The motivic height zeta function is the series

$$\zeta_H^{\mathrm{mot}}(\mathbf{T}) = \sum_{\delta\in\mathrm{Pic}(X_\Sigma)^\vee} \left[\mathbf{Hom}_k^\delta(\mathscr{C}, X_\Sigma)_U\right] \mathbf{T}^\delta.$$

Using the dual exact sequence

$$0 \to \mathrm{Pic}(X_\Sigma)^\vee \to \mathrm{PL}(\Sigma)^\vee \xrightarrow{\gamma^\vee} \mathcal{X}_*(U) \to 0$$

of (5.3), we can see any $\delta \in \mathrm{Pic}(X)^\vee$ as a tuple of integers $\boldsymbol{d} \in \mathrm{PL}(\Sigma)^\vee$ via the identification $\mathrm{PL}(\Sigma) \simeq \oplus_{\alpha\in\Sigma(1)}\mathbf{Z}\mathfrak{D}_\alpha$. Since the D_α are effective divisors, the image of $\mathbf{deg}(f) \in \mathrm{Pic}(X)^\vee$ lies in $\mathrm{PL}(\Sigma)_+^\vee$ for every $f : \mathscr{C} \to X$. More precisely, for every $\boldsymbol{d} \in \mathrm{Pic}(X)^\vee \cap \mathrm{PL}(\Sigma)_+^\vee$ there is a constructible morphism

$$\mathbf{Hom}_k^{\boldsymbol{d}}(\mathscr{C}, X_\Sigma)_U \longrightarrow \mathbf{Sym}_{/k}^{\boldsymbol{d}}(\mathscr{C})$$

sending f to the tuple of divisors $(f^*D_\alpha)_{\alpha\in\Sigma(1)}$. We can rewrite $\zeta_H^{\mathrm{mot}}(\mathbf{T})$ as

$$\zeta_H^{\mathrm{mot}}(\mathbf{T}) = \sum_{\boldsymbol{d}\in\mathrm{PL}(\Sigma)_+^\vee\cap\mathrm{Pic}(X)^\vee} \sum_{\boldsymbol{E}\in\mathbf{Sym}_{/k}^{\boldsymbol{d}}(\mathscr{C})} \left[f \in \mathbf{Hom}_k^{\boldsymbol{d}}(\mathscr{C}, X_\Sigma)_U \mid (f^*D_\alpha)_{\alpha\in\Sigma(1)} = \boldsymbol{E}\right] \mathbf{T}^{\boldsymbol{d}}.$$

Using the decomposition of $\mathcal{X}_*(U)$ given by (5.4), the previous expression can be refined once more thanks to the following remark: for every closed point $v \in \mathscr{C}$, if the $\Sigma(1)$-tuple of local intersection degrees with the D_α is non-trivial, then it belongs to the relative interior of a unique cone $\sigma \in \Sigma - \{\mathbf{0}\}$. Concretely, we compose with the constructible morphism $\gamma^\vee$ from (5.6) to get

$$\zeta_H^{\mathrm{mot}}(\mathbf{T}) = \sum_{\boldsymbol{d}\in\mathrm{PL}(\Sigma)_+^\vee\cap\mathrm{Pic}(X)^\vee} \mathbf{Sym}_{/k}^{\gamma^\vee(\boldsymbol{d})}(\mathscr{C}) \left[\mathbf{Hom}_k^{\boldsymbol{d}}(\mathscr{C}, X_\Sigma)_U \to \mathbf{Sym}_{/k}^{\gamma^\vee(\boldsymbol{d})}(\mathscr{C})_*\right] \mathbf{T}^{\boldsymbol{d}}.$$

Of course, since we are summing over $\boldsymbol{d} \in \mathrm{PL}(\Sigma)_+^\vee \cap \mathrm{Pic}(X)^\vee$, we have $\gamma^\vee(\boldsymbol{d}) = \mathbf{0}$ for every term appearing in the sum, but this will not be the case on the Fourier side. It may also be convenient to set $\mathbf{Hom}_k^{\boldsymbol{d}}(\mathscr{C}, X_\Sigma)_U := \varnothing$ whenever $\boldsymbol{d} \notin \mathrm{Pic}(X)^\vee$ (equivalently we could also have defined these moduli spaces starting from any $\boldsymbol{d}$).

Lemma 5.5. *The morphism* $\mathbf{Hom}_k^{\boldsymbol{d}}(\mathscr{C}, X_\Sigma)_U \to \mathbf{Sym}_{/k}^{\gamma^\vee(\boldsymbol{d})}(\mathscr{C})$ *is a piecewise fibration in* U.

Proof. For any extension k'/k, the fibre above a k' points of $\mathbf{Sym}_{/k}^{\gamma^\vee(\boldsymbol{d})}(\mathscr{C})$ it the set of F' points of U, where $F' = k'(\mathscr{C} \times_k k')$, that is to say n-tuples of non-zero elements of F', that share the same n-tuple of divisors of zeros and poles. Such elements only differ by a k'-point of U. □

Definition 5.6. For any $(\pi_\sigma)_\sigma$ as before, let

$$\mathbf{1} : \mathbf{Sym}_{/k}^{(\pi_\sigma)_\sigma}(\mathscr{C})_* \longrightarrow \mathbf{Sym}_{/k}^{(\pi_\sigma)_\sigma}(\underline{\mathcal{X}_*(U)}_{\mathscr{C}})_*$$

be the morphism sending any

$$\boldsymbol{E} = \sum_v \boldsymbol{m}_v[v]$$

to

$$\boldsymbol{E} = \sum_v \boldsymbol{m}_v(\boldsymbol{m}_v, [v]).$$

Combining this with (5.5) and the summation operator from Section 4.4, we get the following lemma. We may sometimes write π for $(\pi_\sigma)_\sigma$ to shorten our notations.

Lemma 5.7. *For any $\boldsymbol{d} \in \mathrm{PL}(\Sigma)_+^\vee$ the relation*

$$\left[\mathbf{Hom}_k^{\boldsymbol{d}}(\mathscr{C}, X_\Sigma)_U\right] = [U]_k \sum_{(\pi_\sigma)_\sigma \vdash \gamma^\vee(\boldsymbol{d})} \sum_{\mathrm{PDiv}^\pi} \left[\mathbf{1} : \mathbf{Sym}_{/k}^{(\pi_\sigma)_\sigma}(\mathscr{C})_* \longrightarrow \mathbf{Sym}_{/k}^{(\pi_\sigma)_\sigma}(\underline{\mathcal{X}_*(U)}_{\mathscr{C}})_*\right]$$

holds in $\mathscr{M}_{\mathbf{Sym}_{/k}^{\gamma^\vee(\boldsymbol{d})}(\mathscr{C})_*}$.

Proof. Recall that we also have a morphism

$$\mathbf{Sym}_{/k}^{(\pi_\sigma)_\sigma}(\underline{\mathcal{X}_*(U)}_{\mathscr{C}})_* \longrightarrow \mathbf{Sym}_{/k}^{(\pi_\sigma)_\sigma}(\mathscr{C})_*$$

which can be composed with

$$\mathbf{1} : \mathbf{Sym}_{/k}^{(\pi_\sigma)_\sigma}(\mathscr{C})_* \longrightarrow \mathbf{Sym}_{/k}^{(\pi_\sigma)_\sigma}(\underline{\mathcal{X}_*(U)}_{\mathscr{C}})_*$$

to get

$$\boldsymbol{\varepsilon} : \mathbf{Sym}_{/k}^{(\pi_\sigma)_\sigma}(\underline{\mathcal{X}_*(U)}_{\mathscr{C}})_* \to \mathbf{Sym}_{/k}^{(\pi_\sigma)_\sigma}(\underline{\mathcal{X}_*(U)}_{\mathscr{C}})_*.$$

Now we can interpret $\mathbf{Sym}_{/k}^{(\pi_\sigma)_\sigma}(\mathscr{C})_* \longrightarrow \mathbf{Sym}_{/k}^{(\pi_\sigma)_\sigma}(\underline{\mathcal{X}_*(U)}_{\mathscr{C}})_*$ as the motivic function sending $\boldsymbol{D} \in \mathbf{Sym}_{/k}^{(\pi_\sigma)_\sigma}(\underline{\mathcal{X}_*(U)}_{\mathscr{C}})_*$ to one if $\boldsymbol{\varepsilon}(\boldsymbol{D}) = \boldsymbol{D}$ and zero otherwise. Relatively to a divisor $\boldsymbol{E} \in \mathbf{Sym}_{/k}^{(\pi_\sigma)_\sigma}(\mathscr{C})_*$, this give the indicative function $\mathbf{1}_{\boldsymbol{E}}$ of $\boldsymbol{E}$ on the set of divisors having support contained in the one of $\boldsymbol{E}$, hence the choice of notation:

$$\mathbf{1} = (\mathbf{1}_{\boldsymbol{E}})_{\boldsymbol{E} \in \mathbf{Sym}_{/k}^{(\pi_\sigma)_\sigma}(\mathscr{C})_*}.$$

Now relatively to $\boldsymbol{E} \in \mathbf{Sym}_{/k}^{(\pi_\sigma)_\sigma}(\mathscr{C})_*$ it makes sense to rewrite the fibre above $\mathbf{0}$ of the Abel-Jacobi morphism (5.5) as

$$\sum_{\mathrm{PDiv}^\pi(\mathscr{C})} \mathbf{1}_{\boldsymbol{E}}$$

hence the lemma. □

5.5. **Adelic height zeta function as a motivic Euler product.** Recall the notation for characteristic functions given in Notation 4.2.

Definition 5.8. The motivic adelic height function is the motivic Euler product

$$H(\cdot,\mathbf{T}) = \prod_{v\in\mathscr{C}} \left(\sum_{\boldsymbol{m}_v\in\mathcal{X}_*(U)} \mathbf{1}_{t^{\boldsymbol{m}_v}U(\mathcal{O})} \mathbf{T}^{\langle\,\cdot\,,\boldsymbol{m}_v\rangle_\Sigma} \right) \in \mathscr{M}_{\mathbf{Sym}^\bullet(\underline{\mathcal{X}_*(U)}_{\mathscr{C}})}.$$

which, using the definition of the pairing $\langle\cdot,\cdot\rangle_\Sigma$ becomes

$$H(\cdot,\mathbf{T}) = \left(1 + \sum_{\sigma\in\Sigma\setminus\{\mathbf{0}\}} \sum_{\mathbf{m}_v\in\sigma^\circ} \mathbf{1}_{t_v^{\mathbf{m}}U(\mathcal{O})} \prod_{\alpha\in\sigma(1)} T_\alpha^{\langle\mathfrak{D}_\alpha,\mathbf{m}_v\rangle_\Sigma} \right)$$

It is really important for what comes next to understand $H(\cdot,\mathbf{T})$ as a collection of families of motivic functions, as in Example 4.8. We can then rewrite Lemma 5.7 in a compact way as follows.

Proposition 5.9. *The motivic height zeta function may be rewritten as*

$$\zeta_H^{\mathrm{mot}}(\mathbf{T}) = [U]_k \sum_{\mathrm{PDiv}^n} H^{\mathrm{mot}}(\cdot,\mathbf{T}). \tag{5.8}$$

5.6. **Application of the Poisson formula.** Taking the Fourier transform of the motivic Euler product $H(\cdot,\mathbf{T})$ and applying Example 4.5, we obtain

$$(\mathscr{F}H)(\cdot,\mathbf{T}) = \prod_{v\in\mathscr{C}} \left(\sum_{\boldsymbol{m}_v\in\mathcal{X}_*(U)} \mathrm{ev}(\boldsymbol{m}_v,\cdot)\mathbf{T}^{\langle\,\cdot\,,\boldsymbol{m}_v\rangle_\Sigma} \right) \in \mathcal{C}har_{\mathbf{Sym}^\bullet(\mathcal{X}_*(U))}\mathscr{M}_{\mathbf{Sym}^\bullet(\mathscr{C})}.$$

in the sense of Definition 2.12.

An application of Theorem 4.16 to each of the coefficients of $\zeta_H^{\mathrm{mot}}(\mathbf{T})$, which are given by Lemma 5.7, provides the following compact expression for $\zeta_H^{\mathrm{mot}}(\mathbf{T})$.

Proposition 5.10. *The multivariate motivic height zeta function $\zeta_H^{\mathrm{mot}}(\mathbf{T})$ can be rewritten*

$$\zeta_H^{\mathrm{mot}}(\mathbf{T}) = (\mathbf{L}-1)^n \int_{\mathrm{PDiv}^\perp} (\mathscr{F}H)(\boldsymbol{\chi},\mathbf{T})\mathrm{d}\boldsymbol{\chi} \in \mathscr{M}_{\mathbf{Sym}^\bullet(\mathscr{C})}. \tag{5.9}$$

5.7. **The local factors of the Fourier transform.**

Definition 5.11. As in [BT98, Definition 3.6] and [Bou11a] we define a polynomial $Q_\Sigma(\mathbf{X})$ with $\mathbf{X} = (X_\alpha)_{\alpha\in\Sigma(1)}$ via the relation

$$\sum_{\sigma\in\Sigma} \prod_{\alpha\in\sigma(1)} \frac{X_\alpha}{1-X_\alpha} = \frac{Q_\Sigma(\mathbf{X})}{\prod_{\alpha\in\Sigma(1)}(1-X_\alpha)}.$$

The following proposition is uniform in v, hence it holds in our rings of varieties over $\mathscr{C}$. It is the motivic analogue of [BT95, Theorem 2.2.6] quoted as Théorème 4.17 in [Bou11a]. Remember that algebraic characters on the group scheme $\mathbf{Z}^n_{\kappa(v)}$ play the role of characters on the quotient $U(F_v)/U(\mathcal{O}_v)$. Recall also that $\rho_\alpha \in \mathcal{X}_*(T)$ is the generator of the ray given by $\alpha\in\Sigma(1)$, so the evaluation by $\boldsymbol{\chi}_v$ makes sense.

Proposition 5.12. *We have*

$$(\mathscr{F}H_v(\mathbf{T}_v))(\boldsymbol{\chi}_v) = \frac{Q_\Sigma(\boldsymbol{\chi}_v\mathbf{T})}{\prod_{\alpha\in\Sigma(1)}(1-\boldsymbol{\chi}_v(\rho_\alpha)T_\alpha)}$$

relatively to $\boldsymbol{\chi}_v \in \widehat{\mathcal{X}_*(T)_{\kappa(v)}} = \mathbf{G}^n_{m,\kappa(v)}$ and $v\in\mathscr{C}$.

Proof. Indeed, by definition of the local Fourier transform,

$$\begin{aligned}(\mathscr{F} H_v(\mathbf{T}_v))(\boldsymbol{\chi}_v) &= \sum_{\boldsymbol{m}_v \in \mathcal{X}_*(T)_{\kappa(v)}} H_v(\mathbf{T}_v, \boldsymbol{m}_v)\boldsymbol{\chi}_v(\boldsymbol{m}_v) \\ &= \sum_{\sigma\in\Sigma} \sum_{(n_\alpha)\in\mathbf{Z}_{>0}^{\sigma(1)}} H_v\left(\mathbf{T}_v, \sum_\alpha n_\alpha \rho_\alpha\right) \prod_{\alpha\in\sigma(1)} \boldsymbol{\chi}_v(\rho_\alpha)^{n_\alpha} \\ &= \sum_{\sigma\in\Sigma} \prod_{\alpha\in\sigma(1)} \frac{\boldsymbol{\chi}_v(\rho_\alpha)T_{v,\alpha}}{1-\boldsymbol{\chi}_v(\rho_\alpha)T_{v,\alpha}}\end{aligned}$$

hence the claim, applying the definition of Q_Σ to $X_\alpha = \boldsymbol{\chi}_v(\rho_\alpha)T_{v,\alpha}$. □

Proposition 5.13. *The multivariate motivic height zeta function* $\zeta_H^{\mathrm{mot}}(\mathbf{T})$ *can be rewritten*

$$\zeta_H^{\mathrm{mot}}(\mathbf{T}) = (\mathbf{L}-1)^n \int_{\mathrm{PDiv}^\perp} \prod_{v\in\mathscr{C}} \frac{Q_\Sigma(\boldsymbol{\chi}_v\mathbf{T})}{\prod_{\alpha\in\Sigma(1)}(1-\boldsymbol{\chi}_v(\rho_\alpha)T_\alpha)} \mathrm{d}\boldsymbol{\chi}. \tag{5.10}$$

5.8. **Splitting of** $\mathrm{PDiv}(\mathscr{C})^\perp$**.** We separate the part of characters depending on the degree from the part depending only on the degree-zero linear class. This manipulation is very classical but we have to explain how it fits into our motivic setting.

Recall that we fixed once and for all a k-divisor $\mathfrak{D}_1 = \sum \mathfrak{d}_v[v]$ of degree one on $\mathscr{C}$. Any support S being given as in Section 3.3, we can always enlarge it to include the support of $\mathfrak{D}_1$. Characters $\boldsymbol{z} : \boldsymbol{m} \mapsto \boldsymbol{z}^{\boldsymbol{m}}$ on $M = \mathcal{X}_*(U) \simeq \mathbf{Z}^n$ will be identified with

$$\boldsymbol{m} \mapsto \boldsymbol{\chi}(\boldsymbol{m}\mathfrak{D}_1) = \left(\prod_{v\in|\mathfrak{D}_1|} \boldsymbol{\chi}_v(\mathfrak{d}_v)\right)^{\boldsymbol{m}}$$

via

$$\boldsymbol{z} = \prod_{v\in|\mathfrak{D}_1|} \boldsymbol{\chi}_v(\mathfrak{d}_v\cdot).$$

We are able to write any divisor D supported on S as

$$D = (D - \deg(D)\mathfrak{D}_1) + \deg(D)\mathfrak{D}_1$$

so that for any family of characters $\boldsymbol{\chi}$ we have

$$\boldsymbol{\chi}(D) = \boldsymbol{\chi}(D - \deg(D)\mathfrak{D}_1)\boldsymbol{z}^{\deg(D)}.$$

In other words we perform the change of variables

$$\boldsymbol{\chi} \in \mathrm{D}(\mathrm{Div}_{\mathsf{S}}(\mathscr{C})) \mapsto \left(\underbrace{\boldsymbol{\chi}(\cdot - \deg(\cdot)\mathfrak{D}_1))}_{=:\boldsymbol{\chi}'}, \underbrace{\boldsymbol{\chi}(\deg(\cdot)\mathfrak{D}_1)}_{=:\boldsymbol{z}}\right) \in \mathrm{D}(\mathrm{Div}^0_{\mathsf{S}}(\mathscr{C})) \times \mathrm{D}(M).$$

Moreover, Example 2.16 allows us to move $\boldsymbol{z}$ outside the motivic Euler product.

This way we are also able to use Section 1.4 and write:

$$\int_{\mathrm{PDiv}^\perp} = \int_{D((\mathrm{Div}(\mathscr{C})/\mathrm{Div}^0(\mathscr{C}))^n} \circ \int_{(\mathrm{PDiv}^0)^\perp}.$$

Lemma 5.14. *Under the previous splitting,*

$$(\mathscr{F} H_v(\mathbf{T}_v))(\boldsymbol{\chi}'_v, \boldsymbol{z}) = \frac{Q_\Sigma(\boldsymbol{\chi}'_v \prod_\alpha \langle \boldsymbol{z}, \rho_\alpha\rangle T_\alpha)}{\prod_{\alpha\in\Sigma(1)}(1-\boldsymbol{\chi}'_v(\rho_\alpha)\prod_\alpha\langle \boldsymbol{z},\rho_\alpha\rangle T_\alpha)}.$$

In the sequel we write $\boldsymbol{\chi}$ for $\boldsymbol{\chi}'$. The previous discussion leads to the following global statement.

Proposition 5.15. *The multivariate motivic height zeta function* $\zeta_H^{\mathrm{mot}}(\mathbf{T})$ *can be rewritten*

$$\zeta_H^{\mathrm{mot}}(\mathbf{T}) = (\mathbf{L}-1)^n \int_{D(\mathcal{X}_*(U))} \int_{(\mathrm{PDiv}^0)^\perp} \prod_{v\in\mathscr{C}} \frac{Q_\Sigma(\boldsymbol{\chi}_v \prod_\alpha \langle \boldsymbol{z}, \rho_\alpha\rangle T_\alpha)}{\prod_{\alpha\in\Sigma(1)}(1-\boldsymbol{\chi}_v(\rho_\alpha)\langle \boldsymbol{z},\rho_\alpha\rangle T_\alpha)} \mathrm{d}\boldsymbol{\chi}\mathrm{d}\boldsymbol{z}. \tag{5.11}$$

6. Analysis of the integral over the Fourier domain

6.1. **Motivic $\mathfrak{L}$-functions of free Z-modules and indicator functions of cones.** We develop an elementary motivic analogue of the theory from [Bou11a, Chap. 6]. We keep most of Bourqui's notations here. Some of the proofs are very similar to the ones in the classical setting; some of them are included for the reader's convenience.

Conventions are the following: in the whole section, N is a free $\mathbf{Z}$-module, and we denote by

$$\langle\cdot,\cdot\rangle : N^\vee \times N \to \mathbf{Z}$$

the pairing with $N^\vee$. For a cone $\Lambda \subset N_{\mathbf{R}}$, we denote by $\overset{\circ}{\Lambda}$ its relative interior. In what follows, Υ will always denote a rational polyhedral cone in $N_{\mathbf{R}}$ (i.e. generated by a finite number of elements of N). Such a cone may be written as the support of a regular fan.

6.1.1. *$\mathfrak{L}$-function associated to a cone.*

Definition 6.1. Let S be a scheme, N a free $\mathbf{Z}$-module of finite rank and N_S its associated constant group scheme over S.

For every strictly convex rational polyhedral cone Υ in $N_{\mathbf{R}}$, we define the $\mathfrak{L}$-function associated to N relative to S to be the formal power series

$$\mathfrak{L}_{N,\Upsilon}(\mathbf{T}) = \sum_{y\in N\cap\Upsilon} \mathbf{T}^y \in \mathrm{K}_0\mathbf{Var}_S[[N\cap\Upsilon]] = \mathrm{K}_0\mathbf{Var}_S[[\mathbf{T}]].$$

It is the formal power series associated to the motivic function $\mathbf{1}_{N\cap\Upsilon} \in \mathrm{K}_0\mathbf{Var}_{N_S}$.

This definition naturally extends to all the variants and localisations of the Grothendieck ring of varieties we use in the present work. In particular,

$$\mathfrak{L}_{N,\Upsilon}(\boldsymbol{\chi}\mathbf{T}) = \sum_{y\in N\cap\Upsilon} \boldsymbol{\chi}(y)\mathbf{T}^y \in \mathrm{K}_0\mathbf{Char}_N\mathbf{Var}_S[[N\cap\Upsilon]].$$

More generally, if

$$\mathfrak{a} \in \mathrm{K}_0\mathbf{Char}_N\mathbf{Var}_{N_S}$$

is a motivic function with characters on N_S and A is a subset of $N_{\mathbf{R}}$, we set

$$\mathfrak{L}_{N,A,\mathfrak{a}}(\mathbf{T}) = \sum_{y\in N\cap A} \mathfrak{a}(y)\mathbf{T}^y.$$

If $\mathfrak{a}$ is identically 1, we simply write $\mathfrak{L}_{N,A}$.

Lemma 6.2. *If λ_0 lies in the relative interior of $\Upsilon^\vee$, the series*

$$\mathfrak{L}_{N,\Upsilon}\left(T^{\lambda_0}\right) = \sum_{y\in N\cap\Upsilon} T^{\langle\lambda_0,y\rangle}$$

converges for $|T|<1$ and admits a pole at $T=1$ of order at most $\dim(\Upsilon)$. Moreover, if $\dim(\Upsilon)=\mathrm{rk}(N)$, there exists a positive integer a_{λ_0} such that

$$\left((1-T^{a_{\lambda_0}})^{\mathrm{rk}(N)}\mathfrak{L}_{N,\Upsilon}\left(T^{\lambda_0}\right)\right)_{T=1} = a_{\lambda_0}^{\mathrm{rk}(N)}\mathfrak{X}_{N,\Upsilon^\vee}(\lambda_0) \in \mathbf{N}$$

where

$$\mathfrak{X}_{N,\Upsilon^\vee}(\lambda) = \int_\Upsilon \exp(-\langle\lambda,y\rangle)\mathrm{d}y$$

with $\mathrm{d}y$ being the Lebesgue measure on $N_{\mathbf{R}}^\vee$ normalised by $N^\vee$.

Proof. We can write Υ as the support of a regular fan Δ to get

$$\mathfrak{L}_{N,\Upsilon}\left(T^{\lambda_0}\right) = \sum_{\delta\in\Delta}\prod_{\ell\in\delta(1)}\left(\frac{1}{1-T^{\langle\lambda_0,\rho_\ell\rangle}}-1\right)$$

where ρ_ℓ is the generator corresponding to $\ell \in \Delta(1)$. Furthermore, one can show [Bou11a, Remarque 5.3] that

$$\mathfrak{X}_{N,\Upsilon^\vee}(\lambda_0) = \sum_{\substack{\delta\in\Delta\\ \dim(\delta)=\mathrm{rk}(N)}}\prod_{\ell\in\delta(1)}\frac{1}{\langle\lambda_0,\rho_\ell\rangle}.$$

Therefore we take

$$a_{\lambda_0} = \mathrm{lcm}_{\ell\in\delta(1)}\langle\lambda_0,\rho_\ell\rangle$$

and check that

$$(1-T^{a_{\lambda_0}})^{\mathrm{rk}(N)}\sum_{\substack{\delta\in\Delta\\ \dim(\delta)=\mathrm{rk}(N)}}\prod_{\ell\in\delta(1)}\left(\frac{1}{1-T^{\langle\lambda_0,\rho_\ell\rangle}}-1\right)$$

takes the value

$$a_{\lambda_0}^{\mathrm{rk}(N)}\mathfrak{X}_{N,\Upsilon^\vee}(\lambda_0)$$

at $T=1$. Moreover it is clear from the definitions that $a_{\lambda_0}^{\mathrm{rk}(N)}\mathfrak{X}_{N,\Upsilon^\vee}(\lambda_0)$ is a non-negative integer. □

We assume that we are given an exact sequence of $\mathbf{Z}$-modules, all free of finite rank,

$$0\longrightarrow M\xrightarrow{i}N\xrightarrow{j}\Gamma\longrightarrow 0.$$

This exact sequence induces an exact sequence of dual modules

$$0\longrightarrow\Gamma^\vee\xrightarrow{j^\vee}N^\vee\xrightarrow{i^\vee}M^\vee\longrightarrow 0$$

and an exact sequence of algebraic tori

$$0\longrightarrow D(\underline{\Gamma}_S)\xrightarrow{D(j_S)}D(\underline{N}_S)\xrightarrow{D(i_S)}D(\underline{M}_S)\longrightarrow 0.$$

Given a cone $\Lambda\subset N_{\mathbf{R}}$, this allows us to give a meaning to:

$$\mathfrak{L}_{N,\Lambda\cap M_{\mathbf{R}}}(\mathbf{T}) = \sum_{y\in\Lambda\cap i(M)\cap N}\mathbf{T}^y = \sum_{y\in\Lambda\cap M}\mathbf{T}^{i(y)} = \sum_{y\in\Lambda\cap M}i^\vee(\mathbf{T}^y) = \mathfrak{L}_{M,\Lambda\cap M_{\mathbf{R}}}(i^\vee(\mathbf{T})).$$

The following is a motivic version of Bourqui's "lemme technique : forme jouet" [Bou11a, Lemme 6.14].

Lemma 6.3. *Let S be a scheme. We have*

$$\int_{D(\underline{\Gamma}_S)}\mathfrak{L}_{N,\Lambda}(D(j_S)(\boldsymbol{\chi})\mathbf{T})\mathrm{d}\boldsymbol{\chi} = \mathfrak{L}_{N,\Lambda\cap i(M_{\mathbf{R}})}(\mathbf{T})$$

in $\mathrm{K}_0\mathbf{Var}_S[[N\cap\Upsilon]]$.

Proof. The proof is completely formal and goes exactly as the one of [Bou11a, Lemma 6.14]. We can directly compute the left hand side to get

$$\begin{aligned}\int_{D(\underline{\Gamma}_S)} \mathfrak{L}_{N,\Lambda}(D(j_S)(\boldsymbol{\chi})\mathbf{T})\mathrm{d}\boldsymbol{\chi} &= \int_{D(\underline{\Gamma}_S)} \sum_{y\in\Lambda\cap N} \boldsymbol{\chi}(j(y))\mathbf{T}^y \mathrm{d}\boldsymbol{\chi} \\ &= \sum_{y\in\Lambda\cap N} \mathbf{T}^y \int_{D(\underline{\Gamma}_S)} \boldsymbol{\chi}(j(y))\mathrm{d}\boldsymbol{\chi} \\ &= \sum_{\substack{y\in\Lambda\cap N\\ j(y)=0}} \mathbf{T}^y \qquad\qquad \text{(by def. of } \textstyle\int_{D(\underline{\Gamma}_S)}) \\ &= \mathfrak{L}_{N,\Lambda\cap i(M_{\mathbf{R}})}(\mathbf{T})\end{aligned}$$

which is what we wanted. □

Notation 6.4. From now on, we fix once and for all a basis $(\lambda_i)_{i\in I}$ of N and take Λ to be the cone generated by this basis.

Definition 6.5. An *elementary admissible motivic function of non-negative multiplicity* is a motivic formal series

$$\mathfrak{L}_{N,\Lambda,\mathfrak{a}}(\mathbf{T}) = \sum_{y\in N\cap\Lambda} \mathfrak{a}(y)\mathbf{T}^y$$

converging for $|\mathbf{T}| < \mathbf{L}^{\varepsilon}$ for some $\varepsilon > 0$.

Whenever r is a non-negative integer, an *elementary admissible motivic function of multiplicity at least* $-r$ is a motivic function f on Λ admitting a decomposition

$$f(\mathbf{T}) = g(\mathbf{T})\mathfrak{L}_{N',\overset{\circ}{\Upsilon},\mathfrak{a}}(\mathbf{T})$$

where g is an elementary admissible function of non-negative multiplicity, Υ is a rational polyhedral subcone of $N_{\mathbf{R}}$ of dimension at most r and N' is a subgroup of N.

Admissible motivic functions of multiplicity at least $-r$ are finite linear combinations of such elementary functions.

6.1.2. *Decomposition lemmas and bounds on degrees.* Recall that $\Lambda \subset N_{\mathbf{R}}$ is the simplicial cone generated by $(\lambda_i)_{i\in I}$. Let $(\lambda_i^\vee)_{i\in I}$ be the basis of $N^\vee$ induced by $(\lambda_i)_{i\in I}$ and define $\lambda^\vee = \sum_{i\in I}\lambda_i^\vee$. We also fix a rational polyhedral cone $\Upsilon \subset N_{\mathbf{R}}$ contained in Λ, and write Υ as the support of a regular fan Δ. Following Bourqui [Bou11a, §6.4.1], we define for any subset $K \subset I$, any cone $\delta \in \Delta$ and any element $z \in \Lambda \cap N$,

$$\delta(K,z) = \{y \in \overset{\circ}{\delta} \mid \forall\, i \in K, \quad \langle \lambda_i^\vee, y\rangle < \langle \lambda_i^\vee, z\rangle\}$$

as well as

$$\delta(1)_K = \{\ell \in \delta(1) \mid \forall\, i \in K, \quad \langle \lambda_i^\vee, \rho_\ell\rangle = 0\}.$$

We denote by δ_K the cone spanned by rays in $\delta(1)_K$ and by δ^K the cone spanned by rays of δ not in $\delta(1)_K$.

The following lemma is an immediate extension of [Bou11a, Lemme 6.5] to our setting.

Lemma 6.6. *For any $z \in \Lambda \cap N$ we have*

$$\mathfrak{L}_{N,\Upsilon\cap(z+\Lambda)}(\mathbf{T}) = \sum_{\substack{\delta\in\Delta\\ K\subset I}} (-1)^{|K|}\mathfrak{L}_{N,\delta(K,z)}(\mathbf{T}). \tag{6.1}$$

We will also reemploy the following (see [Bou11a, p. 123] for a proof).

Lemma 6.7 ([Bou11a, Lemme 6.5])**.** *Let δ be a cone of Δ and $K \subset I$.*

(1) *There is a decomposition*

$$\delta(K,z) = \left(\overset{\circ}{\delta}_K \cap N\right) \oplus \left(\delta^K(K,z) \cap N\right).$$

(2) *The set $\delta^K(K,z) \cap N$ is finite, of cardinality bounded by*

$$\langle z, \lambda^\vee \rangle^{\mathrm{rk}(N)}.$$

(3) *Moreover, for any $y \in \delta^K(K,z) \cap N$ we have*

$$\langle y, \lambda^\vee \rangle \leqslant |I| \cdot \langle z, \lambda^\vee \rangle \cdot \mathrm{supp}_{\ell \in \Delta(1)} \langle \rho_\ell, \lambda^\vee \rangle.$$

(4) *If K is empty, then $\delta(1)_K = \delta(1)$ and $\delta^K(K,z) \cap N = \{0\}$.*

Lemma 6.8 ([Bou11a, Lemme 6.6])**.** *Let δ be a cone of Δ. Assume that*

(1) *δ has maximal dimension,*
(2) *K is a non-empty subset of I,*
(3) *and $(N/\langle \Upsilon \rangle)^\vee \cap \lambda^\vee = \{0\}$.*

Then $\delta(1)_K$ is a proper *subset of $\delta(1)$.*

6.1.3. *Convergence lemmas.*

Lemma 6.9. *Let $\mathfrak{a}$ be a motivic function on $\Lambda \cap N$ relatively to a scheme S and $\varepsilon > 0$ be such that $\mathfrak{L}_{N,\Lambda,\mathfrak{a}}$ converges for $|\mathbf{T}| < \mathbf{L}^\varepsilon$. In particular, it implies that the sum*

$$\mathfrak{L}_{N,\Lambda,\mathfrak{a}}(\mathbf{1}) = \sum_{y \in \Lambda \cap N} \mathfrak{a}(y)$$

is well-defined in the dimensional completion.

Let

$$f_1(\mathbf{T}) = \int_{D(\underline{\Gamma}_S)} \mathfrak{L}_{N,\Lambda,\mathfrak{a}}(\hat{j}(\boldsymbol{\chi})\mathbf{T}) \mathfrak{L}_{N,\Lambda}(\hat{j}(\boldsymbol{\chi})\mathbf{T}) \mathrm{d}\boldsymbol{\chi} \in \mathrm{K}_0\mathbf{Var}_S[[\mathbf{T}]].$$

This motivic formal series converges for $|\mathbf{T}| < 1$.

Assume that $\Gamma^\vee \cap \Lambda^\vee = \{0\}$. Then the series

$$f_1(\mathbf{T}) - \mathfrak{L}_{N,\Lambda,\mathfrak{a}}(\mathbf{1}) \mathfrak{L}_{N,\Lambda \cap M_{\mathbf{R}}}(\mathbf{T})$$

is an admissible motivic function of multiplicity at least $-\mathrm{rk}(M) + 1$.

Proof. First, one shows that

$$f_1(\mathbf{T}) = \sum_{y_1 \in \Lambda \cap N} \mathfrak{a}(y_1) \mathfrak{L}_{N,\Lambda \cap M \cap (y_1 + \Lambda)}(\mathbf{T}).$$

Indeed, exactly as in the proof of [Bou11a, Proposition 6.15], we can write

$$\begin{aligned} f_1(\mathbf{T}) &= \int_{D(\underline{\Gamma}_S)} \mathfrak{L}_{N,\Lambda,\mathfrak{a}}(\hat{j}(\boldsymbol{\chi})\mathbf{T})\mathfrak{L}_{N,\Lambda}(\hat{j}(\boldsymbol{\chi})\mathbf{T})\mathrm{d}\boldsymbol{\chi} \\ &= \sum_{(y_0,y_1)\in(\Lambda\cap N)^2} \mathfrak{a}(y_1)\mathbf{T}^{y_0+y_1}\int_{D(\underline{\Gamma}_S)} \boldsymbol{\chi}(j(y_0+y_1))\mathrm{d}\boldsymbol{\chi} \\ &= \sum_{y_1\in(\Lambda\cap N)} \mathfrak{a}(y_1) \sum_{\substack{y_0\in\Lambda\cap N\\ j(y_0+y_1)=0}} \mathbf{T}^{y_0+y_1} \\ &= \sum_{y_1\in(\Lambda\cap N)} \mathfrak{a}(y_1) \sum_{\substack{y\in\Lambda\cap M\\ y\in y_1+\Lambda\cap N}} \mathbf{T}^{y} \\ &= \sum_{y_1\in\Lambda\cap N} \mathfrak{a}(y_1)\mathfrak{L}_{N,\Lambda\cap M\cap(y_1+\Lambda)}(\mathbf{T}). \end{aligned}$$

One can write $\Upsilon = \Lambda\cap M_{\mathbf{R}}$ as the support of a regular fan Δ and use Lemma 6.6 to get

$$f_1(\mathbf{T}) = \sum_{\delta\in\Delta}\sum_{J\subset I}(-1)^{|J|}\mathfrak{L}_{N,\overset{\circ}{\delta}_J}(\mathbf{T}) \sum_{y_1\in\Lambda\cap N}\mathfrak{a}(y_1)\mathfrak{L}_{N,\delta^J(J,y_1)}(\mathbf{T}).$$

The main term of this series is given by $J=\varnothing$ and equals

$$\mathfrak{L}_{N,\Lambda,\mathfrak{a}}(\mathbf{1})\mathfrak{L}_{M,\Lambda\cap M_{\mathbf{R}}}(\mathbf{T}).$$

If δ and $J\neq 0$ are given, we know by Lemma 6.7 that for any $y_1\in\Lambda\cap N$ there are at most $\langle y_1,\lambda^\vee\rangle^{\mathrm{rk}(N)}$ elements in $\delta^J(J,y_1)$ and that for any such element $y\in\delta^J(J,y_1)$, the scalar product $\langle y,\lambda^\vee\rangle$ is bounded by $|I|\cdot\langle y_1,\lambda^\vee\rangle\cdot\sup_{\ell\in\Delta(1)}\langle\rho_\ell,\lambda^\vee\rangle$. It means in particular that the series

$$\sum_{y_1\in\Lambda\cap N} a_{y_1}\mathfrak{L}_{N,\delta^J(J,y_1)}(\mathbf{T})$$

is an elementary admissible function of positive multiplicity. Finally, since we assume that $\Gamma^\vee\cap\Lambda^\vee=\{0\}$, Lemma 6.8 ensures that $\delta(1)_J$ is a proper subset of $\delta(1)$. This implies that poles of $\mathfrak{L}_{N,\overset{\circ}{\delta}_K}(T)$ are controlled up to order $-(\mathrm{rk}(M)-1)$ and finally that

$$\mathfrak{L}_{N,\overset{\circ}{\delta}_J}(\mathbf{T}) \sum_{y_1\in\Lambda\cap N}\mathfrak{a}(y_1)\mathfrak{L}_{N,\delta^J(J,y_1)}(\mathbf{T})$$

is admissible of multiplicity at least $-(\mathrm{rk}(M)-1)$. □

Since the order of the pole of $\mathfrak{L}_{M,\Lambda\cap M_{\mathbf{R}}}(\mathbf{T})$ is exactly $\mathrm{rk}(M)$, we get:

Corollary 6.10. *Let λ_0 be in the interior of $\Lambda^\vee$. Then*

$$\left((1-T^{a_{\lambda_0}})^{\mathrm{rk}(M)}f_1\left(T^{\lambda_0}\right)\right)_{T=1} = a_{\lambda_0}^{\mathrm{rk}(M)}\mathfrak{L}_{N,\Lambda,\mathfrak{a}}(\mathbf{1})\mathfrak{X}_{M^\vee,i^\vee(\Lambda^\vee)}(i^\vee(\lambda_0)).$$

6.2. **Application to the motivic height zeta function.** In our application, the short exact sequence

$$0\longrightarrow M\xrightarrow{i}N\xrightarrow{j}\Gamma\longrightarrow 0$$

is given by the short exact sequence of $\mathbf{Z}$-modules

$$0\longrightarrow \mathrm{Pic}(X_\Sigma)^\vee\xrightarrow{\pi^\vee}\mathrm{PL}(\Sigma)^\vee\xrightarrow{\gamma^\vee}\mathcal{X}_*(U)\longrightarrow 0$$

dual to (5.3), and $\Lambda=(\mathrm{PL}(\Sigma)^\vee_+)_{\mathbf{R}}$. We also get for every scheme S an exact sequence of diagonalisable algebraic groups (here split tori)

$$1\longrightarrow D(\underline{\mathcal{X}_*(U)}_S)\xrightarrow{D(\gamma^\vee)}D(\underline{\mathrm{PL}(\Sigma)^\vee}_S)\xrightarrow{D(\pi^\vee)}D(\underline{\mathrm{Pic}(X_\Sigma)^\vee}_S)\longrightarrow 1.$$

Our cone $\Lambda \subset N_{\mathbf{R}}$ will be the cone generated by the dual basis $\mathfrak{D}_\alpha^\vee$, i.e. $\Lambda = \mathrm{Eff}(X_\Sigma)^\vee$.

Recall from (5.11) that we have the expression

$$\zeta_H^{\mathrm{mot}}(\mathbf{T}) = (\mathbf{L}-1)^n \int_{D(\mathcal{X}_*(U))} \int_{(\mathrm{PDiv}^0(\mathscr{C})^n)^\perp} \prod_{v\in\mathscr{C}} \frac{Q_\Sigma((\boldsymbol{\chi}_v(\rho_\alpha)\langle \boldsymbol{z},\rho_\alpha\rangle T_\alpha)_{\alpha\in\Sigma(1)})}{\prod_{\alpha\in\Sigma(1)}(1-\boldsymbol{\chi}_v(\rho_\alpha)\langle \boldsymbol{z},\rho_\alpha\rangle T_\alpha)} \mathrm{d}\boldsymbol{\chi}\mathrm{d}\boldsymbol{z}$$

where we write $\boldsymbol{\chi}_v(\rho_\alpha)$ for the fibre above $v \in \mathscr{C}$ of the algebraic cocharacter corresponding to ρ_α, that is to say

$$\boldsymbol{\chi}(\rho_\alpha) = \mathrm{ev}(\rho_\alpha, \cdot\,) = \left[\boldsymbol{\chi} \in D(\underline{\mathcal{X}_*(U)}_{\mathscr{C}}) \mapsto \boldsymbol{\chi}(\rho_\alpha) \in \mathbf{G}_m\right] \in \mathrm{K}_0\mathbf{Char}_{\mathcal{X}_*(U)}\mathbf{Var}_{\mathscr{C}},$$

to which the motivic Euler product is applied, and similarly $\langle \boldsymbol{z},\rho_\alpha\rangle$ for the motivic class of the algebraic cocharacter

$$\langle \boldsymbol{z},\rho_\alpha\rangle = \mathrm{ev}(\rho_\alpha, \cdot\,) = \left[\boldsymbol{z} \in D(\underline{\mathcal{X}_*(U)}) \simeq D(\underline{(\mathbf{Z}\mathfrak{D}_1)^n}) \mapsto \boldsymbol{z}(\rho_\alpha)\right] \in \mathrm{K}_0\mathbf{Char}_{\mathcal{X}_*(U)}\mathbf{Var}_k.$$

Using multiplicativity of motivic Euler products, we have

$$\begin{aligned}&\prod_{v\in\mathscr{C}} \frac{Q_\Sigma((\boldsymbol{\chi}_v(\rho_\alpha)\langle \boldsymbol{z},\rho_\alpha\rangle T_\alpha)_{\alpha\in\Sigma(1)})}{\prod_{\alpha\in\Sigma(1)}(1-\boldsymbol{\chi}_v(\rho_\alpha)\prod_\alpha\langle \boldsymbol{z},\rho_\alpha\rangle T_\alpha)}\\ &= \left(\prod_{\alpha\in\Sigma(1)} L(\boldsymbol{\chi}(\rho_\alpha)\langle \boldsymbol{z},\rho_\alpha\rangle, T_\alpha)\right) \prod_{v\in\mathscr{C}} Q_\Sigma((\boldsymbol{\chi}_v(\rho_\alpha)\langle \boldsymbol{z},\rho_\alpha\rangle T_\alpha)_{\alpha\in\Sigma(1)})\end{aligned}$$

which is our motivic analogue of expression (4.3.22) in [Bou11a, Lemme 4.44].

To apply the results from the previous section, let us formally define $P(\mathbf{T}\langle \boldsymbol{z},\rho\rangle)$ as

$$\left(\prod_{\alpha\in\Sigma(1)}(1-T_\alpha)\right) \int_{(\mathrm{PDiv}^0(\mathscr{C})^n)^\perp} \left(\prod_{\alpha\in\Sigma(1)} L(\boldsymbol{\chi}(\rho_\alpha)\langle \boldsymbol{z},\rho_\alpha\rangle, \mathbf{L}^{-1}T_\alpha)\right) \left(\prod_{v\in\mathscr{C}} Q_\Sigma(\boldsymbol{\chi}_v(\rho_\alpha)\langle \boldsymbol{z},\rho_\alpha\rangle \mathbf{L}^{-1}T_{\alpha,v})\right) \mathrm{d}\boldsymbol{\chi}$$

where we normalise by $\mathbf{L}$ each indeterminate T_α to later apply Lemma 6.9 more easily, so that

$$\zeta_H^{\mathrm{mot}}(\mathbf{T}) = (\mathbf{L}-1)^n \int_{D(\underline{\mathcal{X}_*(U)}_k)} \mathfrak{L}_{\mathbf{Z}^{\Sigma(1)},\mathrm{Eff}(X)^\vee}((\langle \boldsymbol{z},\rho_\alpha\rangle T_\alpha)_{\alpha\in\Sigma(1)}) P((\langle \boldsymbol{z},\rho_\alpha\rangle T_\alpha)_{\alpha\in\Sigma(1)}) \mathrm{d}\boldsymbol{z}$$

where we recall that

$$\mathfrak{L}_{\mathbf{Z}^{\Sigma(1)},\mathrm{Eff}(X)^\vee}(\mathbf{T}) = \frac{1}{\prod_{\alpha\in\Sigma(1)}(1-T_\alpha)}.$$

So far we have been working with motivic Euler product defined by (mixed) symmetric products taking (the subdivision (5.4) of) $\mathcal{X}_*(U) - \{\mathbf{0}\}$ as a set of index. It is now time to replace them by a motivic Euler product with coefficients indexed by $\mathbf{N}^{\Sigma(1)}$, which are better understood and reappear naturally. Thanks to the discussion in Section 1.4 applied to the situation of Section 5.8, we can compare the effect of $\int_{(\mathrm{PDiv}(\mathscr{C})^n)^\perp}$ and $\int_{(\mathrm{PDiv}^0(\mathscr{C})^n)^\perp}$ above distinct points of

$$\coprod_{(\pi_\sigma)_\sigma} \mathbf{Sym}_{/k}^{(\pi_\sigma)_\sigma}(\mathscr{C})_*$$

which may not share the same support. In particular, one can easily show that the output only depends on the linear class in the group scheme

$$\mathbf{Hom}_{k-\mathrm{gp}}(\underline{\mathcal{X}^*(U)}_k, \mathbf{Pic}(\mathscr{C})) \simeq \mathbf{Pic}(\mathscr{C})^n$$

by the Abel-Jacobi morphism (5.5). Moreover, this remark is fully compatible with both the algebraic splitting $\mathbf{Pic}(\mathscr{C}) \simeq \mathbf{Pic}^0(\mathscr{C}) \oplus \underline{\mathbf{Z}}_k$ and its abstract counterpart $\mathrm{Div}(\mathscr{C}) \simeq \mathrm{Div}^0(\mathscr{C}) \oplus \mathbf{Z}\mathfrak{D}_1$ that is used fibrewise in Section 5.8.

Using the definitions from Section 5.3, we have for any $\boldsymbol{d} \in \mathbf{N}^{\Sigma(1)}$ a commutative diagram

$$\begin{tikzcd}
\mathbf{Sym}^{\boldsymbol{d}}_{/k}(\mathscr{C})_* \arrow[r, "\gamma^\vee"] \arrow[d] & \mathbf{Sym}^{\gamma^\vee(\boldsymbol{d})}(\mathscr{C})_* \arrow[d] \\
\mathbf{Hom}_{k-\mathrm{gp}}(\underline{\mathrm{PL}(\Sigma)}_k, \mathbf{Pic}(\mathscr{C})) \arrow[r, "\gamma^\vee"] & \mathbf{Hom}_{k-\mathrm{gp}}(\underline{\mathcal{X}^*(U)}_k, \mathbf{Pic}(\mathscr{C}))
\end{tikzcd}$$

and we already described the fibre of the bottom morphism (5.7) : it is isomorphic to n copies of $\mathbf{Pic}^0(\mathscr{C})$.

Putting all this together, we obtain the following lemma. Note that the class of $\mathbf{Pic}^0(\mathscr{C})$ is invertible in $\widehat{\mathscr{M}_k}^{\dim}$, since it can be obtained as a special value of $\frac{\prod_{v\in\mathscr{C}}(1-T_v)}{(1-T)(1-\mathbf{L}T)}$.

Lemma 6.11. *For all but finitely many $\boldsymbol{d} = (d_\alpha) \in \mathrm{PL}(\Sigma)^\vee_+$, the coefficient of $\mathbf{T}^{\langle\cdot,\gamma^\vee(\boldsymbol{d})\rangle}$ in*

$$\int_{(\mathrm{PDiv}^0(\mathscr{C})^n)^\perp} \left(\prod_{\alpha\in\Sigma(1)} L(\boldsymbol{\chi}(\rho_\alpha)\langle \boldsymbol{z}, \rho_\alpha\rangle, \mathbf{L}^{-1}T_\alpha)\right)\left(\prod_{v\in\mathscr{C}} Q_\Sigma(\boldsymbol{\chi}_v(\rho_\alpha)\langle \boldsymbol{z}, \rho_\alpha\rangle \mathbf{L}^{-1}T_{\alpha,v})\right) \mathrm{d}\boldsymbol{\chi}$$

equals the coefficient of $\mathbf{T}^{\boldsymbol{d}}$ in

$$\frac{1}{[\mathbf{Pic}^0(\mathscr{C})]^n}\left(\prod_{\alpha\in\Sigma(1)} Z^{\mathrm{Kapr}}_{\mathscr{C}}(\langle \boldsymbol{z}, \rho_\alpha\rangle \mathbf{L}^{-1}T_\alpha)\right)\left(\prod_{v\in\mathscr{C}} Q_\Sigma\right)((\langle \boldsymbol{z}, \rho_\alpha\rangle \mathbf{L}^{-1}T_\alpha)_\alpha) \in \mathcal{C}har_{\mathcal{X}_*(U)}\mathscr{M}_k[[(T_\alpha)_\alpha]]$$

where in the second expression the motivic Euler products are taken with $\mathbf{N}^{\Sigma(1)} \setminus \{\mathbf{0}\}$ as index set.

Remark 6.12. Note that the relation

$$\sum_{\sigma\in\Sigma}\prod_{\alpha\in\sigma(1)} \frac{X_\alpha}{1-X_\alpha} = \frac{Q_\Sigma(\mathbf{X})}{\prod_{\alpha\in\Sigma(1)}(1-X_\alpha)} \tag{6.2}$$

forces the coefficients of this latter motivic Euler product to be actually mixed symmetric product with respect to the collection of subsets $I_\sigma \subset \mathbf{N}^{\Sigma(1)}$.

Lemma 6.13. *The motivic Euler product*

$$\prod_{v\in\mathscr{C}} Q_\Sigma((T_\alpha)_{\alpha\in\Sigma(1)})$$

converges for $|\mathbf{T}| < \mathbf{L}^{-\frac{1}{2}}$.

Moreover, its value at $T_\alpha = \mathbf{L}^{-1}$ is given by the convergent motivic Euler product

$$\prod_{v\in\mathscr{C}}\left((1-\mathbf{L}^{-1})^{\mathrm{rk}(\mathrm{Pic}(X_\Sigma))}\frac{[X_\Sigma]}{\mathbf{L}^{\dim(X_\Sigma)}}\right) \in \widehat{\mathscr{M}_k}^{\dim}.$$

Proof. Recall that Q_Σ was defined by (6.2). By [BT95, Proposition 2.2.3], the polynomial $Q_\Sigma - 1$ contains only monomials of degree ≥ 2, from which we can deduce the convergence statement.

Moreover,

$$\begin{aligned}
Q_\Sigma(\mathbf{L}^{-1}) &= \sum_{\sigma\in\Sigma}\prod_{\alpha\in\sigma(1)}\mathbf{L}^{-1}\prod_{\alpha\notin\sigma(1)}(1-\mathbf{L}^{-1})\\
&= \mathbf{L}^{-|\Sigma(1)|}\sum_{\sigma\in\Sigma}(\mathbf{L}-1)^{|\Sigma(1)|-|\sigma(1)|}\\
&= \mathbf{L}^{-\operatorname{rk}(\operatorname{Pic}(X_\Sigma))-\dim(X_\Sigma)}(\mathbf{L}-1)^{\operatorname{rk}(\operatorname{Pic}(X_\Sigma)}\underbrace{\sum_{\sigma\in\Sigma}(\mathbf{L}-1)^{\dim(X_\Sigma)-|\sigma(1)|}}_{=[X_\Sigma]}\\
&= \left(1-\mathbf{L}^{-1}\right)^{\operatorname{rk}(\operatorname{Pic}(X_\Sigma))}\frac{[X_\Sigma]}{\mathbf{L}^{\dim(X_\Sigma)}}
\end{aligned}$$

where we used the relation $|\Sigma(1)| = \operatorname{rk}(\operatorname{Pic}(X_\Sigma)) + \dim(X_\Sigma)$. □

By Lemma 6.13, we know that $\prod_{v\in\mathscr{C}} Q_\Sigma((\mathbf{L}^{-1}T_\alpha)_{\alpha\in\Sigma(1)})$ converges for $|\mathbf{T}| < \mathbf{L}^{\frac{1}{2}}$. It is also well-known that $(1-T)(1-\mathbf{L}T)Z^{\mathrm{Kapr}}_{\mathscr{C}}(\mathbf{T})$ is a polynomial, therefore $(1-T)Z^{\mathrm{Kapr}}_{\mathscr{C}}(\mathbf{L}^{-1}T)$ converges for $|T| < \mathbf{L}$. Together with Lemma 6.11, these facts ensures that $\mathfrak{L}_{N,\Lambda,\mathfrak{a}} = P$ satisfies the assumptions of Lemma 6.9 for all $0 < \varepsilon \leqslant \frac{1}{2}$. In particular, thanks to Corollary 6.10 we get the following, where $\alpha^*(X_\Sigma)$ is the value of $\mathfrak{X}_{N,\Upsilon^\vee}(\lambda_0)$ in our setting.

Theorem 6.14. *There exists an $\eta > 0$ such that the series*

$$(1-(\mathbf{L}T)^{a_{\lambda_0}})^{\operatorname{rk}(\operatorname{Pic}(X_\Sigma)}\zeta^{\mathrm{mot}}_H\left(T^{\omega^{-1}_{X_\Sigma}}\right)$$

converges for $|T| < \mathbf{L}^{-1+\eta}$. Moreover,

$$\left((1-(\mathbf{L}T)^{a_{\lambda_0}})^{\operatorname{rk}(\operatorname{Pic}(X))}\zeta^{\mathrm{mot}}_H\left(T^{\omega^{-1}_{X_\Sigma}}\right)\right)_{T=\mathbf{L}^{-1}} = a_{\lambda_0}^{\operatorname{rk}(\operatorname{Pic}(X_\Sigma))}\alpha^*(X_\Sigma)\gamma^{\mathrm{mot}}_H(X_\Sigma)$$

in $\widehat{\mathscr{M}}_k^{\dim}$ where

$$\gamma^{\mathrm{mot}}_H(X_\Sigma) = \mathbf{L}^{n(1-g)}\left(\frac{[\mathbf{Pic}^0(\mathscr{C})]\mathbf{L}^{-g}}{1-\mathbf{L}^{-1}}\right)^{\operatorname{rk}(\operatorname{Pic}(X_\Sigma))}\prod_{v\in\mathscr{C}}\left((1-\mathbf{L}^{-1})^{\operatorname{rk}(\operatorname{Pic}(X_\Sigma))}\frac{[X_\Sigma]}{\mathbf{L}^{\dim(X_\Sigma)}}\right)\in\widehat{\mathscr{M}}_k^{\dim}.$$

To deduce Theorem 2 from the previous discussion, it suffices to apply Lemma 1.24 to

$$F(\mathbf{T}) = \frac{1}{[\mathbf{Pic}^0(\mathscr{C})]^n}\left(\prod_{\alpha\in\Sigma(1)}(1-\mathbf{L}T_\alpha)Z^{\mathrm{Kapr}}_{\mathscr{C}}(T_\alpha)\right)\left(\prod_{v\in\mathscr{C}}Q_\Sigma(\mathbf{T})\right)$$

with $\boldsymbol{\rho} = (1,...,1)$.

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